

Certified Strategic Generative AI Professional (CSGAIP)

Official Training





Welcome to the CSGAIP Official Training

Certified Strategic Generative AI Professional
(CSGAIP) Certification Preparation Course

Introduction to the CSGAIP

This certification is designed for professionals who need to understand, evaluate, and responsibly deploy generative AI technologies to solve business problems and enhance workflows.

This certification demonstrates to potential employers and clients that you are familiar with the practical, real-world applications of Generative AI.

The Certified Strategic Generative AI Professional (CSGAIP) is your immediate mandate to secure future relevance.

- Certification is valid for three years. CEU credits for renewal.
- Certification cost is \$199.00 with two included retakes if needed.

Course Components

- 01 **Domain 1 – AI/ML Fundamentals (20%)**
- 02 **Domain 2 – Prompt Engineering /Model Interactions (15%)**
- 03 **Domain 3 – Enterprise Applications & Workflows (25%)**
- 04 **Domain 4 – Strategic Leadership and Business Value (25%)**
- 05 **Domain 5 – Governance, Ethics and Responsible AI (15%)**

Why Get Certified and take this Course?

Top Reasons to Join in



Why Getting Certified is Important

The CSGAIP can enable you to:

- ☐ Provide notice that your **“Hitting above the belt”** from a *generative AI perspective as a leader who is ready to strategically deploy the power of generative AI models in your organization.*
- ☐ Shows your competence in **managing AI-driven challenges.**
- ☐ Accomplishment in *sharing on social media and with prospective employers.*



Learning Reinforcements

- ☐ AI Cloud Service Demos
- ☐ Whiteboard Design Discussions
- ☐ Topic Discussions
- ☐ Example Scenarios
- ☐ Documentation Links
- ☐ Study Guide eBook
- ☐ Practice Questions
- ☐ Exam Preparation
- ☐ Community Sites



Who Should Take this Course?

This certification course is ideal for:

- ☐ Product Managers
- ☐ Business Leaders
- ☐ Solutions Architects
- ☐ Marketing Strategists
- ☐ Developers and Software Professionals
- ☐ Anyone committed to integrating cutting-edge AI into their professional workflow and career path.



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Importance of AI Strategy

Discussion



Topic Discussion



Course Materials

Course Content to download



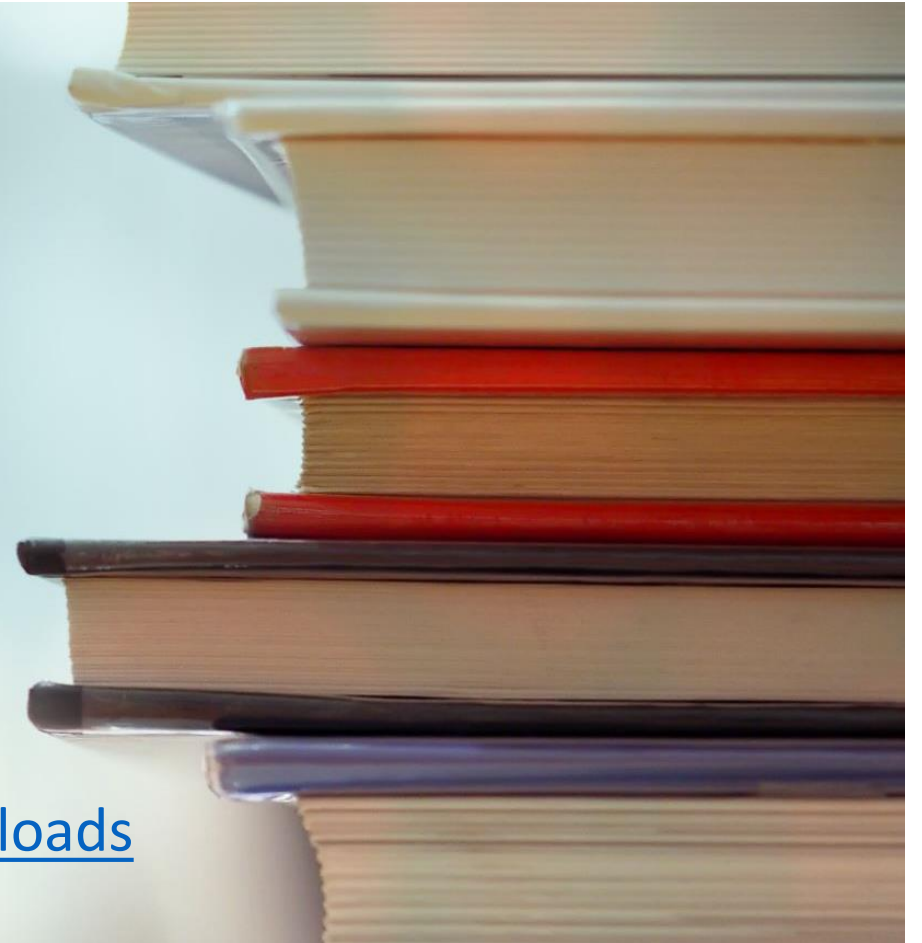
Course Content Download

Welcome to the Course Content!

- Download eBook study guide
- Exam Objectives
- Course Presentation
- Other Resources

We have a lot to cover, so let's get started!

www.digitalcrestinstitute.com/CSGENAIPdownloads



Domain 1 – AI/ML Foundations

Module 1 – CSGAIP Exam

Understanding AI services and concepts



Course Modules/Domains

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Module 1

Sections

01 Section 1.1 - AI/ML Concepts

02 Section 1.2 - Identify Generative AI Technology Stack

03 Section 1.3 - Data and Model Lifecycle Management

AI/ML Concepts

Section 1.1



Section 1.1 Lesson Overview

What is AI/ML?

What are the
Machine Learning
Types

Compare and
Contrasting AI/ML

AI Categories
Generative AI,
Predictive AI, Agentic
AI, and others.

What is Deep
Learning?

What is Natural
Language Processing
(NLP)?

Generative AI and its
Applications

Perceptual Pillars of
AI

Whiteboard - How
Does Google Gemini
Work?



What is AI/ML?

Ensuring the Fundamentals

Lesson Key Objectives

What is AI and ML?

1

What is a comparison for AI vs ML?

2

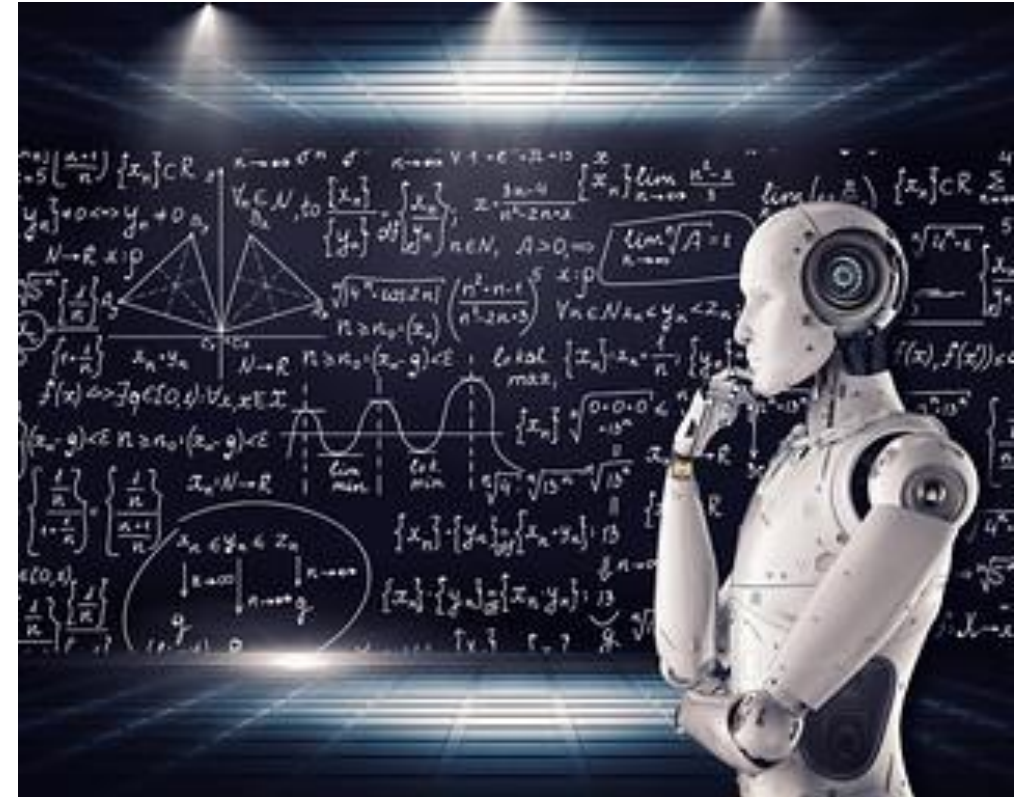
What is the primary goal of AI?

3

What is AI/ML?

Artificial Intelligence (AI) is a broad field that encompasses the creation of machines capable of performing tasks that typically require human intelligence. It aims to create machines that can mimic human cognitive abilities.

- ❑ The **primary goal** of using AI is to create systems that can perform tasks that typically require human intelligence.
- ❑ **Common Examples:** Problem-solving, learning, reasoning, perception, and language understanding.



What is AI/ML?



Machine Learning (ML) is a subfield of AI that enables systems to learn from data without being explicitly programmed. ML provides algorithms that allow computers to improve their performance on a task through experience (data).

- ❑ **ML works as follows:** Data -> Learning
Algorithm -> Model -> Predictions/Decisions.
- ❑ **Examples:** Spam filtering, image classification, recommendation systems.

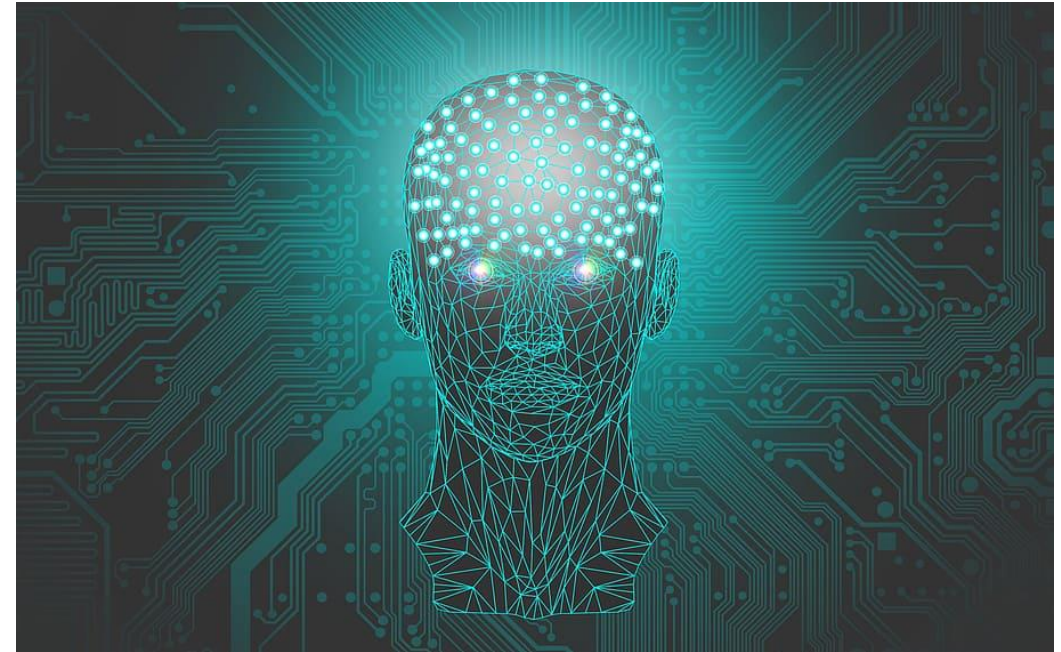
What is AI/ML?

In summary

AI is the broader concept of intelligent machines, **while ML** is a specific technique to achieve AI.

❑ **Goal:** AI aims to mimic human cognitive functions. ML aims to enable machines to learn from data.

❑ **Relationship:** All machine learning is AI, but not all AI is machine learning. There were AI systems before the widespread adoption of ML, and AI approaches still don't heavily rely on it (though ML is dominant today).





What are the Machine Learning Types?

Supervised, Unsupervised and Reinforcement

Lesson Key Objectives

What is Machine Learning?

1

What are Supervised and Unsupervised Learning

2

What is Reinforcement Learning?

3

What are the Machine Learning Approaches?

Machine Learning (ML) is a subfield of Artificial Intelligence (AI) that focuses on enabling computers to learn from data without being explicitly programmed.

- ❑ Instead of writing specific rules for every possible scenario, ML algorithms identify patterns, extract insights, and build models from data.
- ❑ These models can then make predictions, classifications, or decisions on new, unseen data.
- ❑ **Three main Types of ML Learning:** Supervised, Unsupervised, and Reinforcement



What are the Machine Learning Categories?

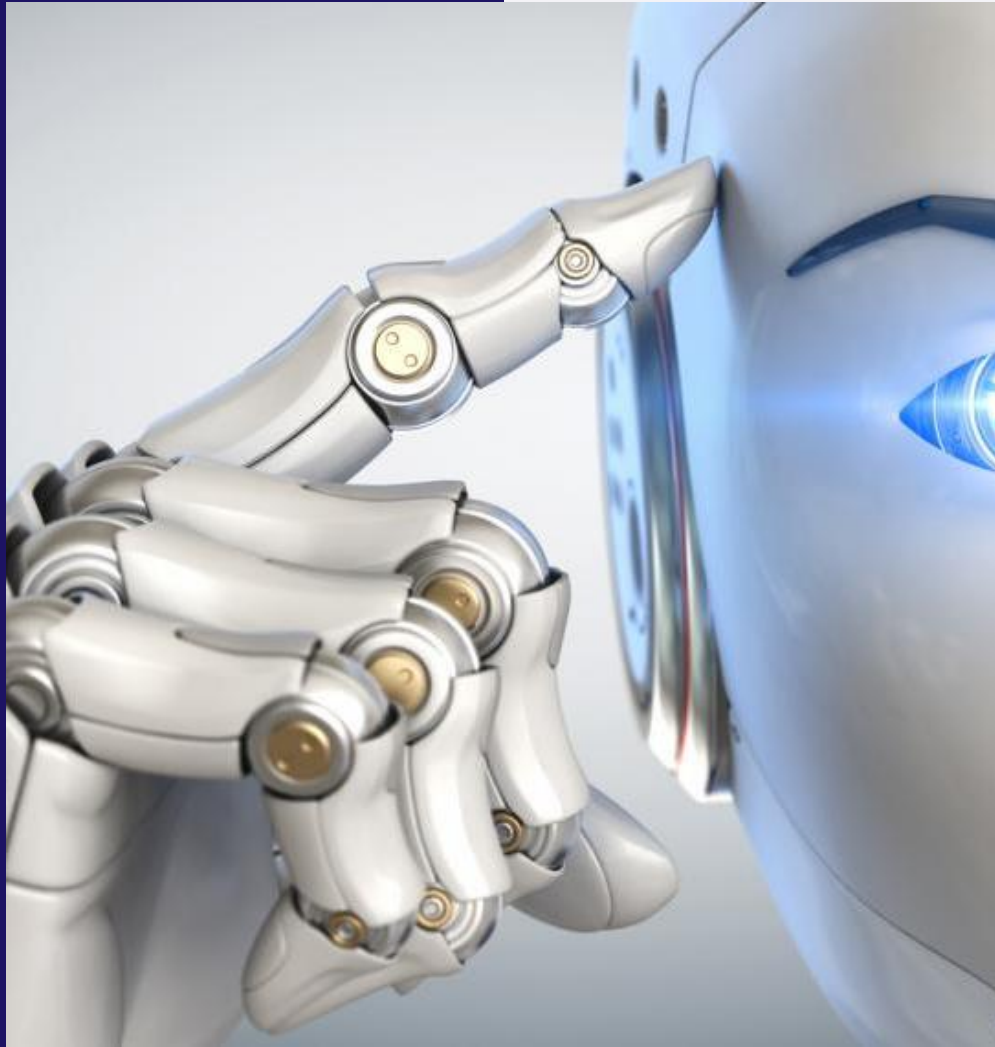


Supervised Learning: The algorithm learns from **labeled data**, meaning it provides the correct output for each input.

- ❑ The goal is to learn a mapping function to predict the output for new, unseen inputs. Examples include image classification (identifying cats vs. dogs) and regression (predicting stock prices).

Unsupervised Learning: The algorithm learns from **unlabeled data**, trying to find hidden patterns or structures without prior guidance.

- ❑ Examples include clustering (grouping customers with similar purchasing behavior) and dimensionality reduction (simplifying complex data).



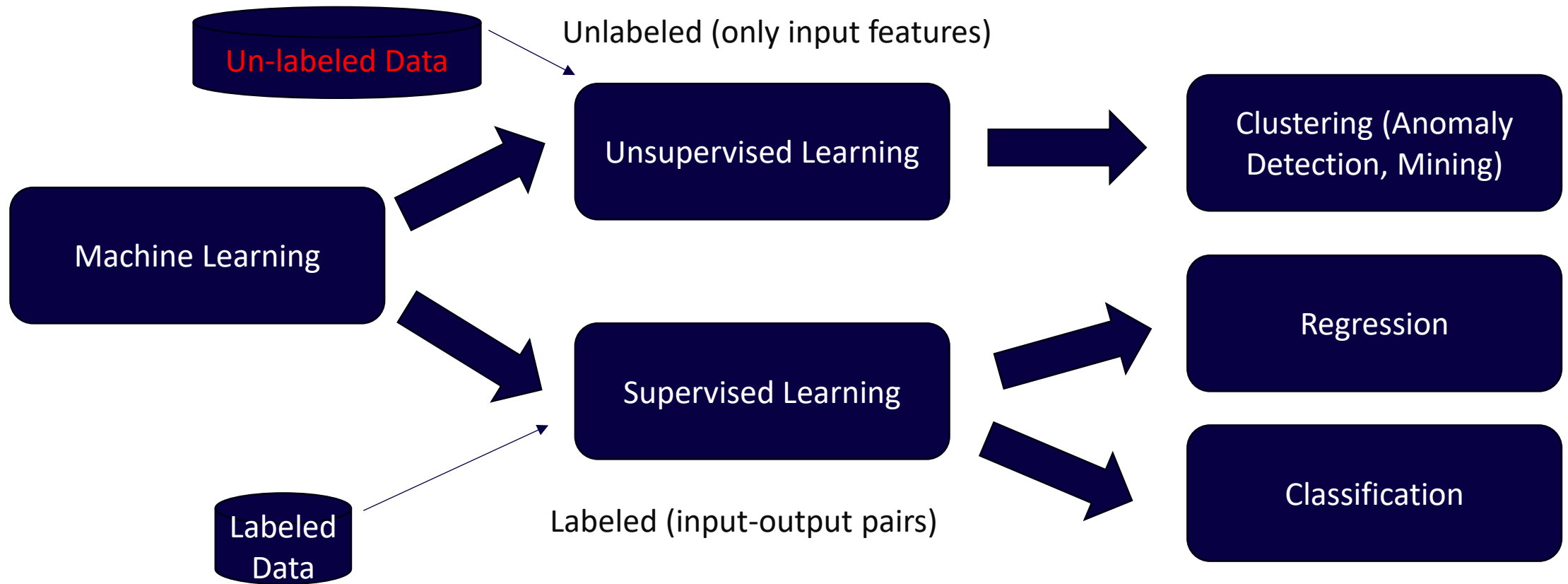
What are the Machine Learning Categories?

When understanding the difference between supervised learning and unsupervised learning, the primary difference is the type of input data used to train the model.

- ☐ **Supervised learning uses labeled training datasets** to try to teach a model a specific, **pre-defined goal**.
- ☐ By comparison, **unsupervised learning uses unlabeled data and operates autonomously** to try and learn the data structure without explicit instructions.

What are the Machine Learning Categories?

Choosing Between Supervised and Unsupervised.



What are the Machine Learning Categories?

Reinforcement Learning: An agent learns to behave in an environment by receiving rewards or penalties for its actions. Regression algorithms predict a real or continuous value, where the algorithm detects a relationship between two or more variables.

The goal is to develop a policy that maximizes the cumulative reward.

- ❑ For example, a regression task might be predicting a salary based on work experience.
- ❑ For instance, a supervised learning algorithm would be fed inputs related to work experience (e.g., length of time, the industry or field, location, etc.) and the corresponding assigned salary amount.
- ❑ After the model is trained, it could be used to predict the average salary based on work experience.

What are Machine Learning Types?

Choosing Between Supervised and Unsupervised.





Compare and Contrast AI and ML

Comparing Side by Side

Lesson Key Objectives

Compare and Contrast AI/ML

1

What is the relationship of ML to AI?

2

What are application examples of AI and ML?

3

Compare and Contrast AI/ML

Artificial Intelligence (AI) and Machine Learning (ML)

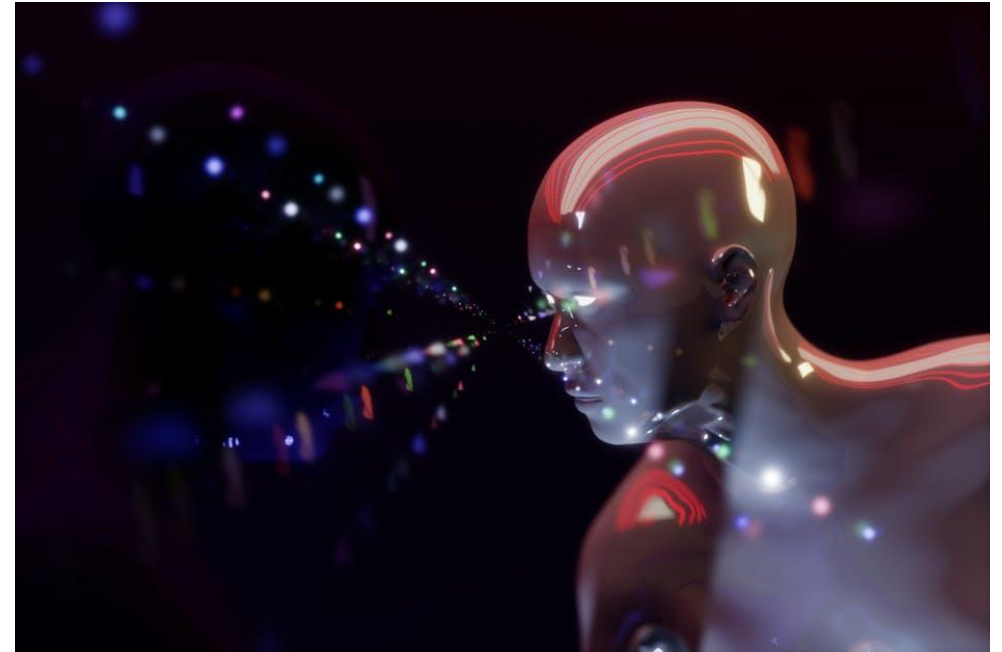
Feature	Artificial Intelligence (AI)	Machine Learning (ML)
Definition	The broad field of creating machines that can perform tasks requiring human intelligence	A subfield of AI that enables systems to learn from data without explicit programming
Scope	A much wider field encompasses various approaches to achieve "intelligent" behavior.	A specific technique within AI focused on learning from data.
Goal	To create machines that can mimic human cognitive abilities (reasoning, learning, problem-solving, perception, language understanding).	To enable machines to learn patterns from data and make predictions or decisions.
Relationship	ML is a subset of AI	Machine learning (ML) is a powerful tool and a common approach used to achieve artificial intelligence (AI).
Examples	Robotics, expert systems, game-playing AI (including those not using ML), natural language understanding systems (some rule-based), computer vision systems (some non-ML).	Spam filtering, image classification, recommendation systems, fraud detection, machine translation (often using deep learning, a subfield of ML), autonomous driving (perception and decision-making).

Compare and Contrast AI/ML

Think of AI as the overarching ambition – creating truly intelligent machines.

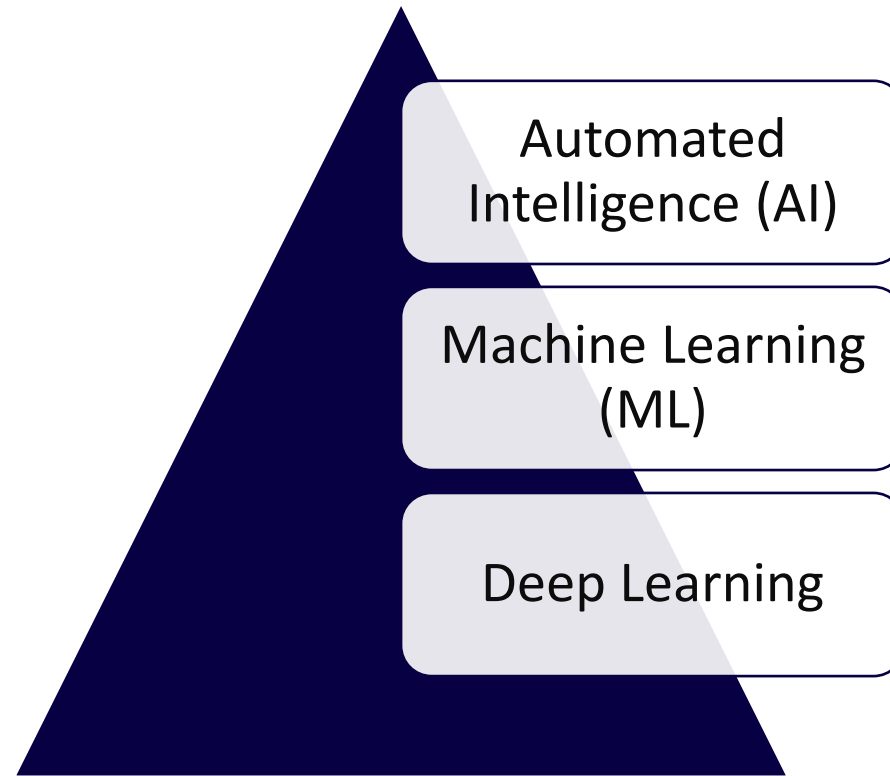
Machine Learning is one of the most successful and widely used pathways to achieve that ambition.

While AI encompasses a broader range of techniques, machine learning (ML) has become a dominant force in modern AI due to its ability to handle complex data and automatically learn intricate patterns.



What are Machine Learning Types?

In Summary





Generative AI, Predictive AI, Agentic AI, and others.

Describe and Identify AI Categories

Lesson Key Objectives

Compare contrast categories of AI

1

Identify core functions for each category of AI.

2

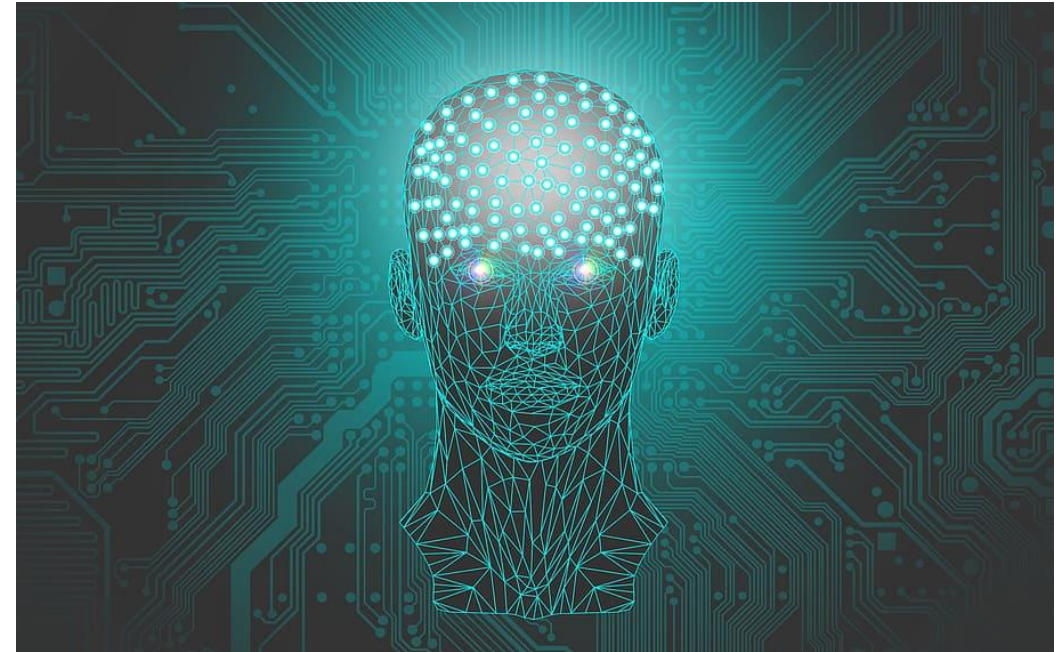
What are application examples for each category?

3

Generative AI, Predictive AI, Agentic AI, and others.

Generative AI is a type of artificial intelligence that can produce new content, such as text, images, audio, and video, by learning from vast amounts of existing data. It identifies patterns and structures within the data and then uses this knowledge to create original outputs that often resemble human-created content.

- ❑ Generative AI has numerous applications, ranging from generating creative content and assisting with writing and coding to powering chatbots and creating personalized experiences.
- ❑ Generative AI utilizes models to generate new data instances that closely resemble the training data.



Generative AI, Predictive AI, Agentic AI, and others.

Predictive AI is a branch of artificial intelligence focused on forecasting future outcomes based on historical and current data. It employs statistical analysis and machine learning techniques to identify patterns, trends, and relationships within data.

- ❑ Predictive AI models can estimate the likelihood of future events, behaviors, or values, enabling proactive decision-making and strategic planning across various industries.
- ❑ AI models that predict future outcomes or classify current data based on historical patterns.



Generative AI, Predictive AI, Agentic AI, and others.

Agentic AI refers to artificial intelligence systems that can operate autonomously to achieve specific goals without constant human oversight.

These systems can perceive their environment, reason about complex situations, make independent decisions, and take actions to reach their objectives.

- ❑ Agentic AI can learn from its experiences and adapt its behavior over time, making it a dynamic and proactive intelligence.
- ❑ AI systems are designed to perceive their environment, make decisions, and take actions to achieve specific goals. Often characterized by autonomy and interaction with the environment.



Generative AI, Predictive AI, Agentic AI, and others.

Explainable AI (XAI): Focuses on making AI decisions transparent and understandable to humans.

Conversational AI: Enables human-like interactions through natural language (chatbots, virtual assistants).

Reasoning and Planning AI: Focuses on logical inference and decision-making in complex scenarios.



Generative AI, Predictive AI, Agentic AI, and others.

Compare AI Categories

AI Category	Generative	Predictive	Agentic	Explainable	Conversational	Reasoning and Planning
Core Function	Creates new data instances that resemble the training data.	Forecasts future outcomes or classifies current data based on historical patterns.	Perceives an environment, makes decisions, and takes actions to achieve goals.	Makes AI decisions and reasoning transparent and understandable to humans.	Simulates human-like conversations through natural language	Solves complex problems through logical inference, deduction, and strategic planning.
Primary Output	New data (images, text, audio, video, code, etc.).	Predictions (numerical values, categories, probabilities).	Actions, decisions, and behaviors within an environment.	Explanations, justifications, visualizations of the AI's decision-making process.	Natural language responses, dialogue flow, understanding of user intent.	Plans, sequences of actions, solutions to problems.

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Compare AI Categories

AI Category	Generative	Predictive	Agentic	Explainable	Conversational	Reasoning and Planning
Focus	Novelty, creativity, data synthesis.	Prediction accuracy, pattern recognition for forecasting or categorization	Autonomy, interaction with an environment, goal achievement	Trustworthiness, accountability, interpretability of AI systems	User experience, natural interaction, understanding and responding to language.	Problem-solving, reasoning, decision-making
Learning Paradigm	Often uses unsupervised learning and self-supervised learning	Primarily uses supervised learning (regression and classification).	Primarily uses reinforcement learning, but can also use supervised and unsupervised learning	Not a specific learning paradigm but a goal applied across different ML/AI techniques.	Often uses supervised learning and reinforcement learning.	Can use various methods including symbolic AI, knowledge representation, search algorithms, and sometimes machine learning.
Applications	Image generation, text generation, music composition, drug discovery, etc.	Sales forecasting, fraud detection, medical diagnosis, credit risk, etc	Autonomous vehicles, robots, game-playing agents,	High-stakes decision-making, regulatory compliance, and auditing AI systems.	Chatbots, virtual assistants, customer service automation, language translation.	Automated theorem proving, logistics planning, scheduling, game playing.



What is Deep Learning

Ensuring the Fundamentals

Lesson Key Objectives

What is Deep Learning?

1

What are Artificial Neural Networks (ANNs)?

2

What are Neurons in Deep Learning?

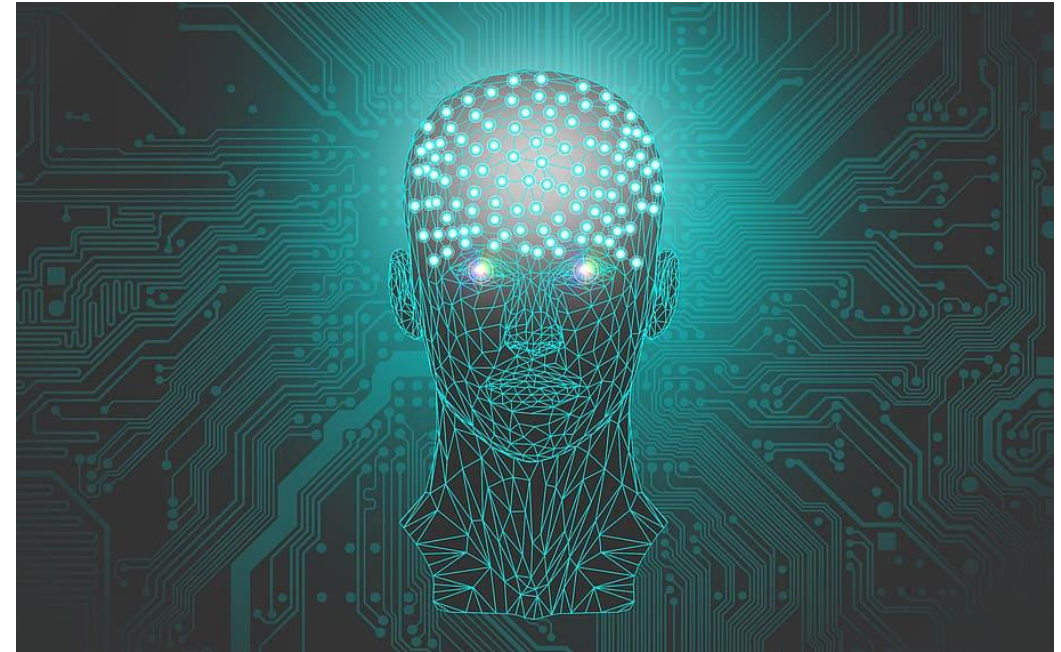
3

What is Deep Learning?

Deep Learning is a subfield of Machine Learning that utilizes artificial neural networks with multiple layers (hence "deep") to extract complex patterns from data.

The structure and function of the human brain inspired Deep Learning.

- ❑ **Applications:** Image recognition, natural language processing, speech recognition, etc.
- ❑ **Key Component:** Artificial Neural Networks (ANNs)



What is Deep Learning?

Neurons (or Perceptrons)—inputs, weights, bias, activation function, and output.

In Artificial Intelligence, specifically neural networks and deep learning, **artificial neurons** serve as the fundamental building blocks for processing information.

Inspired by their biological counterparts, these mathematical functions receive input signals, process them based on their internal weights and an activation function, and then produce an output signal.

By connecting vast numbers of these neurons in layers, AI models can learn intricate patterns and representations from data, enabling them to perform tasks like image recognition, natural language understanding, and much more.

They are the tiny decision-makers allowing AI to "think" and learn.



Natural Language Processing (NLP)

Describe and use cases

Lesson Key Objectives

What is Natural Language Processing (NLP)

1

What applications are use cases for NLP?

2

Natural Language Processing (NLP)

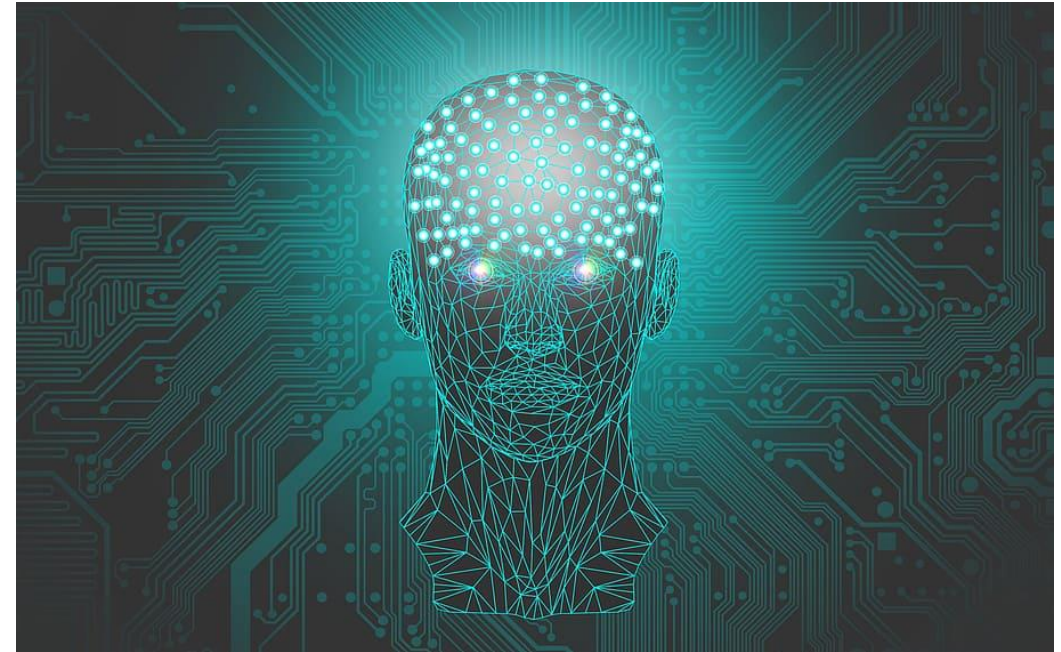
Natural Language Processing (NLP) is a branch of artificial intelligence (AI) that enables computers to understand, interpret, and generate human language.

Key Tasks are:

Text Analysis: Tokenization (splitting text into words), Part-of-speech tagging (identifying nouns, verbs, etc.) Named entity recognition (finding names of places, people, etc.)

Language Generation: Creating human-like text.

Applications used for are Machine Translation, Sentiment Analysis, Chatbots, Voice Assistants, Search Engines





Generative AI Services

Importance of Generative AI

Lesson Key Objectives

What is Generative AI?

1

How does it work?

2

Common Applications

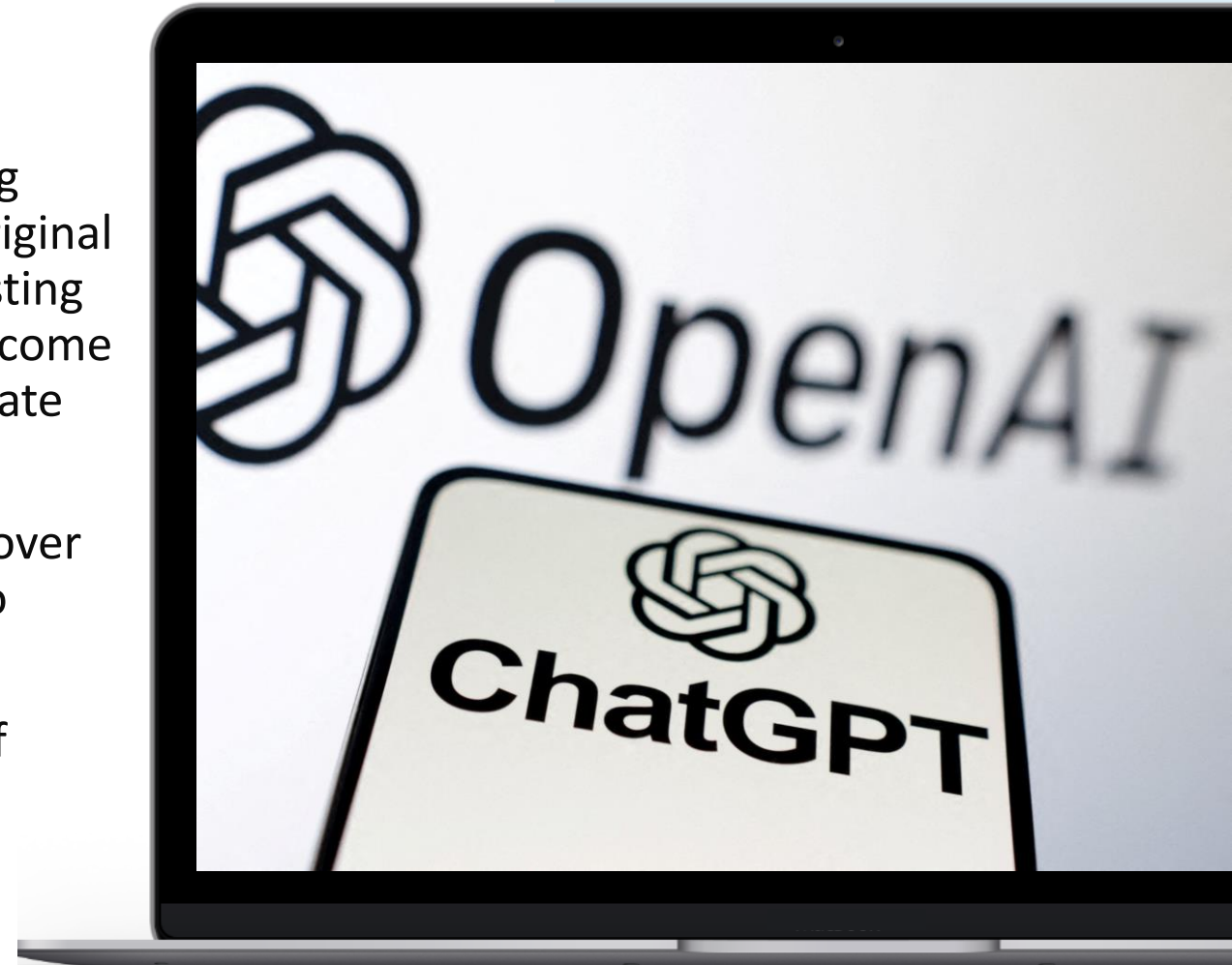
3

Generative AI

What is Generative AI?

Generative AI refers to a robust and rapidly evolving type of artificial intelligence that can create new, original content rather than just analyzing or classifying existing data. Unlike traditional AI that might predict an outcome or categorize an image, generative AI aims to generate something novel.

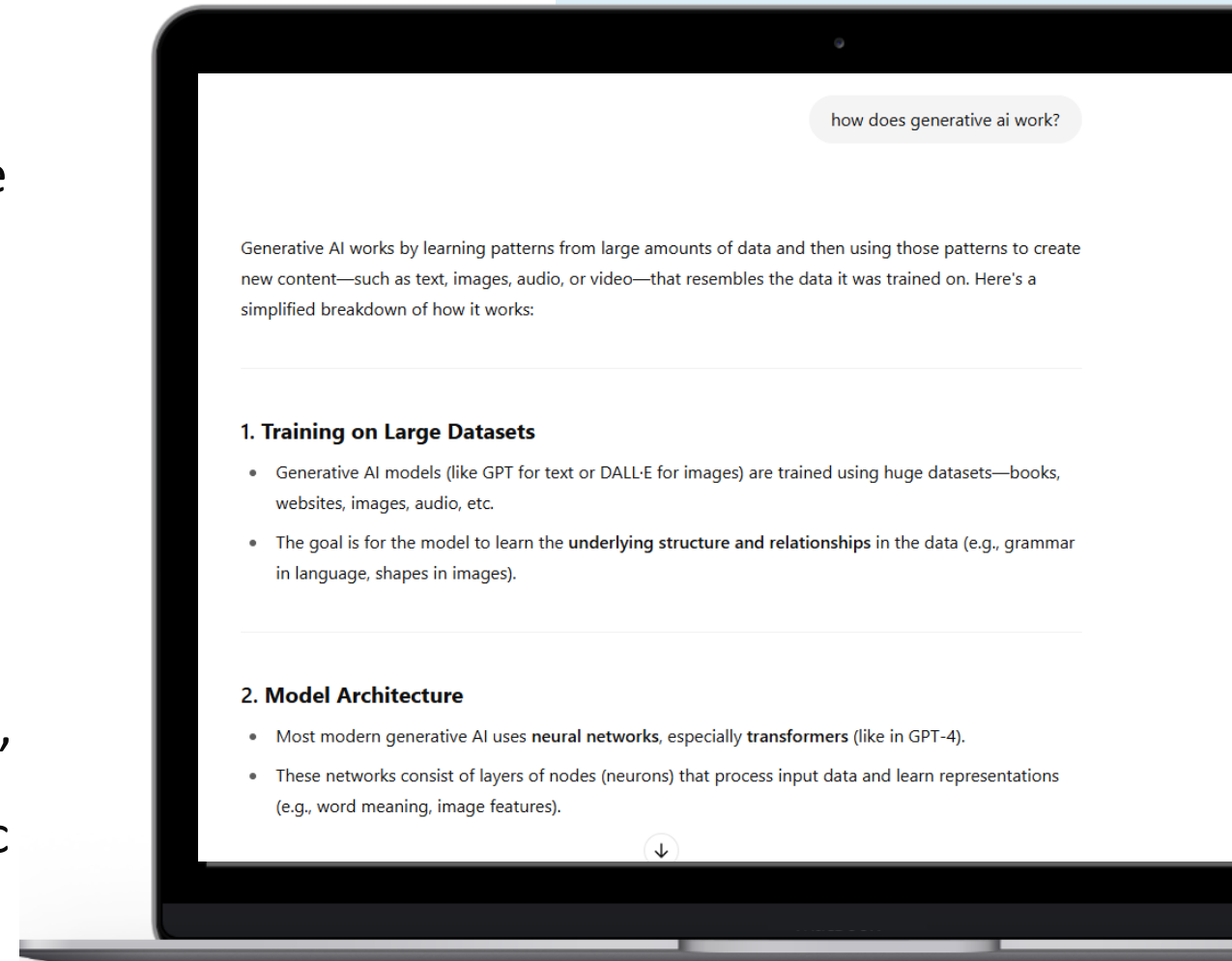
- ☐ If you ask it to "create a picture of a cat jumping over the moon in a watercolor style," it will attempt to generate that image.
- ☐ If you ask it to "write a poem about the beauty of nature," it will produce a unique poem
- ☐ Examples – ChatGPT, Claude AI, Copilot, Gemini



Generative AI

How it works

- ❑ Generative AI models are trained on massive datasets of existing content (text, images, audio, code, etc.). During this training, they learn the patterns, structures, styles, and underlying "rules" of that data
- ❑ A **text-generating** AI learns grammar, syntax, vocabulary, and even common narrative structures by reading countless books, articles, and websites.
- ❑ An **image-generating** AI learns about shapes, colors, textures, lighting, and how different elements combine to form realistic or artistic images by analyzing millions of pictures.



Generative AI

Key Characteristics

- ❑ **Creativity:** Its defining feature is the ability to produce novel outputs.
- ❑ **Contextual Understanding:** It can understand the context and intent of a prompt to generate relevant content.
- ❑ **Diversity of Output:** It can generate a wide range of content types.
- ❑ **Deep Learning Models:** Often powered by advanced neural network architectures



Generative AI

Applications for Generative AI

- ❑ Content Creation
- ❑ Software Development
- ❑ Design and Architecture
- ❑ Healthcare
- ❑ Customer Service
- ❑ Synthetic Data Generation

Generative AI is transforming industries and our interaction with technology, offering unprecedented capabilities for creativity and automation.

GPTs

Discover and create custom versions of ChatGPT that combine instructions, extra knowledge, and any combination of skills.

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Featured

Curated top picks from this week



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Perceptual Pillars of AI

Vision, Speech and Language

Lesson Key Objectives

Identify Vision, Speech, and Language AI (Perceptual Pillars)

1

Identify the integration capabilities

2

Use Case on AWS

3

Vision, Speech, Language (Perceptual AI)

The Three Perceptual Pillars of AI:

- ❑ **Vision AI:** Enabling machines to "see" and interpret the visual world.
- ❑ **Speech AI:** Allowing machines to understand and generate human speech.
- ❑ **Language AI (NLP):** Empowering machines to comprehend, process, and produce human language.

These pillars are at the forefront of AI innovation, transforming how we interact with technology and the world around us.

Vision, Speech, Language (Perceptual AI)

What is Vision AI?

- ❑ The field of AI that trains computers to interpret and understand the visual world from images and videos. It involves teaching machines to recognize patterns, objects, faces, and even emotions.
- ❑ Uses a technique of Convolutional Neural Network (CNN), which is a class of deep neural networks most commonly applied to analyzing visual imagery and Image Processing for enhancing or manipulating digital images.



Vision, Speech, Language (Perceptual AI)

What is Speech AI?

- ☐ The field of AI focused on enabling machines to process, understand, and generate human speech. It bridges the gap between human vocal communication and machine comprehension.
- ☐ Uses Automatic Speech Recognition (ASR), which is the process of converting audio into text.
- ☐ Uses Natural Language Processing (NLP) to understand the meaning and intent behind spoken words.
- ☐ Uses Deep Learning Models : Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTMs), and Transformer models are commonly used.



**Google Cloud
Speech API**

Vision, Speech, Language (Perceptual AI)

What is Language AI?

- ☐ The field of AI that gives computers the ability to understand, interpret, and generate human language in a way that is valuable. It deals with the interaction between computers and human (natural) languages.
- ☐ Tokenization, Parsing, Named Entity Recognition (NER): Fundamental steps in understanding text structure and meaning.
- ☐ Word Embeddings: Representing words as numerical vectors to capture semantic relationships.
- ☐ Recurrent Neural Networks (RNNs), LSTMs, Transformers: Deep learning architectures crucial for sequence processing.
- ☐ Large Language Models (LLMs): Advanced transformer-based models (like GPT, Gemini) capable of highly sophisticated language understanding and generation.



Vision, Speech, Language (Perceptual AI)

The Power of Integration:

These three pillars often work together to create more powerful and intuitive AI systems.

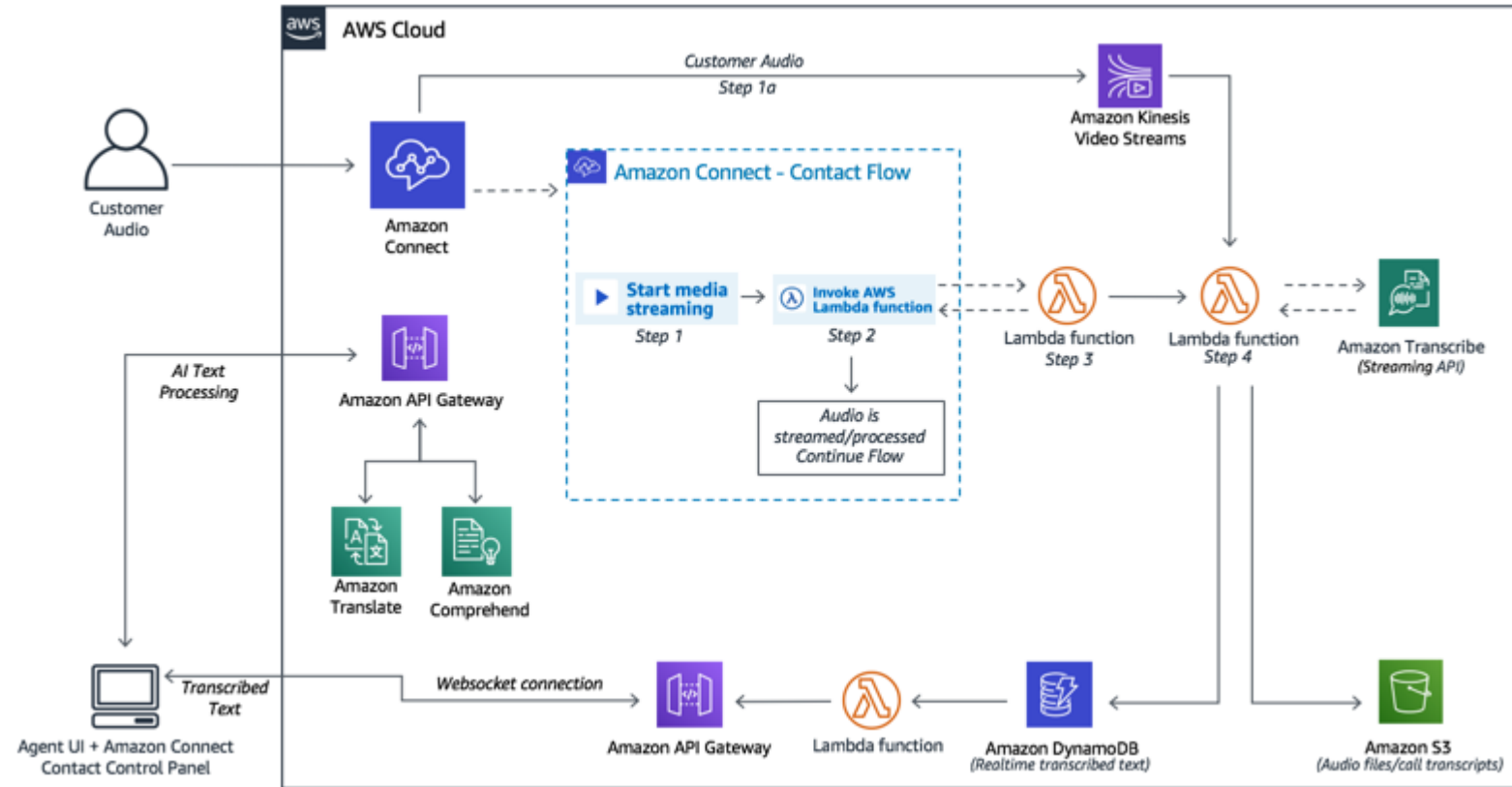
- ❑ **Example:** Voice Assistants: Speech AI (ASR) converts your voice to text, Language AI (NLP) understands your command, and then Language AI (TTS) generates the response. Vision AI might be integrated for context (e.g., "show me restaurants near here").
- ❑ **Example:** Autonomous Vehicles: Vision AI identifies objects and lanes, while Speech AI might handle in-car voice commands for navigation or entertainment. NLP could process map data and traffic reports.
- ❑ **Example:** Video Analysis: Vision AI identifies actions and objects, while Speech AI transcribes spoken dialogue, and Language AI analyzes the sentiment or topics discussed.

Vision, Speech, Language (Perceptual AI)

Use Case Example

❑ AWS Translate

❑ Amazon Comprehend





How Gen AI Works (Gemini)

Whiteboard

Lesson Key Objectives

How does a service like ChatGPT or Google Gemini work?

1

What are the two stages of AI training?

2

Whiteboard Discussion

Generative AI Technology Stack

Section 1.2



Section 1.2 Lesson Overview

Understanding the
AI Tech Stack

Large Language
Models (LLMs) and
Transformers

Breakdown of
Generative AI
Model
Architectures

Generative Pre-
trained
Transformers
(GPTS)

Whiteboard -
Retrieval-
Augmented
Generation (RAG)

Application
Programming
Interfaces (API)

AI Algorithms and
Nural Networks

Demonstration -
GenAI Tools

Key Cloud offerings
(AWS Bedrock,
Google Vertex AI,
Azure OpenAI).

Key Players in AI
Market

Trends and
Advancements in
AI Market

Demonstration -
Google Cloud AI
Ecosystem



Understanding the AI Tech Stack

Holistic view of AI

Lesson Key Objectives

What is the AI Technology Stack?

1

What are the Technology Layers?

2

Identify supporting Technology for each AI Technology Layer

3

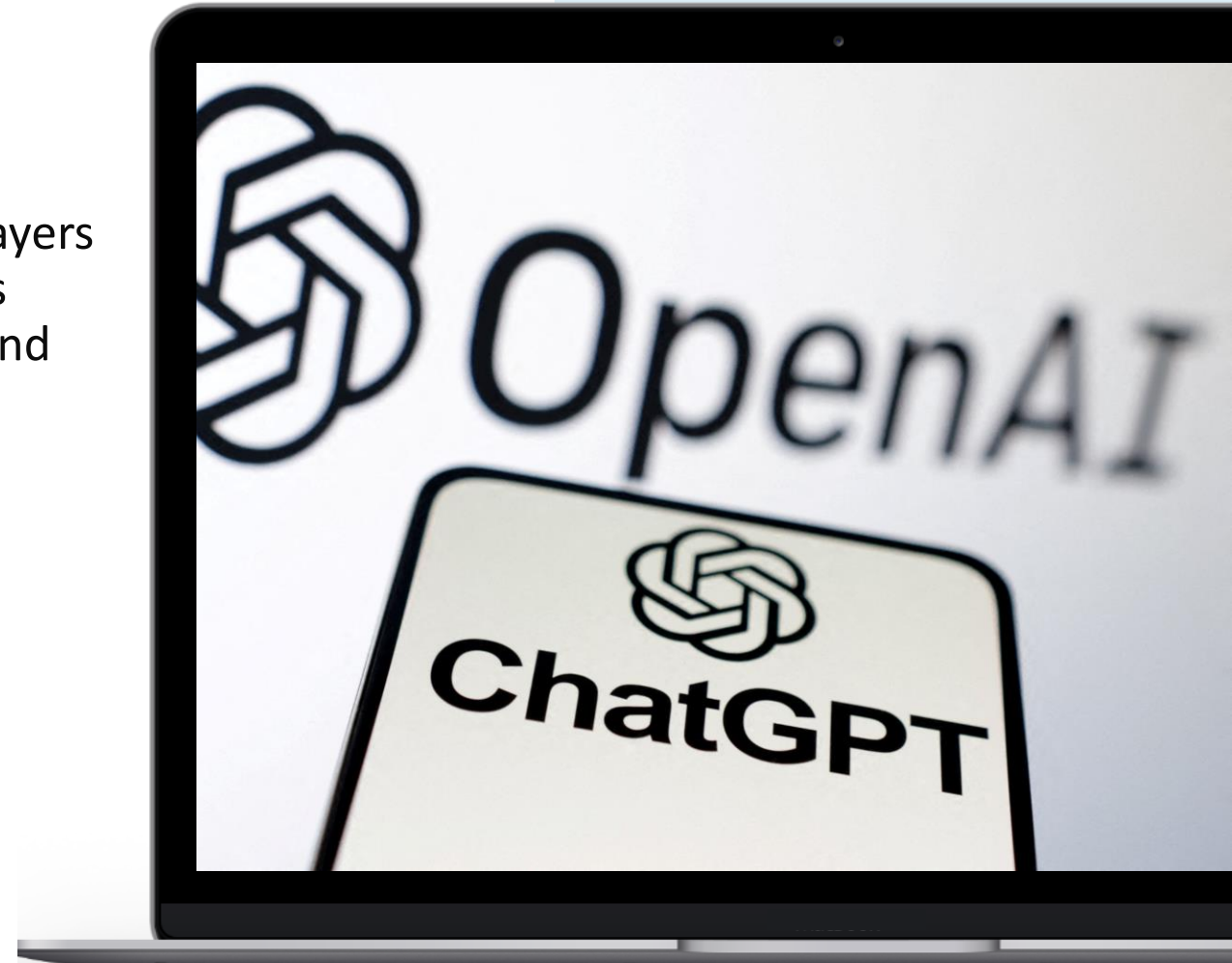
AI Technology Stack

What is Generative AI Technology Stack?

The **Generative AI technology** stack is a complete system of components structured in layers. These layers work together to build, train, and deploy AI systems capable of creating new content like text, images, and code.

We can divide this complex system into three core layers, each with a distinct and vital role.

- ☐ Hardware (Compute)
- ☐ Cloud Platforms
- ☐ Data and Storage



AI Technology Stack

Layer 1 - Compute

Generative AI models are incredibly compute-intensive and rely on specialized hardware.

- ❑ **GPUs (Graphics Processing Units):** Essential for the parallel processing required by deep learning models.
- ❑ **TPUs (Tensor Processing Units):** Google's custom-designed hardware specifically optimized for AI workloads.
- ❑ **Orchestration:** Tools like Docker and Kubernetes are used to package, deploy, and manage these compute resources efficiently.



AI Technology Stack

Layer 2 – The Model - The Intelligence of the Stack

This is where the generative capabilities reside.

This layer includes the foundation models themselves, their underlying architectures, and the software frameworks used to build and train them.

This is considered the “The Brains”

- ☐ **Transformers:** The fundamental architecture behind most modern LLMs (e.g., GPT).
- ☐ **GANs** (Generative Adversarial Networks) and **VAEs** (Variational Autoencoders).
- ☐ **Diffusion Models** (popular for image and video generation).



AI Technology Stack

Layer 2 – The Model - The Intelligence of the Stack (Continued)

- ❑ **Frameworks & Libraries:** Software tools used to build, train, and optimize these models.
- ❑ **Deep Learning Frameworks:** PyTorch, TensorFlow (often with Keras).
- ❑ **Programming Language:** Python is the dominant language due to its extensive ecosystem of machine learning libraries.
- ❑ **Data Pipeline:** Tools and processes for data collection, cleaning, labeling, and preprocessing to ensure the model is trained on high-quality, relevant data.



AI Technology Stack

Layer 2 – The Model - The Intelligence of the Stack (Continued)

Software libraries and languages are used to build, train, and optimize these complex models.

Python: The dominant programming language for the entire AI and machine learning ecosystem.

PyTorch: A popular open-source deep learning framework known for its flexibility and strong community support.

TensorFlow: A comprehensive, end-to-end platform for machine learning developed by Google.



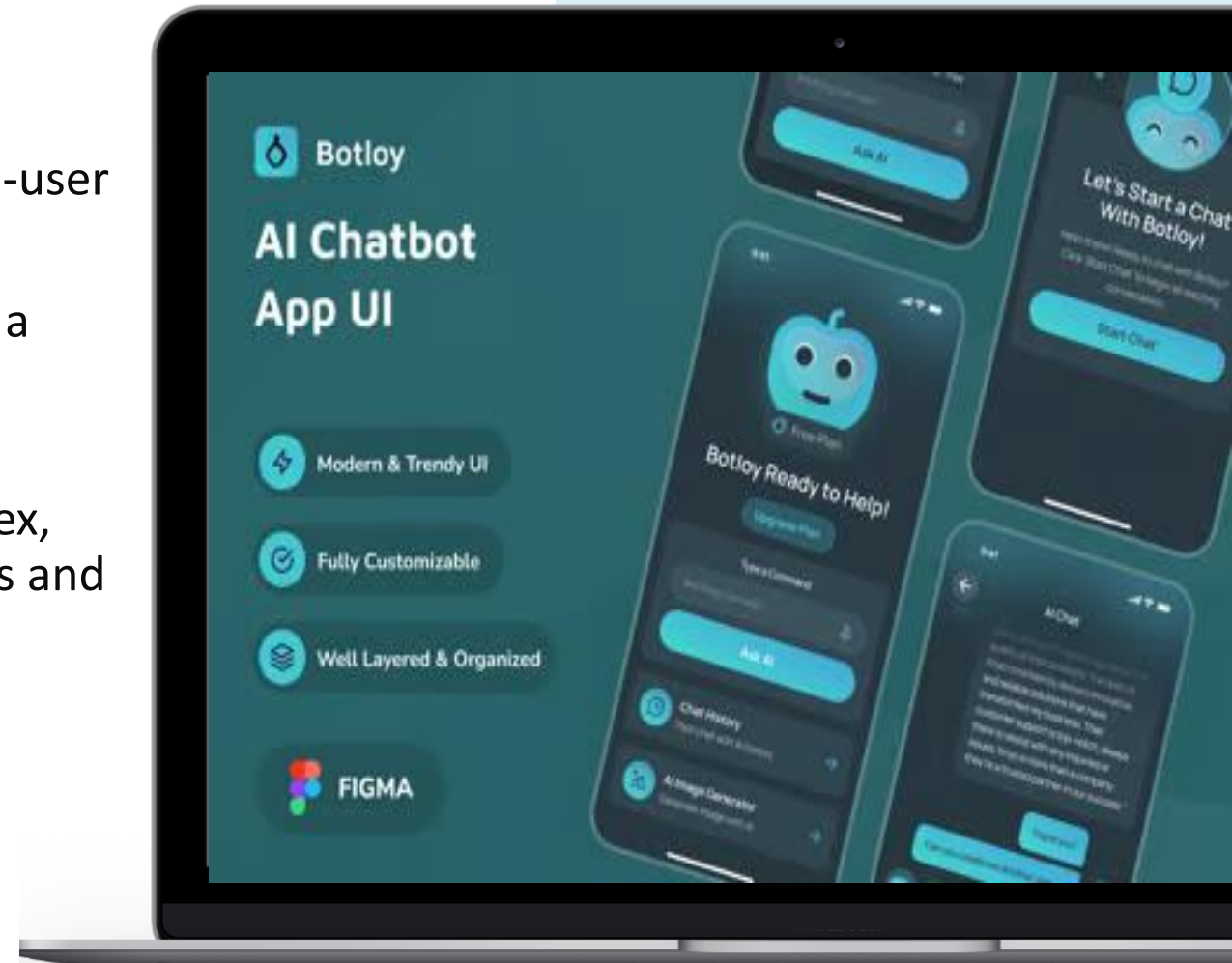
AI Technology Stack

Layer 3 – The Application

This final layer brings the model's power to the end-user through an interface.

It bridges the gap between the complex model and a usable product.

It includes deployment platforms, APIs, and AI application frameworks like LangChain or LlamaIndex, which help connect models to external data sources and perform complex reasoning tasks.



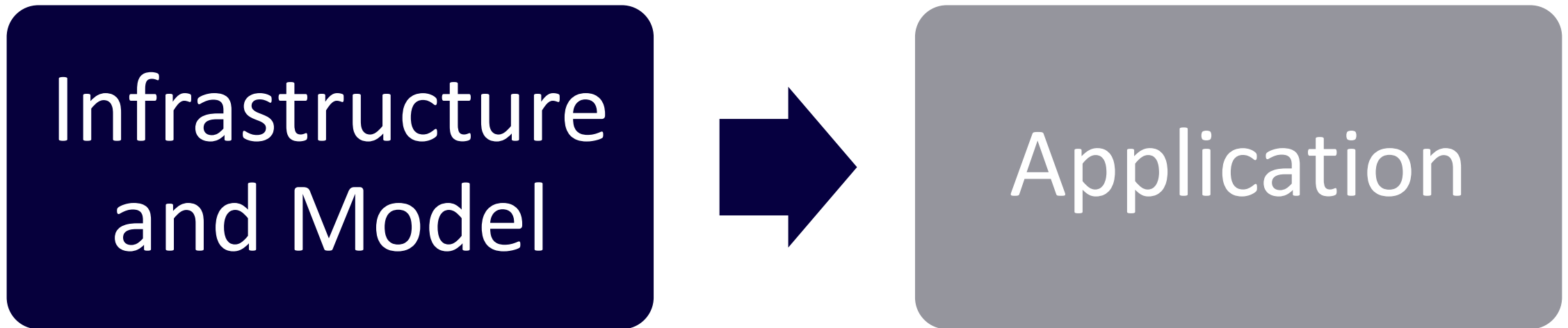
How the Layers Interact

Infrastructure & Model

The "backend" of the stack. This provides the foundational power (GPUs, Cloud) and the core intelligence (LLMs, Diffusion Models) along with the frameworks (PyTorch, TensorFlow) to build it.

Application

The "frontend" of the stack. This delivers the model's capabilities to the user via APIs, user interfaces, and specialized application frameworks like LangChain and LlamaIndex.



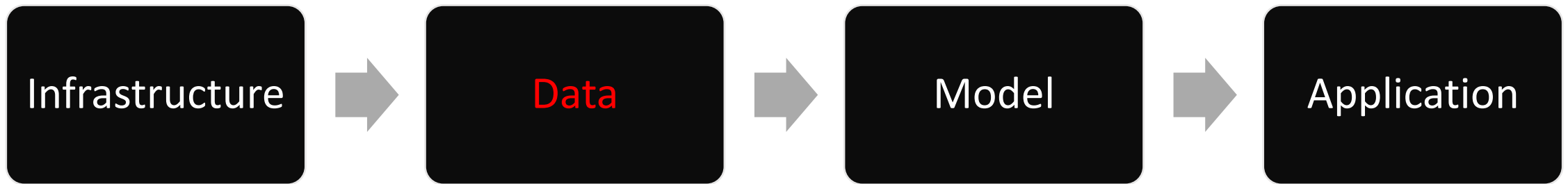
How the Layers Interact

Infrastructure Layer - Provides the foundational (Cloud/GPU) power and storage for data.

Data - Vast datasets are collected and preprocessed to feed the model.

Model - The model is trained on the data and learns to generate new content.

Application - The trained model is deployed and serves the final result to the end-user via a UI or API.





Large Language Models (LLM)

What are they and what are common ones to know

Lesson Key Objectives

What is a Large Language Model (LLM)?

1

How does an LLM work?

2

What is a Transformer?

3

Understanding Large Language Models (LLM)

What is a Large Language Model (LLM)?

A Large Language Model (LLM) is a type of artificial intelligence (AI) that is designed to understand, interpret, and generate human language on a massive scale. Think of it as a super-advanced text predictor and communicator.

Large Language Models:

- ☐ LLMs like ChatGPT or Gemini
- ☐ Sophisticated computer programs built using a method called deep learning.



Understanding Large Language Models (LLM)

What is a Large Language Model (LLM)?

The "Large" part means two things:

- ❑ **Vast Data:** They are trained on huge amounts of text—billions or even trillions of words from books, articles, websites, conversations, and code. This massive library is how they learn how language works.
- ❑ **Billions of Parameters:** Connections in the model that are adjusted during training. The more parameters the model has, the more complex the patterns and connections it can learn.

Understanding Large Language Models (LLM)

How does a Large Language Model (LLM) work?

At its core, an LLM is a giant, extremely sophisticated word predictor.

- ❑ **Prediction:** When you give it a prompt (the input), the LLM reads it, and then its main task is to figure out the most probable next word (or "token") in the sequence.
- ❑ **Repetition:** It repeats this process—it takes the text it just generated, adds it to the original prompt, and predicts the next most likely word, and so on.
- ❑ **Coherent Text:** By repeating this process millions of times, it can generate entire sentences, paragraphs, or even long essays that are grammatically correct and logically flow from your prompt.

Understanding Large Language Models (LLM)

What is a Transformer?

LLMs are powered by a special type of neural network architecture called a **Transformer**.

- ❑ **"Attention" is Key:** Transformers use a mechanism called self-attention. Imagine you are reading a complex sentence. You have to pay attention to specific words to understand the meaning of others.
- ❑ The "attention" mechanism lets the LLM do the same thing: it analyzes how different words in a sentence relate to each other, even if they are far apart.

For example, in the sentence *"The writer put the book on the table, and it had a torn cover,"* the model uses attention to know that "it" refers to the book, not the table.

This is what enables LLMs to comprehend the deeper context and nuances of human language.



Breakdown of Generative AI Model Architectures

(Image/Video Gen) and GANs.

Lesson Key Objectives

What are the different architectures used for Gen AI

1

What are Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), Diffusion Models, and Transformer-based Models

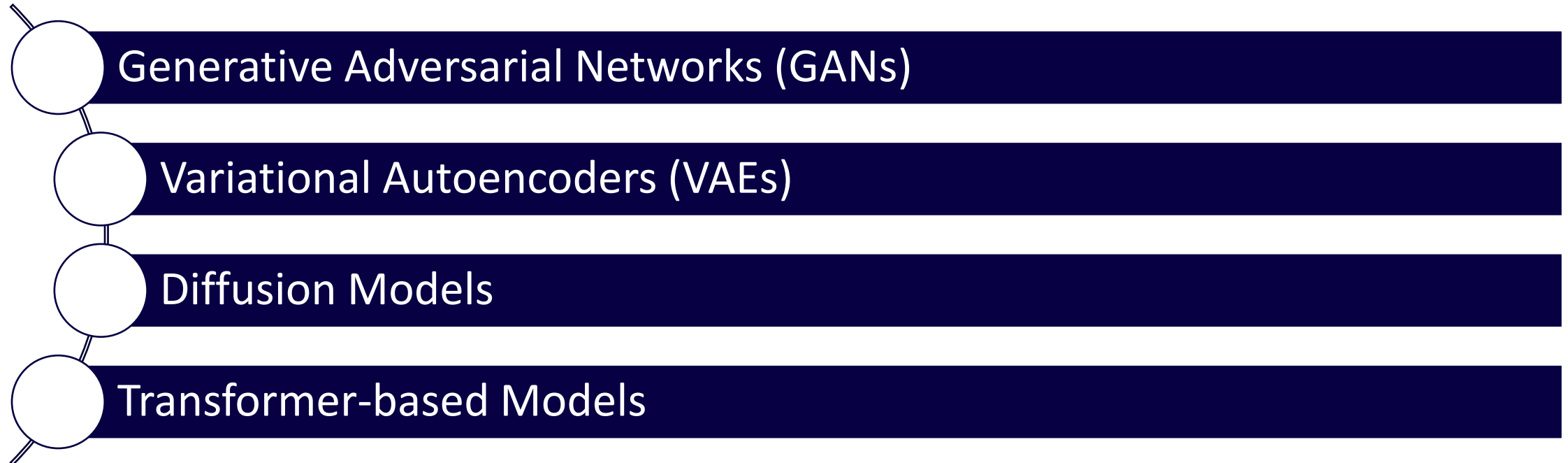
2

How do GANs work?

3

Generative AI Architectures

Generative AI models are primarily categorized by their underlying architecture, which dictates how they learn the data distribution and generate new content. The main architectures are **Generative Adversarial Networks (GANs)**, **Variational Autoencoders (VAEs)**, **Diffusion Models**, and **Transformer-based Models**.



Generative AI Architectures- Generative Adversarial Networks (GANs)

GANs work through a competitive, two-part system known as an adversarial process.

Component	Role	Function
Generator (G)	The Creator	Takes a random noise vector (latent space) as input and attempts to generate new data (e.g., an image) that looks real.
Discriminator (D)	The Critic	Takes both real data and the data generated by the Generator as input and tries to correctly classify whether the sample is real or fake .

Generative AI Architectures - Generative Adversarial Networks (GANs)

The two models are trained simultaneously: the Generator tries to fool the Discriminator, and the Discriminator tries to not be fooled.

This continuous competition pushes the Generator to produce increasingly realistic and high-quality outputs.

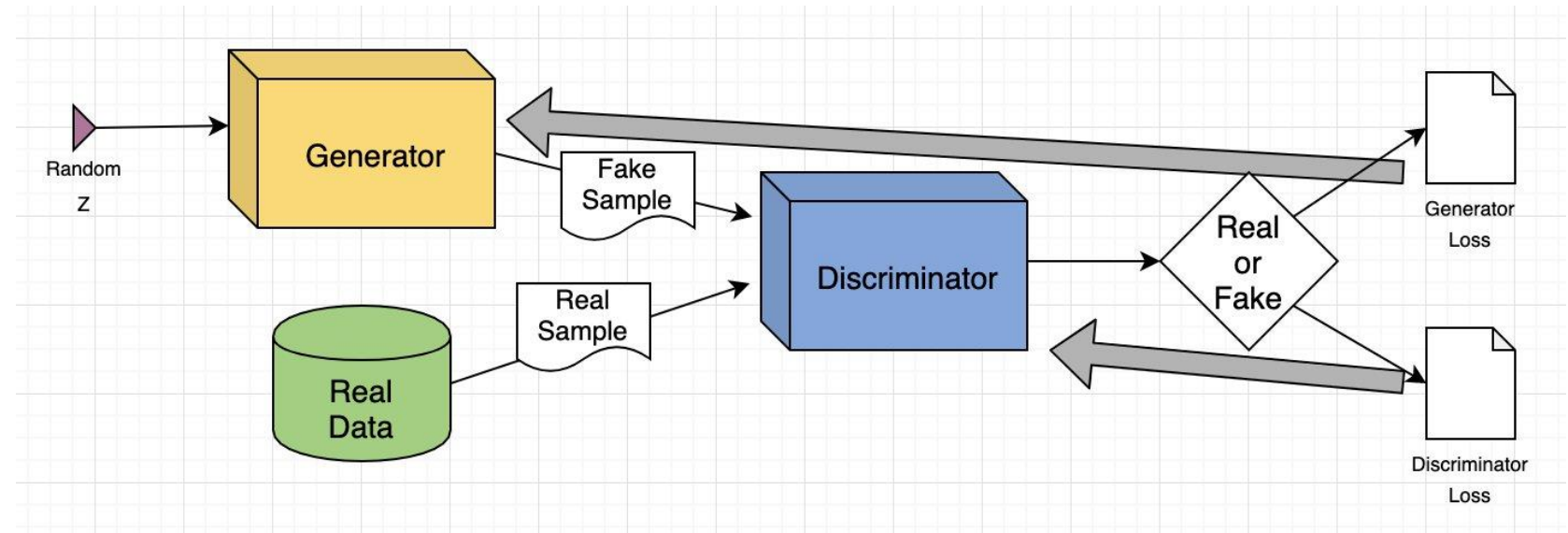
Pros: Produces exceptionally sharp and highly realistic samples (especially images).

Cons: Notoriously difficult to train due to the adversarial process (training instability), and can suffer from mode collapse (generating a limited variety of samples).

Generative AI Architectures - Generative Adversarial Networks (GANs)

A **generative adversarial network** system comprises two deep neural networks—the *generator network* and the *discriminator network*.

Diagram - AWS



Generative AI Architectures - Variational Autoencoders (VAEs)

VAEs are a probabilistic take on the autoencoder architecture, designed to learn a smooth, continuous representation of the input data's distribution.

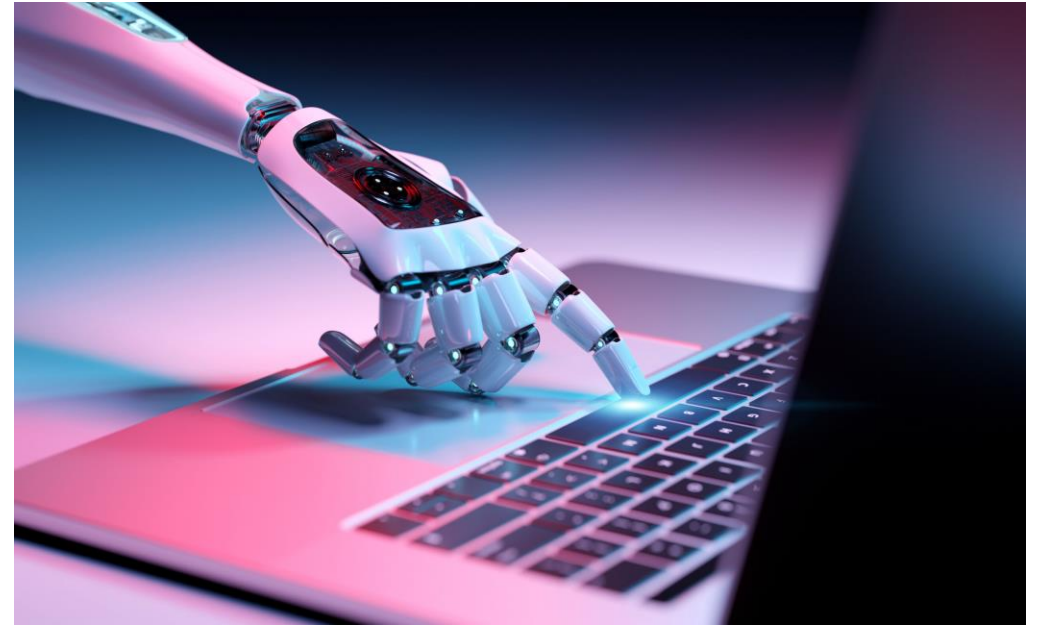
Component	Role	Function
Encoder	The Mapper	Compresses the input data into a latent space by mapping it to a <i>mean</i> (μ) and <i>variance</i> (σ^2) vector, defining a probability distribution (usually Gaussian).
Latent Space	The Code	A lower-dimensional, continuous space where points correspond to different features or characteristics of the data. New samples are generated by sampling a random point from this space.
Decoder	The Reconstructor	Takes a sample from the latent space and attempts to reconstruct it back into a new data point (the generated output).

Generative AI Architectures - Variational Autoencoders (VAEs)

The model is optimized to minimize two objectives: reconstruction loss (how well the output matches the input) and KL divergence loss (a term that keeps the latent space smooth and close to a standard distribution).

Pros: Training is more stable than GANs, and the latent space is well-structured and easy to interpolate, offering more control over generated features.

Cons: Generated samples can often be blurrier compared to GAN outputs.



Generative AI Architectures – Diffusion Models

Diffusion models are the basis for modern high-quality image generators like DALL-E 2 and Stable Diffusion. They work by iteratively adding and then reversing noise.

Forward Diffusion Process: A fixed process that gradually adds Gaussian noise to the input data (e.g., an image) until it is pure noise.

Reverse Diffusion Process: The core of the model. It is trained to reverse the noise process, step-by-step, starting from pure noise and gradually removing it to synthesize a new, coherent data sample. This process is typically conditioned on an input prompt (e.g., text) to guide the generation.

Pros: Outstanding sample quality, high fidelity, and stability in training.

Cons: Can be computationally intensive and slower at sampling (generating an output) compared to GANs.

Generative AI Architectures – Transformers

The Transformer architecture, which relies on the self-attention mechanism, is the foundation for Large Language Models (LLMs) like GPT (Generative Pre-trained Transformer).

Self-Attention Mechanism: Allows the model to weigh the importance of different parts of the input sequence relative to each other, capturing long-range dependencies and context far more effectively than previous architectures (like RNNs).

Generative Flow: For LLMs, they are often structured as decoder-only models. They are trained to predict the next token (word or sub-word) in a sequence based on all the previous tokens, which is called autoregressive generation.



Generative AI Architectures – Transformers



Applications: Primarily used for natural language processing (text generation, translation, summarization) and, when adapted, for image and video generation as well.

Pros: Excellent at capturing long-range context, highly parallelizable (speeds up training), and state-of-the-art for sequence data.

Cons: Requires enormous amounts of data and computational power for pre-training.



Generative Pre-trained Transformers (GPTS)

Generation Process and Tokenization

Lesson Key Objectives

What are Generative Pre-trained Transformers (GPTS)?

1

What are some examples?

2

What is the Generative Process?

3

Generative Pre-trained Transformers (GPTS)

Generative Pre-trained Transformer (GPT) is a type of Large Language Model (LLM) that uses a deep learning architecture called the Transformer to generate human-like text and other content.

It is a foundational technology driving the field of Generative AI.

- ❑ ChatGPT
- ❑ Gemini
- ❑ Claude



Generative Pre-trained Transformers (GPTS)

Term	Meaning
Generative	The model is capable of creating new, novel content—such as text, code, images, or music—rather than just classifying, analyzing, or summarizing existing data.
Pre-trained	The model first undergoes a computationally intensive training process on a massive, diverse dataset of text (and often code and other media) gathered from the internet. This foundational training, which is typically unsupervised, gives the model a broad understanding of language, grammar, and world knowledge.
Transformer	This refers to the neural network architecture that the model is built upon. The Transformer design, introduced in 2017, uses an attention mechanism (specifically self-attention) which allows the model to weigh the importance of different words in an input sequence simultaneously, regardless of their distance. This is crucial for grasping context and relationships across long passages of text.

Generative Pre-trained Transformers (GPTS)

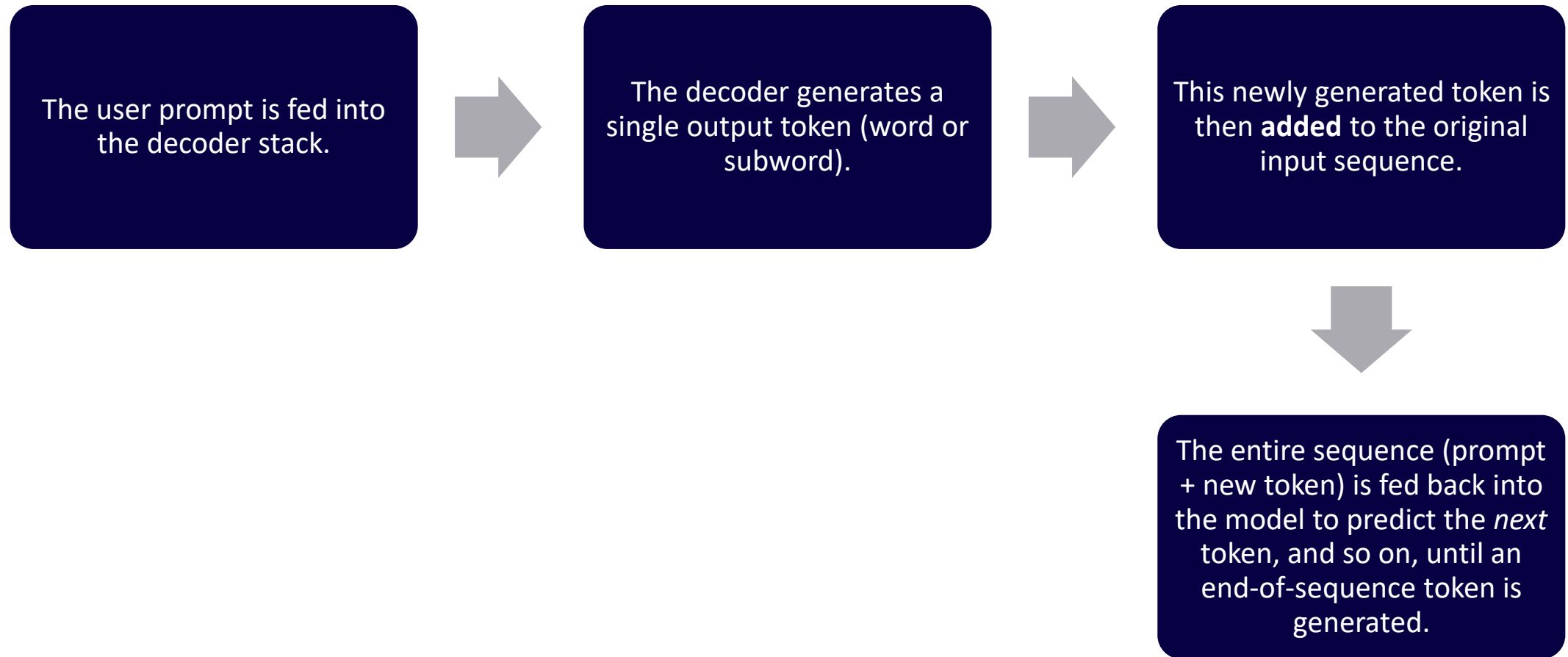
GPT models primarily function by analyzing an input prompt and then predicting the most statistically probable sequence of tokens (words or sub-words) to follow.

This is essentially a sophisticated next-word prediction process.

- ❑ **Input:** A user provides a prompt (e.g., "Write a poem about the ocean.").
- ❑ **Tokenization:** The text is broken down into numerical tokens (sub-words or characters).
- ❑ **Prediction:** The model processes these tokens using its complex network of layers and self-attention mechanisms. Based on its pre-trained knowledge, it calculates the probability of every possible next token and selects the best one.
- ❑ **Generation:** This newly selected token is added to the sequence, and the process repeats, generating the content one token at a time until a complete and coherent response is formed.

Generative Pre-trained Transformers (GPTS)

The generation process is autoregressive: it generates text one token at a time.

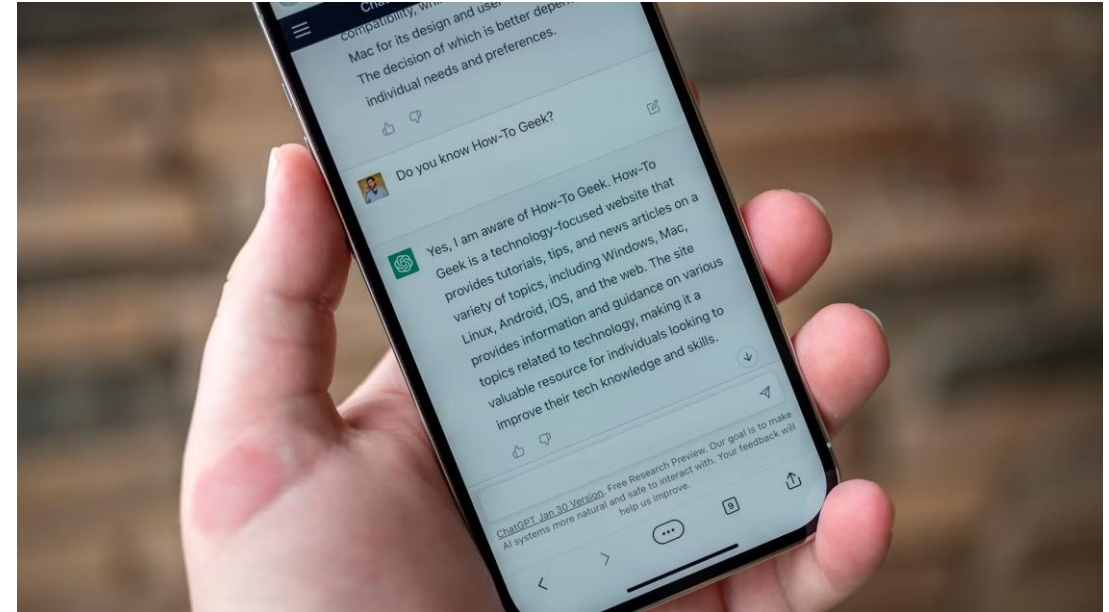


Generative Pre-trained Transformers (GPTS)

Because they are pre-trained on vast amounts of data, GPT models are general-purpose and highly versatile, often requiring only a brief prompt to perform new tasks without specific, task-by-task training.

GPTs are widely used for:

- ❑ **Content Creation:** Writing articles, social media posts, emails, and stories.
- ❑ **Chatbots & Virtual Assistants:** Powering conversational AI systems like ChatGPT.
- ❑ **Code Generation:** Writing, explaining, debugging, and reviewing programming code.
- ❑ **Text Summarization:** Condensing long documents into key points.
- ❑ **Translation:** Translating text between different languages



Generative Pre-trained Transformers (GPTS)

Model Brand Name	Developer	Context/Interface
Gemini (formerly PaLM/LaMDA)	Google	Powers the Gemini conversational assistant (formerly Google Bard), Google Search, and other Google products.
Claude	Anthropic	A family of models (e.g., Claude 3 Opus, Sonnet, Haiku) often cited for strong performance and a focus on safety ("Constitutional AI").
Llama	Meta AI	A series of powerful models (e.g., Llama 3) that Meta often releases as open-source for researchers and developers.
Mistral / Mixtral	Mistral AI	A European AI startup known for fast, high-performing models, including open-source options.
Grok	xAI	The model developed by Elon Musk's xAI, designed to answer questions with wit and a rebellious streak.
Command	Cohere	Models focused on enterprise applications and highly contextual tasks.



Retrieval-Augmented Generation (RAG)

Whiteboard Discussion

Lesson Key Objectives

What is Retrieval-Augmented Generation (RAG)?

1

How does Retrieval-Augmented Generation (RAG) Work

2

Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG):

Simply Put: Systems that automatically search and retrieve relevant documents, data, or facts from external knowledge bases. (Out of LLM)

- ❑ RAG is a technique that significantly enhances the capabilities of Large Language Models (LLMs), but its reliance on external data introduces a unique and critical set of ethical concerns that AI Ethics Officers must proactively address.
- ❑ Hybrid approach designed to overcome two primary limitations of traditional LLMs which are Stale Knowledge and Hallucinations



Whiteboard Discussion



Application Programming Interfaces (API)

Connecting Services

Lesson Key Objectives

What is an Application Programming Interfaces(API)?

1

How does an API work?

2

What are the Key Types of APIs?

3

Application Programming Interfaces (API)

An **API** stands for **Application Programming Interface**.

An Application Programming Interface (API) is a formal contract that defines the precise way one piece of software can interact with and request services from another piece of software.

- ❑ It specifies the methods, data formats, and protocols for communication, acting as a secure and standardized gateway.
- ❑ This interface ensures that the client (the program asking for something) doesn't need to know the inner workings of the server (the program providing the service), promoting modularity and security.
- ❑ Simply put, it's a set of rules and protocols that allows one software application to communicate with another Cloud-based AI/ML Platform APIs

Application Programming Interfaces (API)

APIs define a set of rules and components that govern the interaction between the client and the server

Endpoints: These are the specific URLs that represent the location of a resource or service. For example, in a weather API, /api/v1/forecast might be the endpoint to get the next day's forecast.

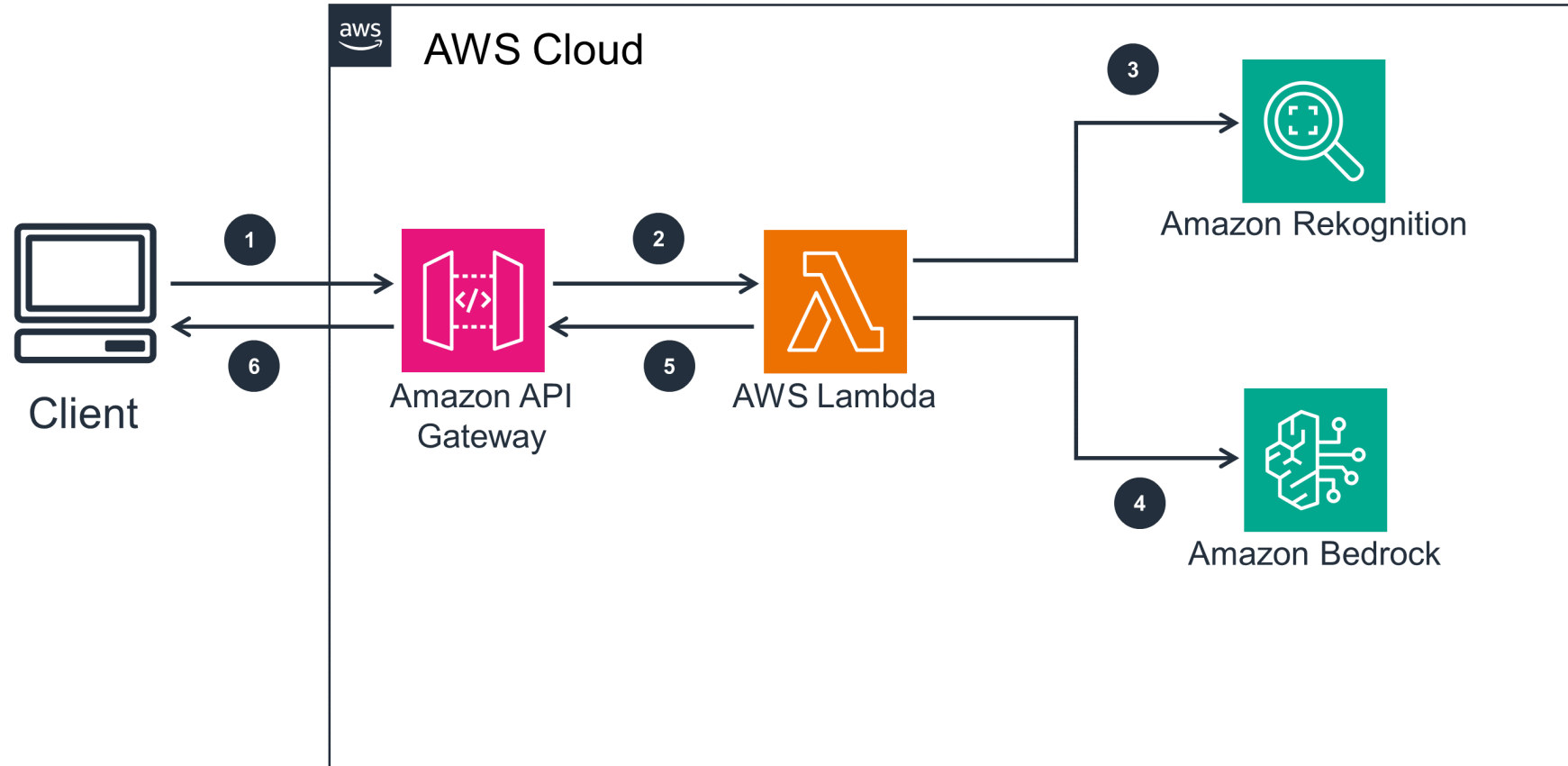
Methods (or Verbs): These define the type of action the client wants to perform on the resource. In Web APIs (like REST), these are usually standard HTTP methods:

- ❑ GET: Retrieve data (e.g., get a list of users).
- ❑ POST: Create a new resource (e.g., post a new user).
- ❑ PUT/PATCH: Update an existing resource (e.g., update a user's address).
- ❑ DELETE: Remove a resource (e.g., delete a user account).

Application Programming Interfaces (API)

APIs connecting to
AWS AI/ML services

Diagram - AWS



Application Programming Interfaces (API)

Requests: The message sent by the client to the server. A typical request includes:

- ☐ The Method (e.g., POST).
- ☐ The Endpoint (URL).
- ☐ Headers: Metadata, like authentication keys or the format the client expects (e.g., Accept: application/json).
- ☐ Body (Payload): The data being sent to the server

Responses: The message sent back by the server to the client. A typical response includes:

- ☐ Status Code: A three-digit number indicating the result of the request
- ☐ Headers: Metadata about the response.
- ☐ Body (Payload): The requested data (e.g., the weather forecast) or a confirmation message, usually formatted as JSON (JavaScript Object Notation) or XML.

Application Programming Interfaces (API)

APIs are essential for integrating AI/ML functionalities into applications and workflows. They allow developers to leverage pre-trained models and access data programmatically. Key types of APIs in the AI/ML landscape include:

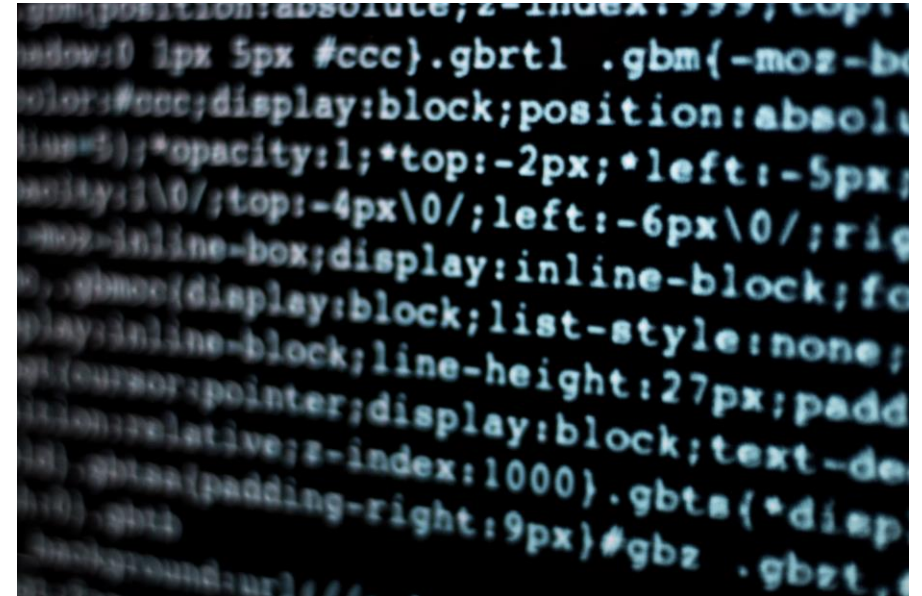
Cloud-based AI/ML Platform APIs

- ❑ **Google Cloud AI APIs (Vertex AI):** Offers APIs for training custom models, using pre-trained models for vision, natural language, speech, and more (including Gemini models).
- ❑ **Amazon AI (SageMaker):** Provides APIs for building, training, and deploying machine learning models, as well as pre-trained AI services.
- ❑ **Microsoft Azure AI Services:** Offers APIs for various AI functionalities like computer vision, natural language processing, speech, and decision-making.

Application Programming Interfaces (API)

Key types of APIs in the AI/ML landscape include:
(Continued)

- ❑ **APIs for Data Access:** Many platforms and data providers offer APIs to access their datasets programmatically. This allows for automated data retrieval and integration into AI/ML pipelines. Examples include APIs for accessing financial, social media, and weather data.
- ❑ **Open-Source AI/ML Libraries with API-like Interfaces:** While not strictly web APIs, libraries like scikit-learn, TensorFlow, and PyTorch provide Python interfaces that function similarly, allowing developers to build and train their own models.



Application Programming Interfaces (API)

Core Function of an API

1. Client Sends Data **Client (Your App) Sends Data:** Your application sends a simple HTTP request (like a piece of text, an image, or an audio clip) to the provider's server.

A large, light blue downward-pointing arrow indicating the flow from step 1 to step 2.

2. API Server Runs the AI Model: The provider's powerful, pre-trained AI model processes the data

A large, light blue downward-pointing arrow indicating the flow from step 2 to step 3.

3. Client Receives the Result: The API returns a response containing the intelligent result in a standard format (JSON, etc).

Application Programming Interfaces (API)

API Type	Stands For	Primary Data Format	Key Characteristic	Typical Use Case
REST	Representational State Transfer	JSON (mostly), XML	Resource-based (uses URLs to identify resources) and Stateless (each request is independent).	Modern public web services, mobile apps, and microservices. Most common type today.
SOAP	Simple Object Access Protocol	XML	Protocol-based with strict formal standards and a defined contract (WSDL).	Enterprise, financial, or legacy systems requiring high security and formal contracts.
GraphQL		JSON	A query language that allows the client to request <i>exactly</i> the data fields they need, avoiding over-fetching.	Complex applications with varied data needs or poor/slow network connections.

Application Programming Interfaces (API)

API Category	What it Does	Example Use Cases	Popular APIs
Generative AI / LLMs	Creates new, original content (text, code, images, video) based on a prompt.	Chatbots, content creation, code generation, generating art/images.	OpenAI (GPT models, DALL-E), Anthropic (Claude), Cohere, Gemini API.
Natural Language Processing (NLP)	Understands, analyzes, and processes human language (text).	Sentiment analysis (positive/negative), text summarization, language translation, extracting keywords.	Google Natural Language AI, Azure AI Language, IBM Watson.
Computer Vision	Enables computers to "see" and interpret visual content (images and video).	Object recognition, face detection/recognition, image tagging, reading text from images (OCR), content moderation.	Google Cloud Vision AI, Microsoft Azure Computer Vision, Clarifai.
Speech	Converts between spoken language and text.	Voice assistants, transcribing meeting recordings, generating natural-sounding voiceovers.	Google Text-to-Speech/Speech-to-Text, AssemblyAI, ElevenLabs.

Application Programming Interfaces (API)

Google Cloud

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Docs

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AI APIs for Google Cloud

Easily integrate AI into your applications with Google Cloud's AI and machine learning APIs. New customers get [\\$300 in free credits](#) to run, test, and deploy workloads.

Go to console

Explore all AI products



USE CASE	APIs	GOOD FOR
Generative AI APIs	Foundation model APIs Pre-trained multitask large models, like Gemini , that can be tuned or customized for specific tasks using Vertex AI. These multimodal models from Google can handle vision, dialog, code generation, code completion, and more.	- Text completion, multi-turn chat, and text embeddings generation - Code completion and generation - Generating and customizing images with Imagen - Universal speech models

OpenAI

Log in

Home

API Platform

API Pricing

Agents

Codex

Open Models

Community

Build leading AI products on OpenAI's platform

Contact sales

Start building


Morgan Stanley


Booking.com




MERCARI

Powered by our frontier models

Our industry-leading models are designed for real-world utility, delivering advanced intelligence and multimodal capabilities.





Understanding Algorithms

What are they and what are common ones to know

Lesson Key Objectives

What is an Algorithm?

1

What are the different Types of Algorithms?

2

What is a Neural Network?

3

Understanding Algorithms

What are AI algorithms?

A computer system that follows rules or instructions to solve a problem.

- ❑ AI algorithms are the foundation of how AI systems learn and make decisions.
- ❑ Machine learning algorithms enable computers to learn from data without explicit programming.



Understanding Algorithms

Types of AI Algorithms



Supervised
Learning
Algorithms



Unsupervised
Learning
Algorithms



Reinforcement
Learning
Algorithms



Deep
Learning
Algorithms



Natural
Language
Processing
(NLP)
Algorithms

Understanding Algorithms

Machine Learning Algorithms:

- ❑ **Supervised learning** uses labeled data to train models for prediction or classification tasks.
- ❑ **Unsupervised learning** discovers hidden patterns and structures in unlabeled data.
- ❑ **Reinforcement learning** trains agents to make optimal decisions in an environment by learning from rewards and penalties.



Understanding Algorithms

What are algorithms? (Continued)

Deep Learning Algorithms (Neural Networks):

Inspired by the human brain, deep learning algorithms use artificial neural networks with multiple layers to extract complex features from data.

- ❑ **Convolutional Neural Networks (CNN)** excel at processing grid-like data, such as images, by using convolutional layers to detect spatial hierarchies of features.
- ❑ **Recurrent Neural Networks (RNNs)** are designed to handle sequential data, like text or time series, by maintaining an internal memory of past information.
- ❑ **Transformers** are a more recent architecture that leverages attention mechanisms to weigh the importance of different parts of the input sequence, proving highly effective for natural language processing and beyond. (Relies on Attention to Weigh Importance)

Understanding Algorithms

What are algorithms? (Continued)

Generative AI Algorithms (2020s and beyond)

These power modern AI creativity and content generation.

- ❑ **Transformer-based Models (GPT, BERT, T5, LLaMA, Claude)**

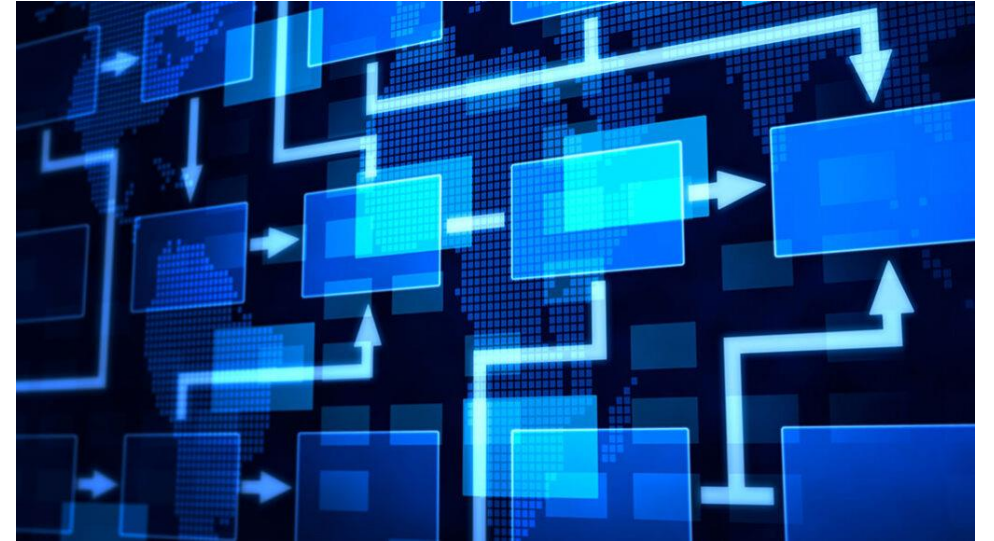
- ❑ Handle natural language understanding and generation.

- ❑ **Diffusion Models (Stable Diffusion, DALL·E, Midjourney)**

- ❑ Generate realistic images, videos, and even 3D models.

- ❑ **Variational Autoencoders (VAEs)**

- ❑ Blend reconstruction and generation — used for creative AI and anomaly detection.



Understanding Algorithms

What are algorithms? (Continued)

These are the core of traditional AI systems — predictive analytics, recommendation engines, fraud detection, etc.

☐ Linear Regression

- ☐ Predicts continuous outcomes (e.g., sales forecasting).
- ☐ Simple, fast, and interpretable.

☐ Logistic Regression

- ☐ Used for binary classification (e.g., spam vs. not spam).

☐ Decision Trees

- ☐ Simple but powerful; models decisions and their consequences.

☐ Random Forest

- ☐ Ensemble of decision trees for higher accuracy and reduced overfitting.

☐ *And several more*



Gen AI Tools

Demonstration

Lesson Key Objectives

Demo Gen AI

1

Demonstration





Key Players in the AI Market

Key Companies and platforms

Lesson Key Objectives

What are the Key Players in AI?

1

What are the Key AI Company Types?

2

Key Players in the AI/ML Market

Identify and briefly describe major companies and organizations involved in AI/ML:

- ❑ **Tech Giants:** Google (Google AI, DeepMind), Microsoft, Amazon (AWS), IBM, Apple, MetaCloud, NVIDIA
- ❑ **Providers:** Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP) Research
- ❑ **Institutions:** MIT, Stanford, Carnegie Mellon University
- ❑ **Model Startups:** OpenAI, Anthropic, Cohere, xAI (Grok)
- ❑ **Specialized AI Startups:** Reflection.Ai, Harvey, Skild.AI, Turing, Shield AI
- ❑ **Data AI Infrastructure:** Databricks, Celestial AI, Nexthop AI, Palantir Technologies



Trends and Advancements in AI/ML

What can we expect in the future?

Lesson Key Objectives

What are the current trends with AI?

1

What are the primary advancements to learn?

2

Trends and Advancements in AI/ML

There are so many trends with AI/ML:

- ❑ Agentic AI
- ❑ AI Reasoning
- ❑ Custom Silicon
- ❑ Custom Migrations
- ❑ Systems to Measure AI Efficacy
- ❑ Focus on Enterprise needs
- ❑ Explainable AI
- ❑ Digital Data Forgetting (Machine Unlearning)

Trends and Advancements in AI/ML

There are so many trends with AI/ML:

- ☐ Interoperability Among Neural Networks
- ☐ Automated Machine Learning (AutoML)
- ☐ Convergence of IoT and Machine Learning (AIoT)
- ☐ Rise of Natural Language Processing (NLP) for Customer Support
- ☐ Emergence of Custom Generative AI Models
- ☐ Increased Focus on AI Ethics and Security Risks
- ☐ Shadow AI and Governance

Trends and Advancements in AI/ML

There are so many advancements with AI/ML:

- ☐ Brain-Inspired AI
- ☐ Robotic Learning from Observations
- ☐ Explainable AI for Robotics
- ☐ AI in the Medical Sector (Diagnosis)
- ☐ AI in Drug Discovery and Biology
- ☐ AI for Code Generation and Optimization
- ☐ AI for Scientific Discovery



GCP AI Ecosystem

Demonstration

Lesson Key Objectives

Brief walkthrough of Google AI Services

1

Demonstration



Data & AI Application Lifecycles

Section 1.3



Section 1.3 Lesson Overview

Explain the
Data Lifecycle

AI Lifecycle
and MLOps

SDLC and AI
Applications

Importance of
Data to AI
Applications

Cloud
Computing
Shared Model

Module
Review
Questions

Module
Review
Summary



Data Lifecycle

Gen AI and the Data Lifecycle

Lesson Key Objectives

What is the Data Lifecycle?

1

Why is the Data Lifecycle important?

2

The Data Lifecycle

The data lifecycle is critically important to Generative AI professionals because the output and quality of any generative model (like GPT, DALL-E, or Midjourney) are almost entirely dependent on the data it is trained on and the data it processes during deployment.

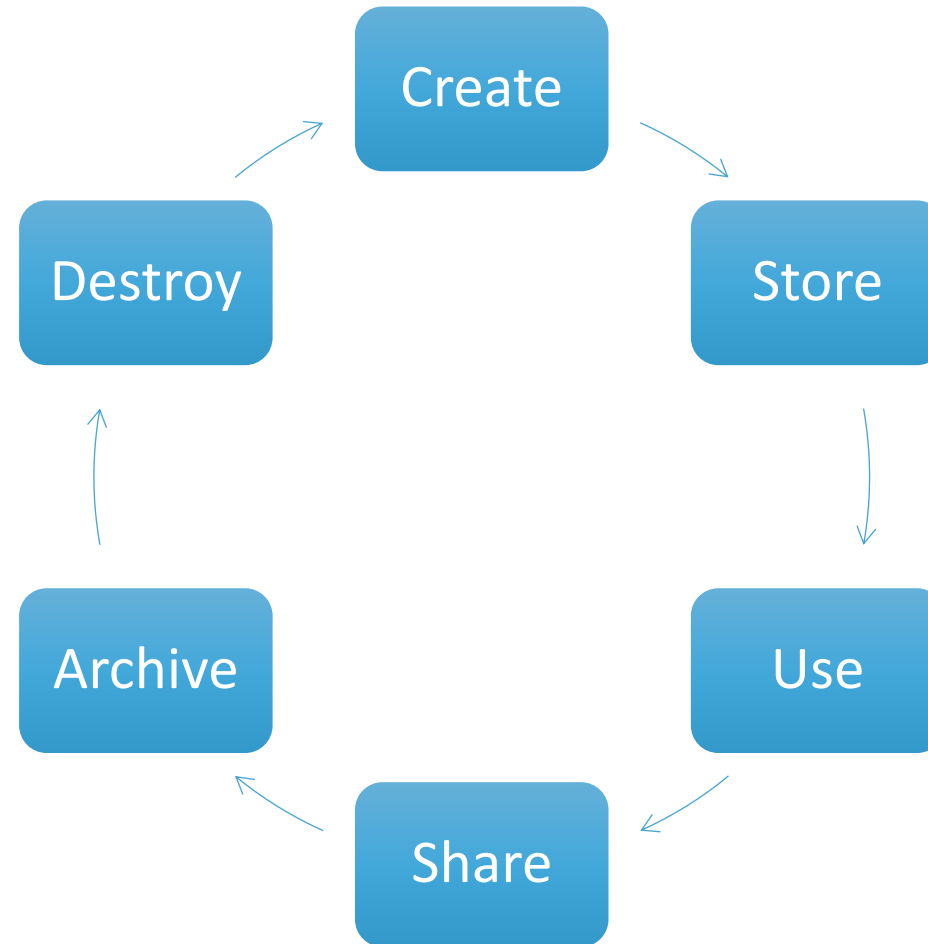
The old computer science adage, "Garbage In, Garbage Out (GIGO)," is never more true than in Generative AI.

Managing the data lifecycle ensures the input is high-quality, relevant, and secure, which directly translates to accurate, useful, and ethical model outputs.



The Data Lifecycle

The Data Lifecycle



The Data Lifecycle

The **data lifecycle** (or Data Lifecycle Management, DLM) is a structured, policy-driven sequence of stages that governs the management of data within an organization from its initial creation or acquisition to its eventual destruction.

It is a framework designed to ensure that data remains accurate, secure, compliant, and accessible for business use, while also optimizing storage costs and mitigating risk.



The Data Lifecycle

The Data Lifecycle and AI Ethics

Data Lifecycle Stage	General Purpose	Specific Challenge & Impact on GenAI
1.  Creation / Collection	Acquire raw data for business use.	Scale & Diversity: Sourcing petabytes of diverse, high-quality, and ethical data (text, images, code) to prevent under-representation and bias in the foundational model.
2.  Processing / Cleaning	Clean, validate, and standardize the data.	Alignment: Filtering out harmful, biased, toxic, or sexually explicit content (safety filtering) to ensure the model's outputs align with safety guidelines. This includes extensive data labeling and transformation.
3.  Storage / Retention	Securely store and organize usable data.	Data Governance: Managing vast, often unstructured, training data sets. For Retrieval-Augmented Generation (RAG), this means efficiently indexing and vectorizing knowledge bases for fast retrieval.
4.  Usage / Analysis	Use data to drive insights and operations.	Fine-Tuning & Prompting: Using internal, processed data to fine-tune pre-trained models. This stage involves defining clear policies on how user prompts and generated outputs are logged and used.
5.  Archival / Disposal	Retain data for compliance; securely destroy it when no longer needed.	Model Retraining: Managing the data needed to perform iterative fine-tuning or full retraining over time. Implementing policies to securely purge sensitive user interaction data to prevent privacy leaks.



What is an AI Lifecycle

Gen AI

Lesson Key Objectives

What is the AI Lifecycle?

1

What are the key phases?

2

What is MLOps?

3

AI Lifecycle

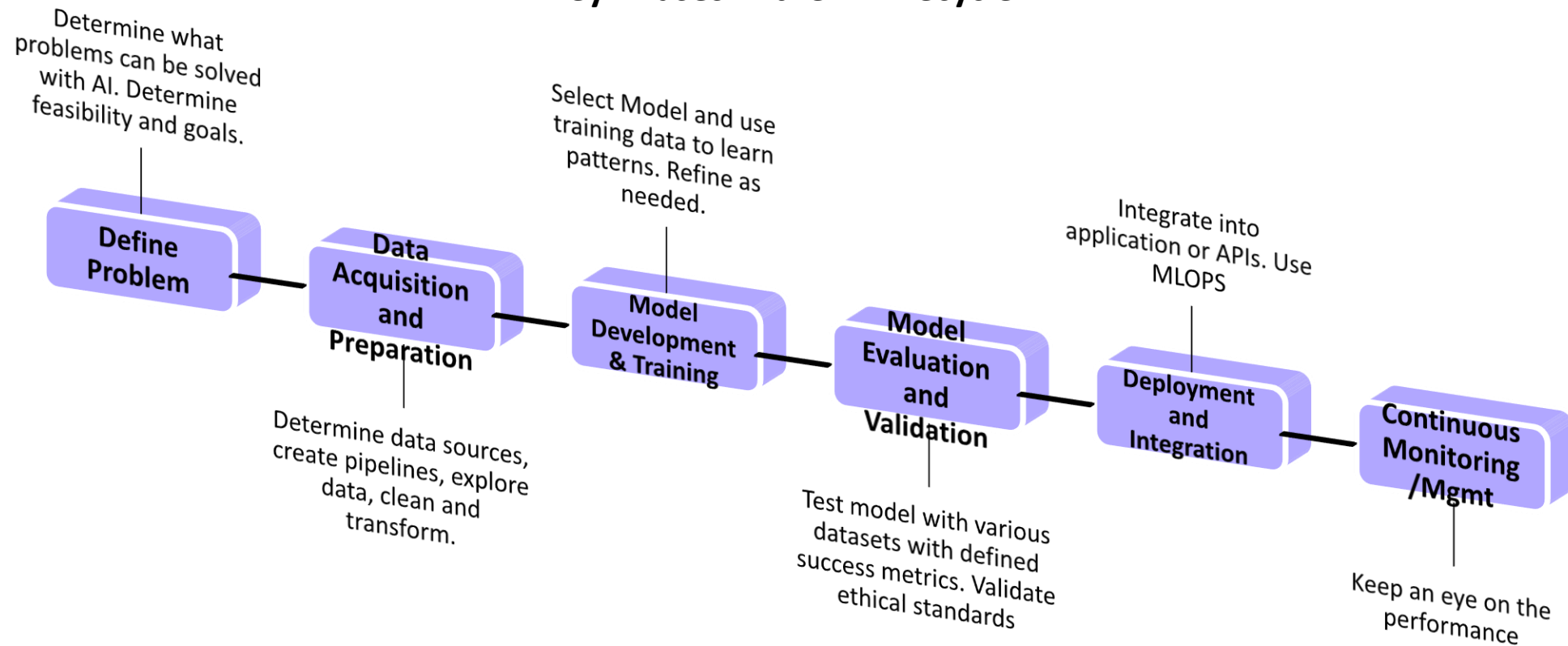
The **AI Lifecycle** is the **iterative, end-to-end process** that guides an Artificial Intelligence project from a defined business problem to a fully operational, monitored, and maintained solution.

Structured methodology ensuring that AI development is purposeful, ethical, and delivers continuous value.



AI Lifecycle

Key Phases in the AI Lifecycle



AI Lifecycle

MLOps

- The application of a structured methodology to AI/ML projects is known as **MLOps** (Machine Learning Operations).
- MLOps effectively creates a continuous loop, combining the **Data Science process** (data, model training, evaluation) with the **Software Engineering/SDLC process** (CI/CD, deployment, monitoring).





Software Development Lifecycle (SDLC) and AI

Ethics and the SDLC Focus

Lesson Key Objectives

What are management frameworks?

1

What are the core steps of a framework?

2

SDLC

The **Software Development Life Cycle (SDLC)** is a structured, systematic process that defines the stages involved in the planning, creation, testing, and deployment of a software system.

Its primary goal is to produce a high-quality product efficiently, on time, and within budget.

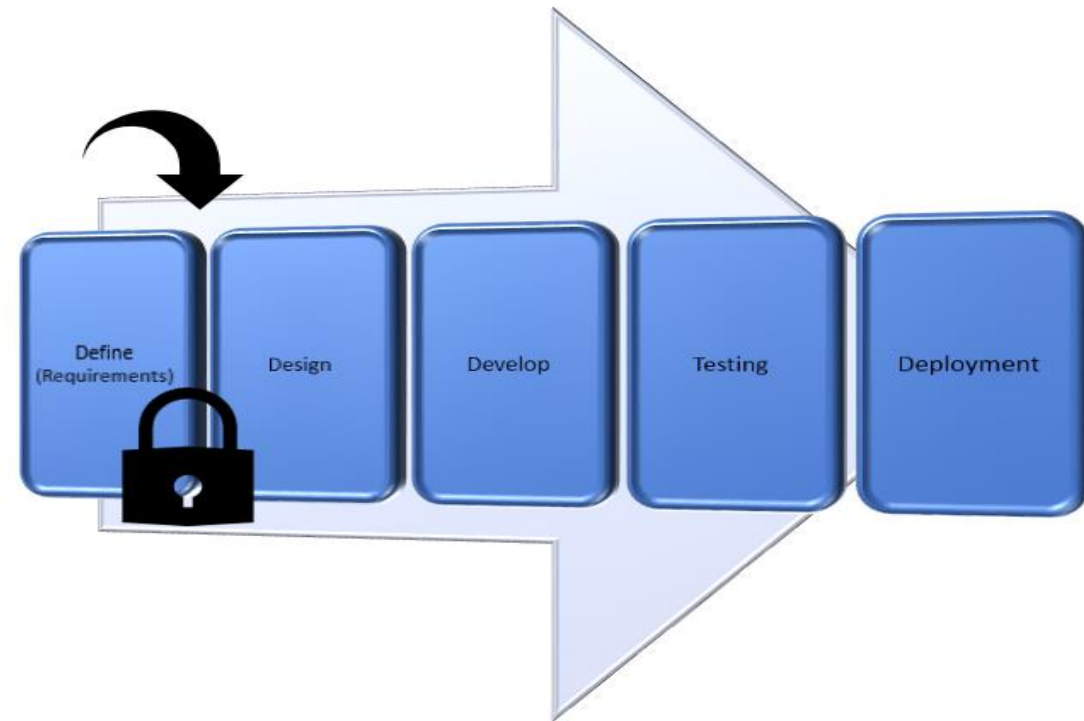
There are Six Main Steps.



Software Development Lifecycle (SDLC) and AI

Software Development Lifecycle (SDLC) is a widely used framework.

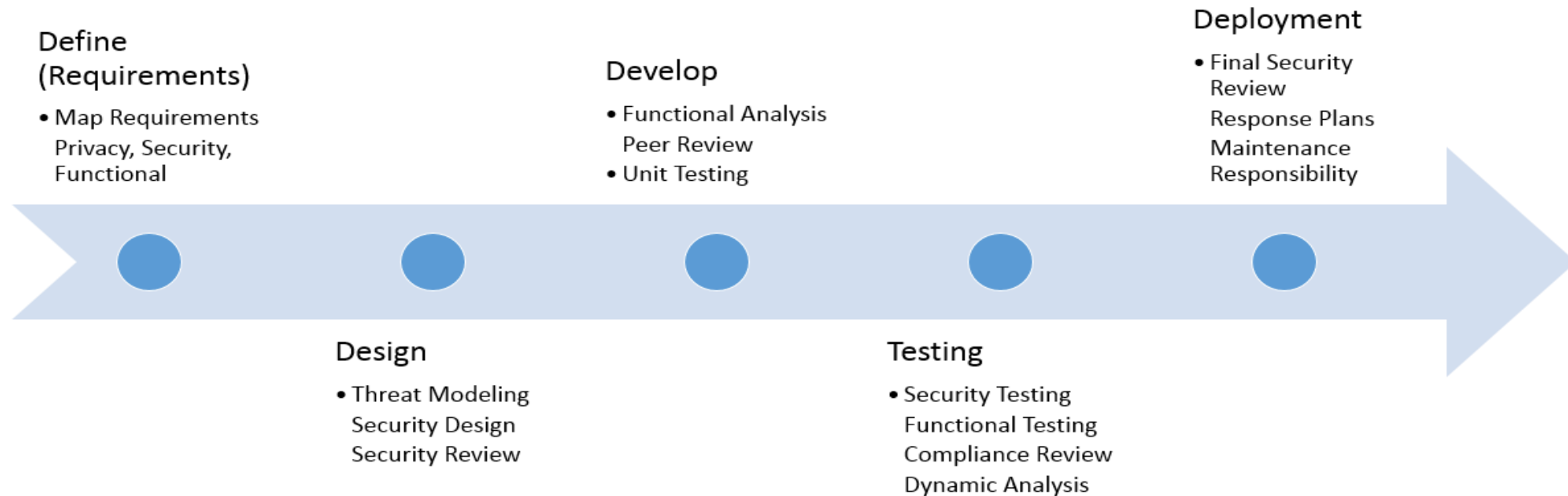
- ❑ AI Best practices as well AI ethical concerns are a challenge to plan for.



Software Development Lifecycle (SDLC)

Software Development Lifecycle (SDLC) and AI

Software Development Lifecycle (SDLC)



Software Development Lifecycle (SDLC) and AI

Ethics should be defined and mapped in the Define/Design phases

Phase	Description	Key Deliverable
1. Planning & Analysis	Define the project's scope, goals, and feasibility (technical, economic, operational). Gather and document all functional and non-functional requirements.	Software Requirement Specification (SRS)
2. Design	Create the blueprint for the software. This involves defining the overall system architecture, database structure, and user interface (UI/UX).	System Design Document, Architecture Diagrams
3. Implementation	The actual coding or programming phase, where developers write the software code according to the design specifications.	Source Code, Integrated Software Modules
4. Testing	Rigorously check the software for defects, functionality, and performance using various methods (unit, integration, system, and user acceptance testing).	Test Reports, Quality Assurance (QA) Approval
5. Deployment	Release the tested and approved software into the production environment for end-users to access and use.	Live, Operational System
6. Maintenance	Ongoing support after launch, including fixing bugs, releasing patches, updating the system to meet new requirements, and monitoring performance.	Updates, Patches, System Monitoring Reports

Software Development Lifecycle (SDLC) and AI

Comparing SDLC to AI/ML Lifecycle

SDLC Phase	AI/ML Lifecycle Phase	Key AI/ML Differences
Planning & Analysis	Problem Scoping & Business Understanding	Explicitly determines if ML is the right solution and defines success metrics based on model performance (e.g., accuracy, F1-score) in addition to business metrics.
Design	Data Engineering & Model Selection	Shifts focus from designing static code architecture to designing the data pipeline and selecting a machine learning model architecture (e.g., neural network vs. decision tree).
Implementation	Model Training & Experimentation	The core "coding" is replaced by training a model on massive datasets. This is highly experimental and iterative, focusing on parameter tuning, not just writing logic.
Testing	Model Evaluation & Validation	Testing is two-fold: Code Testing (traditional SDLC) and Model Testing on validation/test data to check for statistical performance, bias, and fairness .
Deployment	MLOps & Model Deployment	The model is often encapsulated in a prediction service and requires specialized tooling (MLOps) for versioning, infrastructure, and scalability.
Maintenance	Monitoring & Retraining	Critical difference: Not just fixing code bugs, but monitoring for Data Drift and Model Drift (when data patterns or model predictions change over time), which necessitates re-training and redeploying the model.



GenAI Lifecycle Model

Whiteboard

Lesson Key Objectives

What are the core steps in the NIST AI RMF?

1

What are the four implementation tiers ?

2

GenAI Lifecycle Model



A **Generative AI (GenAI) lifecycle model** is a structured framework that guides the entire process of developing, deploying, and maintaining a generative AI application, such as a large language model (LLM), a chatbot, or an image generator.

It is an adaptation of the traditional Machine Learning (ML) lifecycle (or MLOps) that includes specialized steps to address the unique requirements and risks of models that create content.

The primary goal of the GenAI lifecycle is to ensure the resulting application is effective, scalable, and most importantly, aligned with ethical, legal, and safety standards throughout its operation.

GenAI Lifecycle Model

The GenAI lifecycle is essential for AI because Generative AI introduces unique and amplified risks:

- ❑ **Hallucination and Misinformation:** GenAI's ability to create fluent, believable, yet entirely false content means integrity and truthfulness must be monitored throughout the lifecycle.
- ❑ **Copyright and Data Provenance:** The large, often unsourced, datasets used for training FMs create new legal and ethical concerns around intellectual property that must be addressed in the Selection and Data Preparation phases.
- ❑ **Bias Amplification:** Training data biases can be deeply embedded and amplified in the creative outputs, making the Alignment and Evaluation stages critical to check for and mitigate discriminatory or harmful content generation.
- ❑ **Prompt Hacking and Misuse:** The professional must anticipate and guard against attempts to force the model to generate harmful, illegal, or unethical content during the Customization and Deployment stages.



Putting it all together

Whiteboard Discussion





Explain the Importance of Data in AI/ML Solutions

Data drives organizations and AI.

Lesson Key Objectives

What is data so important to AI?

1

Data serves as what in an AI system?

2

What does “garbage in, garbage out mean?

3

Explain the Importance of Data in AI Solutions

Data is the **lifeblood** of AI/ML solutions.

The **quality, quantity, and characteristics** of the data directly determine the capabilities, accuracy, and reliability of these intelligent systems.

- ❑ Without good data, even the most sophisticated algorithms will be ineffective.
- ❑ Therefore, **a strong focus** on data acquisition, cleaning, preprocessing, and management is a prerequisite for building successful AI/ML applications.
- ❑ **Data serves as the fuel** that powers these intelligent systems, enabling them to learn patterns, make predictions, and draw insights. (**Garbage in – Garbage Out**)



Explain the Importance of Data in AI Solutions

Key Areas why Data is so important:

- ❑ **Model Learning and Training** - AI algorithms identify patterns, relationships, and insights from the data they are fed. (This process is called "training")
- ❑ **Model Performance and Accuracy** - The quality and quantity of training data directly impact the AI's performance (Accurate and relevant data is essential for valid results)
- ❑ **Objectives and Features** - AI lacks inherent innovation. It relies on the data to understand objectives and how to achieve them. (Patterns and historical data)
- ❑ **Evaluation and Validation** - Accurate and relevant data is essential for valid results
- ❑ **Continuous Improvement** - AI systems are designed to continuously learn and improve over time as they are exposed to new data. (Adapt and Maintain)

Data Sources and API's

Data sources provide the raw material for training and evaluating AI/ML models. At the same time, APIs (Application Programming Interfaces) offer programmatic ways to access and utilize data and pre-trained AI/ML capabilities.

Data Sources for AI/ML are available for AI/ML systems training, catering to different tasks and domains.

These can be broadly categorized as follows

- ☐ Open Data Portals (Data.gov, data.world, Kaggle, OpenML, HuggingFace, etc)
- ☐ Academic Datasets (Papers with Code)
- ☐ Cloud-based Data Marketplaces (AWS, GCP, Azure)
- ☐ Domain-Based Datasets (Physio.net)
- ☐ Synthetic Data
- ☐ Website Scraping

Application Programming Interfaces (API)

In Summary

- ❑ Data provides AI systems with the knowledge they need to function intelligently.
- ❑ Without sufficient, high-quality, and representative data, AI cannot learn effectively, make accurate predictions, or provide valuable insights. (AI algorithms learn from data)
- ❑ Therefore, data management, quality assurance, and ethical considerations around data are fundamental to the successful development and deployment of AI systems.
- ❑ Accurate and relevant data is essential for valid results. If the data is flawed, the AI's output will likely be flawed as well – **a concept known as "garbage in, garbage out."**
- ❑ APIs are used for programmatic access to data. This can be on premises or in the cloud.

Module Review Summary

Short Summary of what we covered in this module



Module Review Summary

- ❑ **Artificial Intelligence(AI)** is commonly demonstrated by machines, as opposed to natural intelligence displayed by animals, including humans.
- ❑ The **goal** of using AI is to create systems that can perform tasks that typically require human intelligence.
- ❑ **Machine Learning (ML)** is a subfield of Artificial Intelligence (AI) that focuses on enabling computers to learn from data without being explicitly programmed.
- ❑ **Three main types** of ML Learning, which are Supervised, Unsupervised, and Reinforcement
- ❑ **Deep Learning** is a subfield of Machine Learning that utilizes artificial neural networks with multiple layers (hence "deep") to extract complex patterns from data.
- ❑ **Generative AI** is a type of artificial intelligence that can produce new content, such as text, images, audio, and video, by learning from vast amounts of existing data.
- ❑ **Predictive AI** is a branch of artificial intelligence focused on forecasting future outcomes based on historical and current data.
- ❑ **Agentic AI** refers to artificial intelligence systems that can operate autonomously to achieve specific goals without constant human oversight.

Module Review Summary

- ❑ **Explainable AI (XAI)** focuses on making AI decisions transparent and understandable to humans.
- ❑ **Conversational AI** enables human-like interactions through natural language (chatbots, virtual assistants).
- ❑ **Reasoning and Planning AI** focuses on logical inference and decision-making in complex scenarios
- ❑ **Natural Language Processing (NLP)** is a branch of AI that enables computers to understand, interpret, and generate human language.
- ❑ **AI algorithms** are the foundation of how AI systems learn and make decisions.
- ❑ **Machine learning algorithms** enable computers to learn from data without explicit programming
- ❑ The **quality, quantity, and characteristics of the data** directly determine these intelligent systems' capabilities, accuracy, and reliability. (**Data serves as the fuel**)
- ❑ **RAGs** are used to automatically search and retrieve relevant documents, data, or facts from external knowledge bases. (Out of LLM)
- ❑ The **AI Lifecycle** is the iterative, end-to-end process that guides an Artificial Intelligence project from a defined business problem to a fully operational, monitored, and maintained solution.
- ❑ **The Software Development Life Cycle (SDLC)** is a structured, systematic process that defines the stages involved in the planning, creation, testing, and deployment of a software system.

Module Review Summary

- ❑ **The main architectures** for Generative AI models are Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), Transformers, and Diffusion Models.
- ❑ GANs operate through an adversarial training process, which involves two competing neural networks: a **Generator** and a **Discriminator**.
- ❑ **VAEs** are probabilistic models that learn a compressed, structured representation of the data, allowing for controlled generation and manipulation.
- ❑ **Transformers** are the dominant architecture for sequence data, utilizing a mechanism that allows the model to weigh the importance of different parts of the input.
- ❑ **Diffusion models** are a newer class of models that have recently achieved state-of-the-art results in image generation, known for their high fidelity and diversity
- ❑ An **Application Programming Interface (API)** specifies the methods, data formats, and protocols for communication, acting as a secure and standardized gateway.
- ❑ The term **Generative Pre-trained Transformer (GPT)** is both a general term for a class of models and a specific brand name owned by OpenAI.
- ❑ We learned about different cloud providers and their services such as **AWS Bedrock** and **GCP Vertex AI**.

Review Questions

Let's Test our knowledge



Module Review Questions

1. What would be the main goal for a company to want to use AI? (Select One)

- A. Perform tasks that do not require human intervention.
- B. Perform tasks that commonly require human intelligence.
- C. Perform cost-effective tasks.
- D. Perform tasks that are knowledge-based tasks.

Module Review Questions

1. What would be the main goal for a company to want to use AI? (Select One)

- A. Perform tasks that do not require human intervention.
- B. Perform tasks that commonly require human intelligence.**
- C. Perform cost-effective tasks.
- D. Perform tasks that are knowledge-based tasks.

Explanation: The main goal of using AI is to create systems that can perform tasks that typically require human intelligence. AI can perform tasks that are either knowledge based or manual and these tasks may or not always be cost effective.

Module Review Questions

2. Which type of AI focuses on making AI decisions transparent and understandable to humans.

(Select One)

- A. Agentic AI
- B. Generative AI
- C. Explainable AI
- D. Predictive AI

Module Review Questions

2. Which type of AI focuses on making AI decisions transparent and understandable to humans?

(Select One)

- A. Agentic AI
- B. Generative AI
- C. Explainable AI**
- D. Predictive AI

Explanation: Explainable AI focuses on making AI decisions transparent and understandable to humans. Explainable AI aims to make the decision-making process of AI systems transparent and interpretable, addressing the "black box" nature of many AI models.

Module Review Questions

3. Which of the following is a subfield of Machine Learning that utilizes artificial neural networks with multiple layers to extract complex patterns from data. (Select One)

- A. Neural Networks
- B. Deep Learning
- C. Algorithms
- D. Natural Language Processing

Module Review Questions

3. Which of the following is a subfield of Machine Learning that utilizes artificial neural networks with multiple layers to extract complex patterns from data. (Select One)

- A. Neural Networks
- B. Deep Learning**
- C. Algorithms
- D. Natural Language Processing

Explanation: Deep Learning is a subfield of Machine Learning that utilizes artificial neural networks with multiple layers (hence "deep") to extract complex patterns from data.

The structure and function of the human brain inspired Deep Learning.

Module Review Questions

4. The four pillars of an effective AI strategy provide a comprehensive framework for organizations looking to leverage artificial intelligence to achieve their business goals. Which Pillar would be the first pillar to start with on a new client engagement? (Select One)

- A. Risk Management
- B. Value
- C. Vision
- D. Adoption Planning

Module Review Questions

4. The four pillars of an effective AI strategy provide a comprehensive framework for organizations looking to leverage artificial intelligence to achieve their business goals. Which Pillar according to Gartner would be the first pillar to start with on a new client engagement? (Select One)

- A. Risk Management
- B. Value
- C. Vision**
- D. Adoption Planning

Explanation : The four pillars of an effective AI strategy provide a comprehensive framework for organizations looking to leverage artificial intelligence to achieve their business goals. Vision is when we “Establish a clear and compelling vision” for how AI will be integrated into the organization and aligned with its overarching business strategy.

According to Gartner, these pillars are in this order: Vision > Value > Risk Management > Adoption

Domain 2 – Prompt Engineering and Model Interaction

Module 2 – CSGAIP Exam

Prompting your Models



Module 2

Sections

- 01 Section 2.1 – Prompt Engineering and Optimization
- 02 Section 2.2 – Demonstrate Advanced Prompting Strategies
- 03 Section 2.3 – Model Selection and Model Limitations

Prompt Engineering and Optimization

Section 2.1



Section 2.1 Lesson Overview

What is Prompt Engineering?

What is Context Engineering

Prompt Engineering /Context Engineering

Demonstration - Role:
Assigning a persona to ChatGPT

Demonstration -
Task/Goal: Clearly
defining what you want
the AI to do.

Demonstration -
Context: Providing
relevant background
information, audience,
purpose.

Demonstration -
Format: Specifying the
desired output
structure

Demonstration -
Tone/Style: Guiding the
AI's voice

Demonstration -
Constraints/Limitations:
Setting boundaries for
the output

Demonstration - Clarity,
Specificity, and
Affirmative Directives



Prompt Engineering

What is it?

Lesson Key Objectives

What is Prompt Engineering?

1

What is a Prompt?

2

How does prompting work?

3

Prompt Engineering

What is Prompt Engineering?

Prompt engineering is the art and science of designing and refining the instructions (prompts) given to a generative artificial intelligence (AI) model, especially large language models (LLMs), to achieve a desired and optimal output.

- ❑ Prompting is a way to guide the AI service to your desired outcome.
- ❑ Roadmap



Prompt Engineering

Key Components of an effective prompt

An effective prompt often includes one or more of the following elements:

- ☐ **Instruction:** The core task you want the model to perform. (Write, Translate, List, etc)
- ☐ **Context:** Background information or specific details to help the model understand the broader scenario ("As an AI Ethics Officer review this policy and identify...").
- ☐ **Input Data:** The specific text, numbers, or information the model needs to process (A paragraph to be summarized).
- ☐ **Output Indicator/Format:** Guidance on the style, tone, or structure of the response ("Write in the style of Shakespeare," "Use a bulleted list," "Limit the response to 200 words").
- ☐ **Role-Playing:** Instructing the AI to adopt a specific persona to provide a tailored response ("Act as a financial advisor," "You are a friendly chatbot")

Prompt Engineering

Example Prompts

- ❑ Prompt Example(Role): "Act as a content strategist. Brainstorm 10 blog post topics about remote work productivity for small businesses. For each topic, suggest a unique angle and a clear unique selling proposition (USP)."
- ❑ Prompt Example (Headlines): "Suggest five catchy and SEO-friendly headlines for a blog post about 'Beginner's Guide to Investing in Cryptocurrencies'."
- ❑ Prompt Example: (Podcast Outline): "Create a podcast outline for an episode discussing 'The Impact of AI on the Job Market.' Include talking points for introduction, main segments, and conclusion"

Prompt Engineering

Example Prompts

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- ❑ Prompt Example: (Podcast Outline): "Create a podcast outline for an episode discussing 'The Impact of AI on the Job Market.' Include talking points for introduction, main segments, and conclusion"

Prompt Engineering

When to Use?

Prompt Engineering:

- ☐ Copywriting variations
- ☐ Simple on demand prompts “Write me a promotional email”
- ☐ One-shot code generation
- ☐ Flashy demonstrations





Prompt Engineering /Context Engineering

Comparing and contrasting

Lesson Key Objectives

Compare/Contrast Prompt Engineering and Context Engineering

1



Prompt Engineering /Context Engineering

Comparing Prompt Engineering to Context Engineering

- ❑ Context engineering is a **superset** of prompt engineering.
- ❑ While prompt engineering is focused on **what you say** to the model at a single moment, context engineering is focused on **what the model knows** when you say it and how that knowledge is managed over time

Prompt Engineering /Context Engineering

Relationship Between the Two

Prompt engineering is a key component of context engineering.

- ❑ Context Engineering determines what fills the LLM's entire input context window (e.g., historical data, search results, tool definitions).
- ❑ Prompt Engineering determines what the core instruction or question is inside that window.

Think of the LLM's context window as a piece of paper:

- ❑ Prompt Engineering carefully writes the main instruction on the page.
- ❑ Context Engineering involves deciding what additional information (notes, books, reference materials, past conversation transcripts) to include on that page alongside the main instruction.

Prompt Engineering /Context Engineering

Comparing Prompt Engineering to Context Engineering

	Prompt Engineering	Context Engineering
Focus	Crafting effective instructions	Design a multiple turn AI system
Scope	Text of immediate question/instruction	Ecosystem (memory, tools, data, and prompt).
Goal	To get a specific, high-quality response for one request.	To ensure consistent, accurate, and personalized performance over time and across different users.
Mindset	Linguistic tuning and creative wording.	Systems design and architectural thinking.
Components	System instructions, user request, few-shot	Includes the prompt plus: • Memory • RAG (Retrieved-Augmented Generation) • Tool/Functions • Structured output formats (JSON, XML) • Guardrails and system rules
Key Question	How can I phrase this instruction to get the best immediate result?	What information, tools, and memory does the LLM need at this exact moment to accomplish the user's overall goal?



Prompts and Context

Demonstration

Lesson Key Objectives

Prompting ChatGPT

1

Context and Prompting

2

Demonstration





Context Engineering

What is it?

Lesson Key Objectives

What is Context Engineering?

1

Important Components.

2

Context Engineering

What is Context Engineering?

Context engineering is a systematic discipline in AI, particularly with large language models (LLMs), that focuses on designing and managing the entire informational environment an AI system uses to generate intelligent, accurate, and personalized responses.

- ❑ Described as the "art and science of filling the context window with just the right information for the next step."
- ❑ Holistic approach to prompting



Context Engineering

Context engineering is a holistic approach that goes beyond simply writing a good prompt.

Components include:

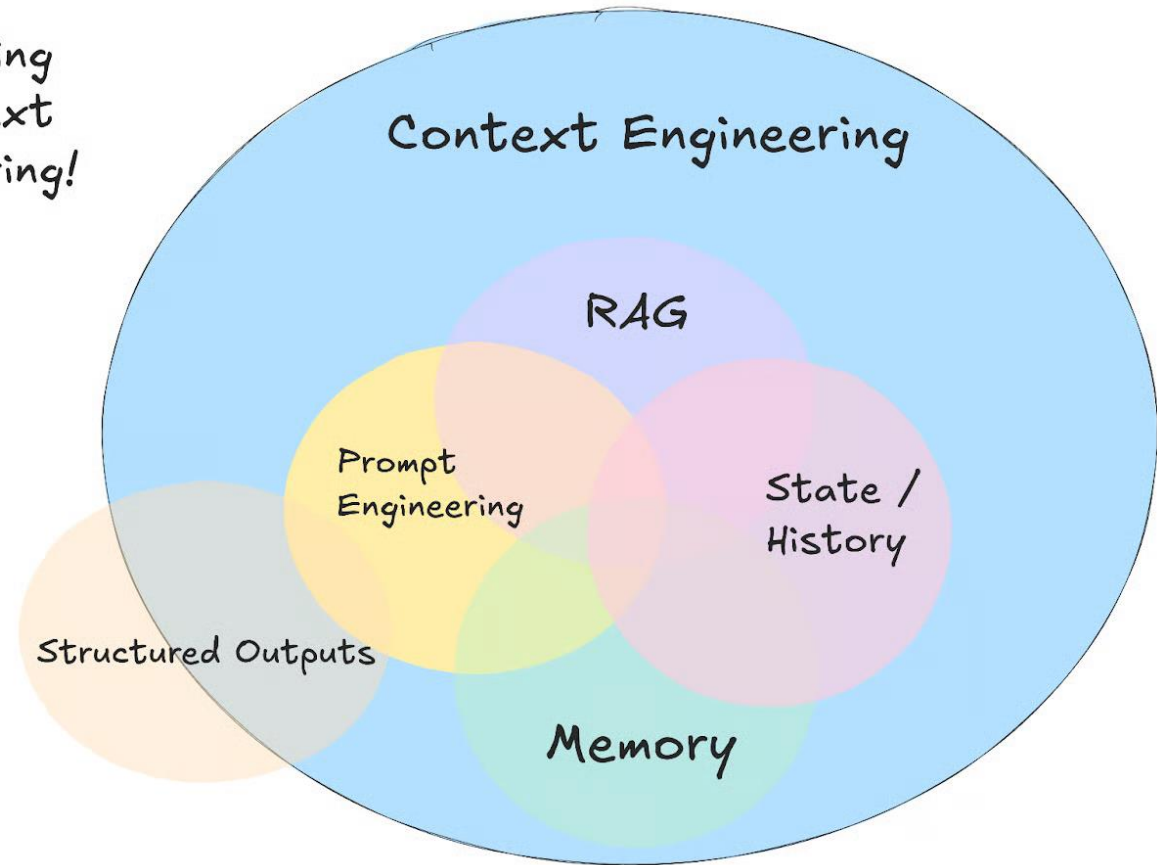
- ☐ System Instructions
- ☐ Retrieval-Augmented Generation (RAG)
- ☐ Memory Management
- ☐ Tool Integration
- ☐ Data and Format Optimization



Prompt Engineering

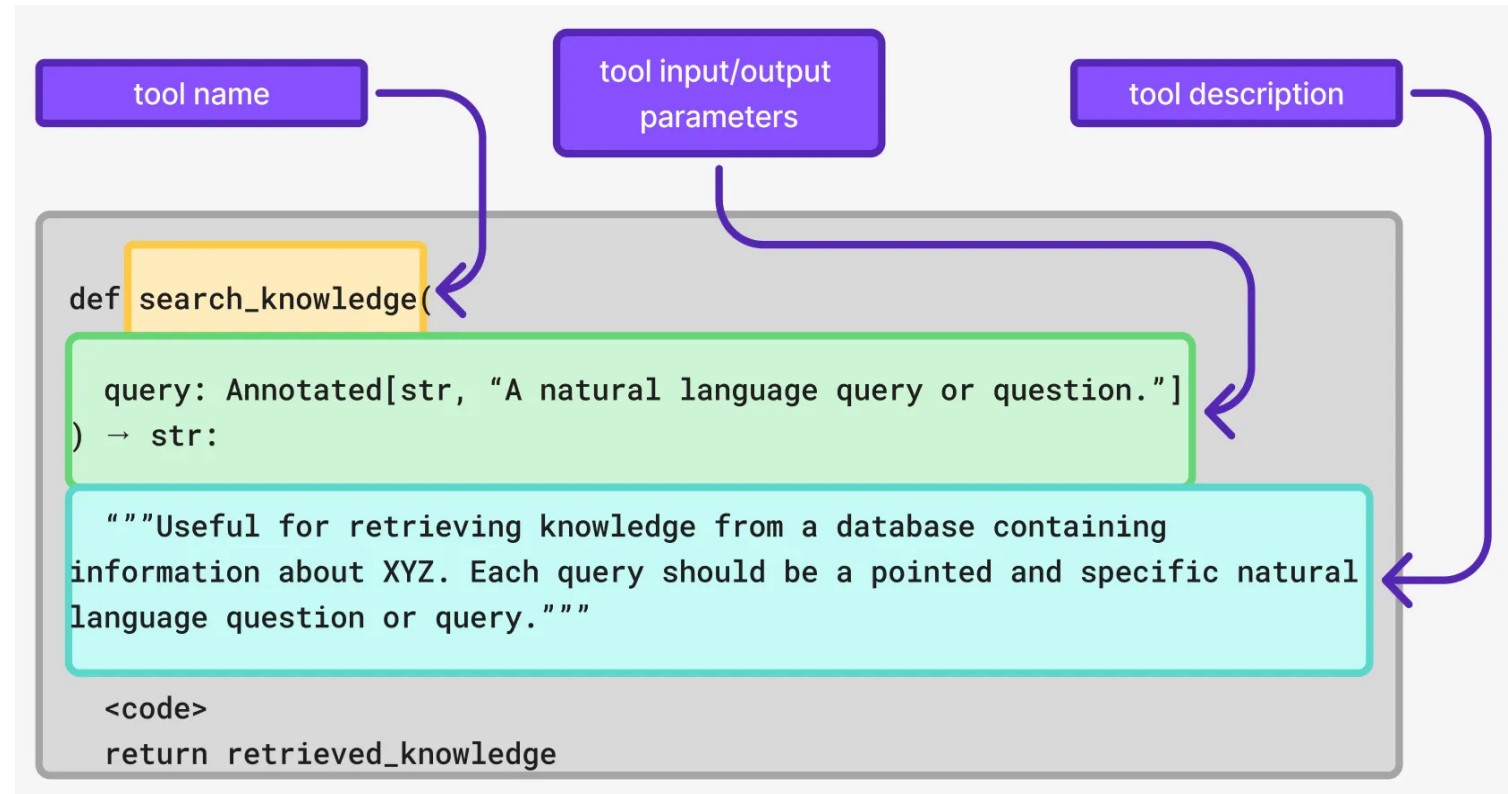
- ❑ Source: 12-factor-agents
- ❑ Used for Conversational AI, document analysis tools, coding assistants

Everything
is Context
Engineering!



Prompt Engineering

- ❑ Source: Llamaindex
- ❑ Look at Context Engineering as effectively “coding” your prompts.



Context Engineering

When to Use?

Context Engineering:

- ☐ LLM agents with memory
- ☐ Customer support bots that don't hallucinate
- ☐ Multi-turn flows
- ☐ Production systems that need predictability





Roles

Demonstration

Lesson Key Objectives

Assigning a persona to ChatGPT.

1

Demonstration





Tasks/Goals

Demonstration

Lesson Key Objectives

Clearly define what you want GPT to do.

1

Demonstration





Provide Context

Demonstration

Lesson Key Objectives

Providing relevant background information, audience and prurpose

1

Demonstration





Provide Format

Demonstration

Lesson Key Objectives

Specify the desired output structure.

1

Demonstration





Tone and Style

Demonstration

Lesson Key Objectives

Guiding the AI's voice.

1

Demonstration





Constraints and Limitations

Demonstration

Lesson Key Objectives

Setting the boundaries for the Output

1

Demonstration





Clarity, Specificity, and Affirmative Directives

Demonstration

Lesson Key Objectives

Setting the Clarity, Specificity, and Affirmative Directives

1

Demonstration



Prompting Strategies

Section 2.2



Section 2.2 Lesson Overview

Demonstration -
Iterative refinement:
How to improve
prompts based on
outputs.

Demonstration - Zero-
Shot Prompting

Demonstration -
Feedback Loops

Demonstration -
Generate Content

Demonstration -
Selecting Prebuilt GPTS

Module Review
Questions

Module Review
Summary



Iterative Refinement

Demonstration

Lesson Key Objectives

Improve prompts based on outputs and adjustments.

1

Demonstration





Zero-Shot and Few Shot Prompting

Demonstration

Lesson Key Objectives

Zero-Shot Prompting

1

Demonstration





Feedback Loop

Demonstration

Lesson Key Objectives

Few-Shot Prompting (In-Context Learning)

1

Demonstration





Detailed Content Outlines

Demonstration

Lesson Key Objectives

Chain-of-Thought (CoT) Prompting

1

Demonstration





Selecting and Creating GPTs

Demonstration

Lesson Key Objectives

Select and Create GPTS in ChatGPT

1

Demonstration



Module Review Summary

Short Summary of what we covered in this module



Module Review Summary

- ❑ **Prompt Engineering** is the discipline of designing and refining input prompts (instructions) to efficiently guide a large language model (LLM) to produce desired, accurate, and high-quality outputs.
- ❑ **Context Engineering** is a broader approach that focuses on designing the entire system and flow around the prompt.
- ❑ **Prompt Engineering** is focused on the immediate instruction for a single input/output while Context Engineering is focused on the overall architecture and what the model knows to maintain consistency, memory, and scalability across complex, multi-turn applications.
- ❑ **Iterative Refinement** is the process of starting with a basic prompt, analyzing the AI's initial output, and then adjusting or adding details to the prompt in a new turn to improve the result. It's a continuous feedback loop until the desired quality is achieved.
- ❑ **Zero Shot Prompting** is asking the model to perform a task without providing any examples in the prompt. It relies solely on the model's pre-trained knowledge and general understanding of the instruction
- ❑ **Selecting prebuilt GPTs** is the practice of choosing a specialized, pre-configured version of a Large Language Model that has been specifically designed or tuned for a particular task, role, or knowledge domain.

Module Review Questions

Let's Test our knowledge



Module Review Questions

1. Which of the following best describes the primary focus of Context Engineering in an LLM application? (Select One)

- A. Specifying the desired style and tone of the AI's response.
- B. Perform tasks that are knowledge-based tasks.
- C. Utilizing few-shot examples to improve the model's accuracy on a specific task.
- D. Crafting the specific instruction for a single, immediate request.

Module Review Questions

1. Which of the following best describes the primary focus of Context Engineering in an LLM application? (Select One)

- A. Specifying the desired style and tone of the AI's response.**
- B. Perform tasks that are knowledge-based tasks.
- C. Utilizing few-shot examples to improve the model's accuracy on a specific task.
- D. Crafting the specific instruction for a single, immediate request.

Explanation: Context Engineering focuses on designing the entire system flow and managing the dynamic information the LLM uses for long-term consistency and scalability.

Module Review Questions

2. You want to ensure a Large Language Model (LLM) uses terminology and knowledge appropriate for a financial analyst, which prompt component should be prioritized?(Select One)

- A. Role
- B. Task/Goal
- C. Format
- D. Constraints/Limits

Module Review Questions

2. You want to ensure a Large Language Model (LLM) uses terminology and knowledge appropriate for a financial analyst, which prompt component should be prioritized?(Select One)

A. Role

B. Task/Goal

C. Format

D. Constraints/Limits

Explanation: The Role component (e.g., "Act as a financial analyst") assigns a specific persona, dictating the model's expertise, vocabulary, and perspective.

Module Review Questions

3. A user enters the prompt: "Classify the following customer review as positive, negative, or neutral: 'The new interface is fast and intuitive.'"

This is an example of what kind of prompting strategy? (Select One)

- A. Iterative
- B. One Shot
- C. Zero-Shot
- D. Chain of Thought

Module Review Questions

3. A user enters the prompt: "Classify the following customer review as positive, negative, or neutral: 'The new interface is fast and intuitive.'"

This is an example of what kind of prompting strategy? (Select One)

- A. Iterative
- B. One Shot
- C. Zero-Shot**
- D. Chain of Thought

Explanation: Zero-Shot Prompting gives a direct instruction for a task with zero examples provided in the prompt, relying on the model's general training.

Domain 3 – Enterprise Applications and Workflow

Module 3 – CSGAIP Exam

Applying Strategy and Resources for AI Application Success



Module 3

Sections

- 01 Section 3.1 – Enterprise Application Planning & Integration
- 02 Section 3.2 – Use Generative AI for Data-Driven Tasks

Enterprise Application Planning and Integration

Section 3.1



Section 3.1 Lesson Overview

General
Application
Integration
Patterns

Evaluation Criteria
of Gen AI Services

Whiteboard -
AI/ML Planning
with GCP

AI Integration
Patterns

Whiteboard -
Deploying AI
Services on AWS

AI Adoption
Challenges

AI Adoption Best
Practices



Evaluation Criteria of Gen AI Services

Important Criteria

Lesson Key Objectives

What are evaluation factors for Gen AI?

1

Identify Quality and Performance Metrics.

2

Evaluation Criteria of Gen AI Services

Safety and Ethics

Evaluating Gen AI services is incomplete without assessing their compliance with safety, fairness, and ethical standards.

- ☐ **Harmful Content:** Testing for and tracking the rate of generating prohibited outputs (e.g., hate speech, violence, illegal activity).
- ☐ **Bias and Fairness:** Evaluating the model's tendency to perpetuate stereotypes or show unequal quality of service across different demographic groups (**Representation Bias**).
- ☐ **Hallucination Rate:** Measuring the percentage of generated content that is factually incorrect, fabricated, or ungrounded from the source material.
- ☐ **Adversarial Robustness:** Testing the model's resistance to "jailbreaking" or inputs designed to elicit problematic or unintended responses.

Evaluation Criteria of Gen AI Services

Operational and Business Metrics (KPIs)

These metrics focus on the system's performance, efficiency, and business impact in a real-world setting.

- ❑ **Latency:** The time it takes for the service to generate a response. This is often broken down into **Model Latency** (inference time) and **Retrieval Latency** (for RAG systems).
- ❑ **Throughput:** The volume of requests or tokens the system can handle per unit of time.
- ❑ **Uptime and Error Rate:** Standard system reliability metrics, essential for enterprise-grade services.
- ❑ **Adoption Rate and Frequency of Use:** Metrics that reflect user satisfaction and the business value of the service.
- ❑ **Cost Efficiency:** Measuring the resource utilization (GPU/TPU) and overall **Return on Investment (ROI)**.

Evaluation Criteria of Gen AI Services

Metric Type	Use Case	Key Metrics
Automatic (Text)	Text Generation, Translation, Summarization	BLEU Score (measures N-gram overlap with a reference text; higher is better). ROUGE (Recall-Oriented Understudy for Gisting Evaluation, measures overlap, especially for summarization). Perplexity (measures how well a model predicts a sequence of words; lower is better).
Automatic (Image)	Image Generation, Style Transfer	FID (Fréchet Inception Distance) (compares the distribution of generated and real images; lower is better). Inception Score (IS) (measures image quality and diversity; higher is better).
AI-Assisted (LLM as a Judge)	General Content Quality	Groundedness/Faithfulness (fact-checking against source material). Relevance (how well the response addresses the query). Coherence/Fluency (linguistic quality, flow, and grammar).
Human Evaluation	Subjective Quality	Direct human ratings on categories like creativity, novelty, usefulness, clarity, and tone/style alignment with the prompt.

Quality and Performance Metrics



AI Adoption Best Practices

Criteria and process

Lesson Key Objectives

Why the need for AI Adoption Best Practices?

1

Identify the Ten AI Adoption Best Practices.

2

AI Adoption Best Practices



AI Adoption

- ☐ AI adoption is no longer a futuristic concept but a critical driver of innovation across industries in 2025.
- ☐ Businesses increasingly leverage AI for improved efficiency, decision-making, and customer experiences.
- ☐ To navigate this transformative landscape successfully, organizations should adhere to best practices for AI adoption.

AI Adoption Best Practices



AI Adoption Best Practices

Best Practice	Overview
Define Clear Goals and Objectives	Establish measurable objectives and set KPIS
Assess Data Readiness	Conduct a data audit and identify quality, structure, governance, security etc and implement a data lifecycle.
Develop a Comprehensive AI Implementation Strategy	Define Scope, Identify stakeholders, Identify Business Use Cases, Plan Rollout as a project.
Prioritize Governance, Transparency, and Ethics	Provides a strong Framework, guidelines for accountability, define AI ethical principles and provide for transparency.
Enhance Data Management	Adopt advanced data management techniques, such as anonymization, differential privacy, and encryption, to protect sensitive data while maintaining its usability for AI-driven insights.

AI Adoption Best Practices

Best Practice	Overview
Choose Right AI Strategy	Select the right combination of AI tools, platforms, and deployment models and evaluate whether to deploy AI on your infrastructure or leverage third-party platforms.
Build an AI Proficient Team	Assess your team's existing skills and identify any gaps in AI expertise. Invest in talent development and foster a culture of continuous learning.
Manage Risks and Build Ethical Frameworks:	Identify and mitigate potential risks associated with AI adoption and develop an ethical framework with clear guidelines.
Test and Evaluate Models	Provides a strong Framework, guidelines for accountability, define AI ethical principles and provide for transparency.
Craft a Compelling Case for Investment	Quantify the expected Return on Investment (ROI) using pilot projects and real-world data to estimate potential cost savings, productivity gains, and risk reduction.

Using Generative AI for Data-Driven Tasks (Use Cases)

Section 3.2



Section 3.2 Lesson Overview

Use Case - Data
Science and
Business
Intelligence

Use Case - Data
Augmentation and
Synthesis

Use Case - Code
and Content
Generation for
Data Analysis

Use Case- FedEx
Data Analysis and
Interpretation

Module Review
Questions

Module Review
Summary



Data Science and BI Use Case

Morgan Stanley

Data Science and BI Use Case – Morgan Stanley

A critical use case across the financial industry (including firms like J.P. Morgan) is using Generative AI (specifically models like GANs) to create **synthetic data**.

The Challenge:

Financial institutions deal with highly sensitive data (PII, transaction details). Strict regulations (like GDPR) make it difficult to share, test, or train complex DS models.

Technology: They use a Large Language Model (LLM) trained on Morgan Stanley's vast, internal knowledge base, which includes decades of market research, economic reports, analyst notes, and company-specific data.



Data Science and BI Use Case – Morgan Stanley

The GenAI Solution: Generative AI models learn the statistical properties and correlations of the real, sensitive data and then **generate an entirely new, artificial dataset** that is statistically identical but contains no real customer information.

DS Impact: This synthetic data can be used for:

- ☐ Training new fraud detection models faster.
- ☐ Testing and quality assurance (QA) of trading systems.
- ☐ Developing new risk management models in a privacy-compliant sandbox.

Data Science and BI Use Case – Morgan Stanley

Traditional BI/DS

Manual Data Retrieval: Search databases, download reports, skim PDFs.

Descriptive Only: Reports show what *did* happen in the market last quarter.

High Barrier to Entry: Requires expertise in SQL or proprietary BI tools to drill into data.

Generative AI Enhancement

Generative Retrieval (RAG): Ask a question; receive a direct, synthesized answer instantly.

Inferred Insights: AI synthesizes reports to suggest *why* something happened and what the next potential strategic step is.

Natural Language Interface: Any advisor can interact with deep data simply by talking to the AI, democratizing access to BI.



Automated Content Generation

Use Case - Shopify

Content Generation - Shopify

A critical use case across the ecommerce industry allows companies to rapidly create high-quality, engaging, and unique text and visual content for their vast product catalogs.

The Challenge:

Listing a new product requires a compelling, unique description and title. Manually writing thousands of these is slow, costly, and difficult to keep consistent.

Technology: Input: A product manager or merchant uploads basic product data (e.g., product name, a few bullet points of features, and technical specifications).

Example Input: "Product: Hydrating Vitamin C Serum. Features: Brightens skin, reduces fine lines, made with natural ingredients. Tone: Professional and enthusiastic."

Content Generation - Shopify

Continued

Process: The generative AI model (a Large Language Model, or LLM) analyzes the input and, in real-time, crafts a full-length, narrative product description.

Output: A polished, human-quality product description, along with variations for different marketing channels (e.g., a short version for social media ads, a longer version for the product page, or a version translated into multiple languages).



Content Generation - Shopify

Benefit

Operational Efficiency

Scalability

Improved SEO & Visibility

Content Quality & Consistency

Description

Drastically reduces the time required for copywriters. What took hours now takes seconds, allowing products to be listed faster (speed to market).

Allows companies to manage and list massive inventories (thousands of products) without having to hire a large team of writers.

The AI automatically embeds relevant, high-ranking search terms into the copy, improving the product's visibility on Google and within the company's own site search.

Ensures a uniform, high-quality, and on-brand message across the entire product catalog, eliminating variations in tone and style between different human writers.



AI/ML Use Case - Finance

Chase AI Fraud Detection

Lesson Key Objectives

How does Chase use AI

1

What are the efficiencies and benefits that Chase realizes?

2

Logistics Use Case – JPMorgan Chase

JPMorgan Chase Scenario: JPMorgan Chase, like other large financial institutions, processes an enormous volume of transactions daily, making it a prime target for fraudulent activities ranging from credit card fraud and wire transfer scams to account takeovers.

Traditional rule-based fraud detection systems often struggle with the sophistication and speed of modern fraud, leading to:

- ☐ **High False Positive Rates:** Legitimate transactions can be flagged as suspicious, causing inconvenience for customers and increasing operational overhead for the bank in investigating these false alarms.
- ☐ **Slow Detection of Novel Fraud Schemes:** Rule-based systems are often reactive, requiring manual updates based on known fraud patterns. They struggle to identify new and evolving fraud techniques.
- ☐ **Significant Financial Losses:** Despite existing measures, substantial financial losses occur due to undetected fraudulent activities.
- ☐ **Increased Operational Costs:** A large team of analysts must manually review alerts and investigate suspicious transactions.

Logistics Use Case – JPMorgan Chase

Solution: JPMorgan Chase implements an advanced AI-powered fraud detection system that leverages machine learning and deep learning algorithms to analyze vast amounts of transaction data in real-time.

This system goes beyond traditional rule-based approaches by:

- ❑ **Anomaly Detection:** AI algorithms learn individual customers' and accounts' normal behavior patterns. By continuously monitoring transactions, the system can identify deviations from these established patterns that may indicate fraudulent activity, even if they don't match known fraud rules
- ❑ **Behavioral Biometrics:** The AI analyzes patterns in how customers interact with their banking platforms, such as typing speed, mouse movements, and navigation patterns. Unusual deviations in these behavioral biometrics can signal that an account has been compromised.

Logistics Use Case – JPMorgan Chase

This system goes beyond traditional rule-based approaches by: (Continued)

- ❑ **Network Analysis:** The AI can analyze the relationships between different accounts and transactions, identifying suspicious networks or clusters of activity that might be indicative of organized fraud rings
- ❑ **Natural Language Processing (NLP) for Contextual Analysis:** For transactions involving free-form text, such as wire transfer instructions or customer service interactions, NLP can identify keywords or phrases often associated with fraudulent activities.
- ❑ **Predictive Modeling:** Machine learning models are trained on historical fraud data to predict the likelihood of future fraudulent transactions based on various features and patterns. This allows the bank to proactively identify and potentially block high-risk transactions.

Logistics Use Case – JPMorgan Chase

The AI Solution: The implementation of AI in fraud detection has led to significant benefits and improvements for JPMorgan Chase:

- ☐ Reduced False Positives
- ☐ Faster Detection
- ☐ Increased Detection Rates
- ☐ Optimized Analyst Productivity
- ☐ Scalability
- ☐ Cost Savings

Topic Discussion





AI/ML Use Case – Logistics

Fedex Orion

Logistics Use Case - FedEx

FedEx provides a compelling use case for how a large logistics company can significantly improve efficiency by implementing AI in route optimization.

The Challenge:

Before AI-powered solutions, FedEx relied on more traditional methods for planning delivery routes. These methods often couldn't dynamically adjust to real-time factors like traffic congestion, unexpected road closures, or last-minute delivery requests.

This could lead to:

- ☐ Increased mileage driven.
- ☐ Higher fuel consumption and related costs.
- ☐ Delays in deliveries, impacting customer satisfaction.
- ☐ Inefficient allocation of driver time.

Logistics Use Case - FedEx

The AI Solution: FedEx Surround and ORION (On-Road Integrated Optimization and Navigation). FedEx has implemented sophisticated AI-driven systems like FedEx Surround and ORION to tackle these challenges:

Real-time Data Analysis: These systems analyze a vast array of real-time data, including:

- ☐ GPS tracking of vehicles.
- ☐ Live traffic conditions.
- ☐ Weather forecasts.
- ☐ Package destinations and time-sensitive deliveries.
- ☐ Historical delivery patterns.

Logistics Use Case - FedEx

The AI Solution: The implementation of AI in route optimization has led to significant efficiency improvements for FedEx:

Reduced Mileage and Fuel Consumption: By calculating and dynamically adjusting to the most efficient routes, FedEx has drastically reduced the total miles driven by its fleet, leading to substantial fuel savings and a lower carbon footprint.

Faster Delivery Times and Improved Customer Satisfaction: Optimized routes enable drivers to complete their deliveries more quickly and efficiently, leading to improved on-time delivery rates and higher customer satisfaction.

Better Resource Allocation: AI helps FedEx optimize the utilization of its drivers and vehicles, ensuring that resources are deployed effectively to meet delivery demands.

Lower Operational Costs: The combination of reduced fuel consumption, optimized resource allocation, and potentially lower overtime costs for drivers contributes to significant overall operational cost reductions

Topic Discussion



Module Review Summary

Short Summary of what we covered in this module



Module Review Summary

- **Effective evaluation is essential** for choosing the right model, ensuring it meets business requirements, and managing associated risks.
- **Quality of Output:** This is the most direct measure of the model's performance.
- **Fidelity/Coherence:** How realistic, grammatically correct, and logically consistent is the generated content?
- Evaluation often involves a blend of **Automated Metrics** (e.g., BLEU, ROUGE scores for text, FID for images) and **Human Evaluation** (expert review, user studies, and A/B testing) to capture the subjective aspects of quality and usefulness.
- The **specific criteria prioritized** must always be aligned with the intended downstream application of the Gen AI service.
- **Automated Content Generation (ACG)**, leverages Generative AI (Gen AI) to create various forms of marketing, product, and communication content with minimal human intervention.
- **Orion (Optimization, Resources, and Information Online)** is a powerful, AI-driven routing and dispatch platform used by FedEx for package delivery and pickup.

Module Review Questions

Let's Test our knowledge



Module Review Questions

1. Which of the following Gen AI evaluation criteria is primarily concerned with minimizing harmful or offensive output?(Select One)

- A. Diversity/Creativity
- B. Fidelity/Coherence
- C. Latency
- D. Safety

Module Review Questions

1. Which of the following Gen AI evaluation criteria is primarily concerned with minimizing harmful or offensive output?(Select One)

- A. Diversity/Creativity
- B. Fidelity/Coherence
- C. Latency
- D. Safety**

Explanation: Toxicity and Safety: The model's ability to avoid generating harmful, offensive, hateful, or illegal content.

Module Review Questions

2. When evaluating a Gen AI service, which criterion measures the degree to which a user can effectively modify or control the generated output through specific instructions (e.g., tone or style)?(Select One)

- A. Diversity/Creativity
- B. Controllability
- C. Throughput
- D. Fidelity/Coherence

Module Review Questions

2. When evaluating a Gen AI service, which criterion measures the degree to which a user can effectively modify or control the generated output through specific instructions (e.g., tone or style)?(Select One)

- A. Diversity/Creativity
- B. Controllability**
- C. Throughput
- D. Fidelity/Coherence

Explanation: Controllability assesses the model's sensitivity to user-defined constraints, allowing users to effectively steer the generation process, which is critical for application integration.

Domain 4 – Strategic Leadership and Business Value

Module 4 – CSGAIP Exam

Driving Business Value with AI Leadership



Module 4

Sections

- 01 Section 4.1 – Building the Business Case for GenAI
- 02 Section 4.2 – Vendor and Technology Strategy
- 03 Section 4.3 – Strategic AI Proposal Development

Building the Business Case for Generative AI

Section 4.1



Section 4.1 Lesson Overview

4 Pillars of AI
Strategy

Whiteboard - Value
Driven Roadmap

Key Value
Propositions of
Generative AI

Measuring
Generative AI
Value: Strategic
Metrics

Demonstration -
Google Cloud
Value Proposition

Business Use Cases

Uplift Metrics



4 Pillars of AI Strategy

Understanding what the Key Pillars are

Lesson Key Objectives

What are the Four Key Pillars?

1

What is the goal of a Pillar according to Gartner?

2

What pillars emphasize Risk?

3

4 Pillars of an AI Strategy

The four pillars of an effective AI strategy provide a comprehensive framework for organizations looking to leverage artificial intelligence to achieve their business goals.

According to Gartner, these pillars are:

Vision

Value
Realization

Risk
Management

Adoption
Plans

4 Pillars of an AI Strategy

1. **Vision:** Establish a clear and compelling vision for how AI will be integrated into the organization and aligned with its overarching business strategy.
2. **Value Realization:** Focuses on identifying and quantifying the potential value that AI can bring to the organization.
3. **Risk Management:** This pillar emphasizes the importance of proactively addressing the potential risks of AI implementation, including ethical considerations, data privacy and security, regulatory compliance, and possible biases in AI systems.
4. **Adoption Plans** - This involves developing detailed plans for how AI will be adopted and integrated across the organization.



Presenting a Value-Driven Proposition and Roadmap

Whiteboard

Lesson Key Objectives

What is Value and a Value Roadmap?

1

What are common AI value discussion points?

2

Identify how you can present value via whiteboards and other tools

3

Presenting a Value-Driven Proposition and Roadmap

AI Value Proposition and Roadmap

- ❑ **Value** refers to the tangible and intangible benefits a project delivers beyond its cost, measured by the ratio of benefits to resource usage. (Value is the degree to which a project satisfies stakeholder needs and contributes to the overall success of the organization)
- ❑ A **value roadmap** is a strategic document that plots the course of a product or service, highlighting the specific values it delivers over time.
- ❑ AI initiatives can **deliver a range of benefits, from tangible to intangible**, including short-term and long-term gains, as well as strategic and tactical impacts, which influence the ROI model.
- ❑ As a Solutions Architect, we should **thoroughly capture the value and impact of AI initiatives** for our customers.
- ❑ Evaluate across **three distinct ROI categories** to capture their value and impact fully.

Presenting a Value-Driven Proposition and Roadmap

Measurable ROI	This involves direct, quantifiable impacts of AI, such as cost savings and revenue increases.
Strategic ROI	This focuses on AI's role in achieving long-term organizational goals (3-to-5-year period), such as digital transformation initiatives to achieve a competitive advantage.
Capability ROI	This looks at how AI projects improve an organization's overall AI maturity, through skills development, specialized job roles, and cultural readiness.

Presenting a Value-Driven Proposition and Roadmap

Three types of ROI

- ❑ **Measurable ROI**—The AI system reduces inventory carrying costs and decreases lost sales due to out-of-stock items, leading to cost savings and increased revenue. Furthermore, the system's ability to adjust inventory levels in response to changing market conditions improves customer satisfaction, as popular items are consistently in stock.
- ❑ **Strategic ROI**—By automating inventory management, the retail chain supports its strategic goal of operational efficiency and superior customer service. The AI system's capabilities allow the organization to adapt quickly to consumer trends and seasonal fluctuations, strengthening its market position.
- ❑ **Capability ROI**—The AI system enhances the skills of the workforce, as employees learn to interact with and manage advanced AI tools. This improves the organization's overall technological capability, fostering an innovation-driven culture.

Presenting a Value-Driven Proposition and Roadmap

Value Driven Roadmap (AI)

- ❑ To effectively present a value-driven roadmap, **focus on demonstrating how the roadmap aligns with company goals**, addresses stakeholder needs, and delivers quantifiable benefits.
- ❑ Use **clear visuals and metrics, prioritize communication** with key stakeholders, and emphasize the roadmap's adaptability to changing market conditions.
- ❑ **Visualize, Quantify, and Summarize**



Whiteboard Discussion



AI Powered Value Proposition Canvas Generators

Demonstration

Lesson Key Objectives

What is a AI Powered Value Proposition Tool

1

Why use these tools?

2

Demonstration





Google Cloud Value Proposition

Demonstration

Lesson Key Objectives

What are AI/ML services on GCP?

1

Why is Vertex AI so important?

2

Identify what potential value points Google Cloud AI services can bring to the table

3

Demonstration





Business Use Cases

Importance of a business use case

Lesson Key Objectives

What is a business use case?

1

What are the main reasons for a business use case?

2

Business Use Case

What is a Business Case?

A **business use case** is a document that presents the costs, risks, and benefits of a particular initiative, justifies the investment, and pitches because decision-makers should approve of the recommendation.

Why a Business Use Case?

- ☐ Present the need and rationale for the purchase.
- ☐ Obtain buy in for funding
- ☐ Add resources such as staff or contractors
- ☐ Change the scope of a project





Uplift Metrics

Value Quantified

Lesson Key Objectives

What are uplift metrics

1

Identify importance of uplift metrics and best practices

2

Identify and provide for some examples of uplift metrics

3

Uplift Metrics

Uplift Metrics are what we use to “influence” the deal and are the foundation of the business case.

Common Best Practices

- ☐ Always use **customers language** (jargon)
- ☐ Consider **customers demographics**
- ☐ Assemble **conservative estimates**
- ☐ Use proven **case studies**
- ☐ Use **neutral research** sources
- ☐ Partner with client to develop Use Case



Uplift Metrics

Uplift Metrics Examples

- ❑ Reduces TCO by 70%. Current vs Proposed
- ❑ Faster Time To Market is 2 months faster with proposed solution which could increase revenue by 100%.



Vendors and the AI Technology Stack

Section 4.2



Section 4.2 Lesson Overview

Funding Models (CAPEX
and OPEX)

AI Cloud Cost Drivers

Demonstration -
Understanding AI Cloud
Costs on AWS

AI Vendor Selection
Criteria

AI Technology Stack
Strategy

Executive Messaging

Proof of Concepts (PoC)

AI Processors -
GPU/TPU/ASIC/NPU/FPGA



CAPEX and OPEX

Understanding the procurement funding sources.

Lesson Key Objectives

Discuss what is CAPEX and OPEX.

1

Identify common OPEX and CAPEX AI and Cloud expenses.

2

Describe the Hybrid AI Cost Model.

3

Estimating AI and Cloud Costs

CAPEX and OPEX

Cloud computing introduces new ways to consume and pay for IT services.

- ❑ The shift from CAPEX to OPEX financing and cash flow models using cloud
- ❑ The increasing reliance on cloud-based AI-as-a-Service (AlaaS) models has pushed most AI spending toward the OpEx side for greater flexibility and scalability.



Estimating Cloud Costs

CAPEX and OPEX

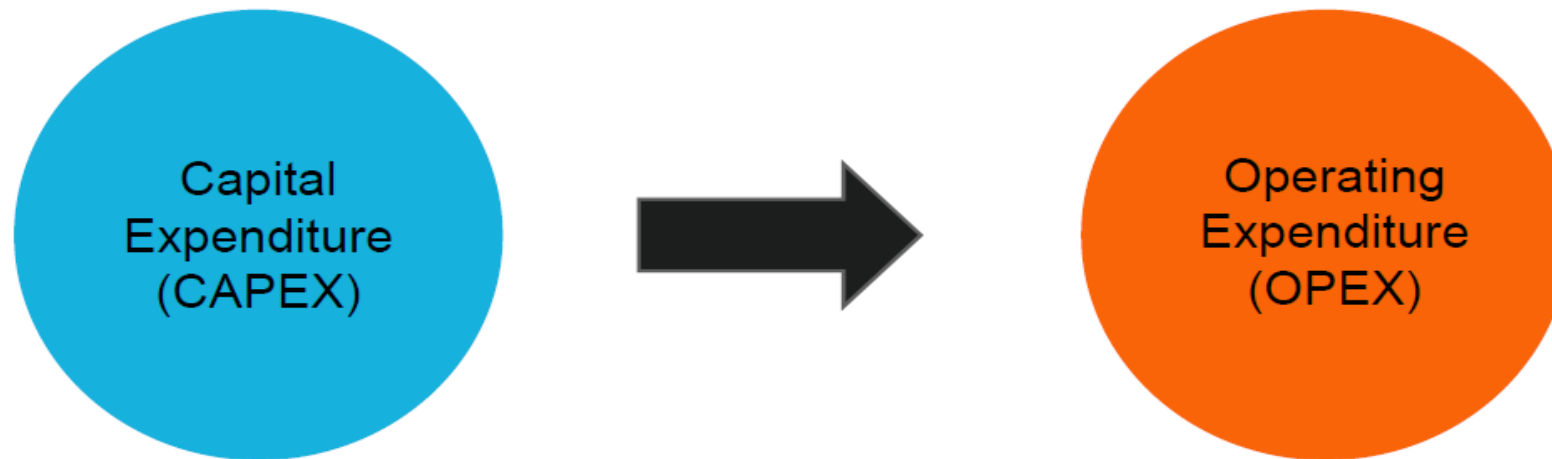
- ❑ **Capital Expenditures (Capex)** are investments made by an organization for long-term benefits in the future.
- ❑ **Operational Expenditures (Opex)** are the ongoing costs related to day-to-day operations

	Capital Expense (CapEx)	Operating Expense (OpEx)
Purpose	Assets purchased with a useful life beyond the current year	Ongoing costs to run a business
When Paid	Lump sum up front	Monthly or annual recurring
When Accounted For	Over a 3 to 10 year lifespan as the asset depreciates	In the current month or year
Listed As	Property or equipment	Operating cost
Tax Treatment	Deducted over time as asset depreciates	Deducted in the current tax year
Example	Buying a laser printer	Buying toner for printer

Estimating Cloud Costs

CAPEX and OPEX

- ❑ Cloud computing is a shift from CAPEX to OPEX financing and payment models. The method of financing IT is changing with cloud computing.
- ❑ CAPEX is about ownership of assets, whereas OPEX is about paying for services.



Capital Expenditure (CapEx) for AI

Category	Specific AI Expenditure	Financial Impact
Infrastructure	Purchasing and installing dedicated servers, high-end GPUs (like NVIDIA chips), and networking equipment for an on-premise AI data center.	Large upfront cost; amortized/depreciated over several years.
Data Center	Construction or significant physical upgrades to a private data center to house AI/ML hardware.	Large, one-time investment; creates a long-term physical asset.
Software Licensing	Acquiring a perpetual license for enterprise AI software or Machine Learning Operations (MLOps) platforms.	High upfront cost for the license itself.
Intangible Assets	Costs associated with in-house development of proprietary AI models, patents, or intellectual property (IP) that provide long-term value.	Costs are capitalized and amortized over the asset's life.

Operational Expenditure (OpEx) for AI

Category	Specific AI Expenditure	Financial Impact
Cloud AI Services	Monthly/annual subscription fees for cloud providers (AWS SageMaker, Azure AI, Google Cloud AI Platform) for model training and inference.	Recurring, predictable cost; deductible in the current year.
Software as a Service (SaaS)	Fees for using third-party AI tools or Large Language Models (LLMs) via an API (e.g., OpenAI, Anthropic, or specialized vector databases).	Cost scales directly with usage/demand; no upfront hardware investment.
Personnel & Maintenance	Salaries for Data Scientists, ML Engineers , and ongoing costs for software maintenance, patches, and support contracts .	Recurring costs of daily operations.
Data Acquisition	Recurring payments for third-party datasets or data labeling services used to train and test AI models.	Ongoing cost for essential operational inputs.

Hybrid Expenditure (OpEx) for AI

The Modern Hybrid AI Cost Model

Strategic Component	Typical Cost Model	Rationale
Core, Proprietary AI Assets	CapEx	Retaining ownership and control over critical, secret-sauce IP (e.g., the base foundational model).
Infrastructure & Scalability	OpEx	Using cloud computing for elastic scaling of compute (GPUs/TPUs) during model training and peak inference periods.
Ongoing Maintenance	OpEx	Day-to-day expenses for cloud usage, specialized software subscriptions, and internal staff salaries.



AI Cloud Cost Drivers

Understanding potential costs and bill items.

Lesson Key Objectives

What are the biggest cost drivers.

1

Identify Gen AI Costs

2

Key Cost Optimization Strategies

3

AI Cloud Cost Drivers

Cost Drivers AI and Cloud

The primary cost drivers for AI workloads in the cloud center around the specialized Compute resources needed for both training and inference (deployment), followed by costs for data storage and data transfer.

- ❑ For advanced generative AI models, API usage (token consumption) is a new, major driver.
- ❑ Costs in the Cloud are variable, meaning your costs will vary month to month based on numerous factors.
- ❑ Costs are broken into Compute, Storage, Network, APIs and Models



AI Cloud Cost Drivers

Compute Resources

Component	AI Phase	Cost Driver	Details
GPU/TPU Time	Training	Duration & Hardware Type	Training deep learning models, especially large language models (LLMs), is extremely compute-intensive . It requires renting high-end accelerators like NVIDIA A100s, H100s, or Google's TPUs for hundreds or thousands of hours. The cost for these can range from \$3 to over \$20+ per hour per instance, with total training costs for frontier models reaching millions.
GPU/TPU Time	Inference	Query Volume & Model Size	After training, serving the model to generate predictions or content (inference) requires compute resources. The cost is driven by the number of user queries (latency requirements) and the size/complexity of the deployed model, which dictates the type of hardware needed for fast response times.
Instance Right-sizing	All	Underutilization	Paying for compute instances that are larger than necessary (over-provisioning) or running development/testing environments 24/7 when they are only used during business hours leads to significant unnecessary expenses.

AI Cloud Cost Drivers

Data Costs

Managing the massive datasets required for modern AI models creates recurring costs.

- ❑ **Data Storage:** Large AI models rely on terabytes or petabytes of data for training. Storing this data in high-performance cloud storage (like S3 or GCS) and keeping multiple copies (for backups or different stages of the pipeline) can quickly add up.
- ❑ **Data Preparation (Hidden Cost):** Although not a direct cloud billing cost, the process of collecting, cleaning, and labeling data is essential.
- ❑ **Data Labeling:** Outsourcing can be a major expense, costing thousands or tens of thousands of dollars, which directly impacts the cloud resources used in later stages.

AI Cloud Cost Drivers

Networking and Data Transfer

Managing the massive datasets required for modern AI models creates recurring costs.

Cloud providers often charge for data moving out of their network or between regions, which is critical for distributed AI training.

- ❑ **Egress Traffic:** This is the cost for data leaving the cloud provider's network (e.g., sending results to a user's on-premises system or a different cloud).
- ❑ **Inter-Region Transfer:** Moving large datasets between different geographic regions for processing or redundancy is charged at a premium rate.

AI Cloud Cost Drivers

Model Deployment and API Usage

For applications built on large, pre-trained models (like LLMs), the cost model shifts from hardware rental to usage-based fees.

- ❑ **Inference API Costs:** Cloud providers and AI companies (like OpenAI, Anthropic) charge per token (a word or part of a word) for using their models via an API.
- ❑ **Input Tokens:** The length of the prompt you send to the model. **Output Tokens:** The length of the response generated by the model.
- ❑ **Context Window:** Models with larger context windows (allowing for longer prompts/histories) are generally more expensive per token.
- ❑ **Managed Services:** Using managed MLOps platforms (like AWS SageMaker or Google Vertex AI) reduces the operational effort but includes a platform fee on top of the base compute costs.

Key Cost Optimization Strategies

Minimizing Your Enterprises Cloud Costs

- ☐ Right Sizing and Scheduling
- ☐ Leverage Spot and Reserved Instances
- ☐ Model Optimization
- ☐ Token Optimization (LLMs)
- ☐ Implement a FinOps Practice





Understanding AI Cloud Costs

Demonstration

Lesson Key Objectives

AWS AI and Cloud Costs

1

Demonstration





AI Vendor Selection Criteria

Important Considerations

Lesson Key Objectives

AI Vendor Selection Criteria

1

Key Criteria

2

AI Vendor Selection Criteria

Selecting the right Artificial Intelligence (AI) vendor is a crucial strategic decision that moves beyond simple features.

Common evaluation pillars are:

- ☐ Technical Capabilities
- ☐ Data & Governance
- ☐ Business Alignment
- ☐ Partnership & Support.



AI Vendor Selection Criteria

Technical Capabilities and Model Quality

Criterion	Description	Key Questions to Ask the Vendor
Performance & Benchmarks	The accuracy, speed (latency), and robustness of the model in real-world conditions relevant to your specific use case.	What are your model's performance metrics (e.g., F1 Score, accuracy) on industry-standard or proprietary benchmarks? Can you provide an ROI projection based on our specific business case?
Model Transparency (XAI)	The ability to understand <i>why</i> the AI made a particular decision (Explainable AI). Critical for regulated industries (finance, healthcare).	How does your solution support explainability (XAI) and auditability for compliance? What mechanisms are in place for human oversight and override?
Scalability and Architecture	The solution's ability to handle increasing data volumes, user queries, and future feature expansion.	How is the solution designed for scalability (e.g., microservices, cloud-agnostic architecture)? What is the typical time-to-deploy a pilot and then scale it across the enterprise?

AI Vendor Selection Criteria

Data Governance, Security, and Compliance

Criterion	Description	Key Questions to Ask the Vendor
Data Privacy & Security	The vendor's protocol for handling sensitive data, including encryption and access controls.	What security certifications (e.g., SOC 2, ISO 27001) do you hold, and are you compliant with GDPR, HIPAA, or CCPA ? How is our data segregated and encrypted , and where is it physically stored?
Data Ownership & Usage	Clarity on who owns the input data, the model output, and any derived insights. Crucial for proprietary information.	Who owns the intellectual property (IP) of the generated output and the data we feed into the model? Do you use our input data for further training of your public models?
Bias and Ethical AI	The vendor's commitment to detecting and mitigating biases within their model and training data.	What is your process for testing and mitigating bias in model outputs? Can you disclose the sources and curation methods for your training data?

AI Vendor Selection Criteria

Partnership, Integration, and Support

Criterion	Description	Key Questions to Ask the Vendor
Integration Capabilities	The ease with which the AI solution connects with your existing enterprise technology stack (e.g., ERP, CRM, data lakes).	What APIs, SDKs, or connectors do you provide for integration with our specific systems? How much custom development effort is typically required for successful integration?
Customer Support & SLA	The vendor's level of dedicated support, ongoing maintenance, and guaranteed uptime.	What is your Service Level Agreement (SLA) for platform uptime and problem resolution timeframes? What kind of training and dedicated support do you provide for our technical and business teams?
Roadmap and Flexibility	The vendor's vision for future development and their adaptability to new AI trends (e.g., new LLM releases).	How often do you release updates and how does your product roadmap align with our future strategic needs? What is your strategy for handling model drift (performance degradation over time)?

AI Vendor Selection Criteria

Business Alignment and Cost

The financial and strategic factors that ensure the investment provides real business value.

Clarity of Use Case: The vendor must demonstrate a deep understanding of your industry and a proven track record (case studies) with measurable results (KPIs) relevant to your business problem.

Total Cost of Ownership (TCO): Look beyond the initial license fee. Include costs for integration, custom fine-tuning, ongoing compute (per-query or per-token), and internal staff training.

Pricing Model: Determine if the pricing is transparent, predictable, and aligned with your expected usage (e.g., subscription, pay-per-use, or token-based).





Executive Messaging

Addressing the C Suite.

Lesson Key Objectives

Discuss why executing messaging is important.

1

Identify common best practices in executive messaging.

2

Executive Messaging

Executive Messaging is focused on how we address the C Suite and the board of directors.

Common best practices are:

- ☐ Be Direct and ensure focus
- ☐ Address the audience correctly and seek agreement
- ☐ Schedule a follow-up action plan as needed
- ☐ Use targeted numbers and metrics
- ☐ Use visuals as needed
- ☐ Avoid technical jargon and speak high level (Macro vs Micro)





Proof of Concepts (POC)

Validating and testing a proposed solution

Lesson Key Objectives

What is Proof of Concept?

1

Why have vendors/integrators perform a Proof of Concept

2

Topic Discussion



Strategic AI Proposal Development

Section 4.3



Section 4.3 Lesson Overview

Components of
a Strategic GEN
AI Proposal?

Discussion:
Defining
Business Value

Defining Key
Performance
Indicators (KPIs)

Solution
Overview &
Outcomes

Whiteboard - AI
Proposal
Workflow

Module Review
Summary

Module Review
Questions



Key Performance Indicators (KPIs)

Important Numbers for Cloud and AI Services

Lesson Key Objectives

What are Key Performance Indicators (KPI)?

1

What is the difference between a KPI and a Metric?

2

What are some important KPIs for Cloud, Operations, and Gen AI?

3

Key Performance Indicators (KPI)

A Key Performance Indicator (KPI) for Cloud and Artificial Intelligence (AI) is essential for measuring effectiveness, financial efficiency, and alignment with business strategy.

Because both domains cover infrastructure and strategic business outcomes, their KPIs are often divided into Financial, Operational/System, and Value categories.

Some frequent Cloud KPIS

- ☐ Availability
- ☐ Workloads
- ☐ Utilization
- ☐ Resource Allocation
- ☐ Response Time



Key Performance Indicators (KPI)

KPIs vs Metrics

Feature	Key Performance Indicator (KPI)	General Metric
Purpose	Measures progress toward a strategic goal (the <i>outcome</i>).	Measures the performance of a process or activity (the <i>output</i>).
Focus	High-level, critical for executive/management decisions.	Tactical, useful for day-to-day operations and analysis.
Example	Customer Lifetime Value (CLV)	Website Page Views
Relationship	If the KPI (CLV) is low, you look at the supporting metrics (Page Views, Conversion Rate, etc.) to figure out why.	

Key Performance Indicators (KPI)

Business Impact KPI'S

KPI	Description	Example Use Case
Return on Investment (ROI)	The financial benefit (e.g., revenue increase, cost savings) minus the total investment (development, infrastructure, operation).	Fraud detection system's ROI = (Amount of fraud prevented - System cost) / System cost.
Cost Reduction	Savings from automating manual tasks, optimizing processes, or reducing errors.	Cost savings from automated customer support (chatbots).
Revenue Uplift	Increase in sales, average order value, or new revenue streams attributed to the AI system.	Revenue increase from personalized product recommendations.
Customer Satisfaction (CSAT/NPS)	Improvement in customer experience due to faster service or more relevant interactions.	Higher CSAT scores after implementing an AI-powered service tool.
Time to Market	The time saved in bringing a product or service to market using AI.	Faster drug discovery using an AI research assistant.

Key Performance Indicators (KPI)

GenAI KPI'S	Metric	Description
	Hallucination Rate	Percentage of generated content that is factually incorrect or ungrounded.
	Groundedness	Measures the percentage of generated statements that can be verified by the source material.
	Coherence/Fluency	Subjective or LLM-based scores of how well the output reads and makes sense.

Key Performance Indicators (KPI)

Regression Model KPI'S	Metric	Description
	Mean Absolute Error (MAE)	The average magnitude of errors in a set of predictions, measured in the same units as the target variable.
	Root Mean Squared Error (RMSE)	Penalizes large errors more heavily, useful when big mistakes are particularly bad.

Key Performance Indicators (KPI)

	KPI	Description
Operational KPI'S	Model Latency (Response Time)	The time it takes for the system to process a request and deliver a result. Critical for real-time applications.
	System Throughput	The number of requests the AI system can handle per unit of time (e.g., requests per second).
	Uptime / Availability	The percentage of time the AI system is available and operational.
	Model Drift Detection	The time taken to detect when the model's predictive performance is declining due to changes in real-world data.
	Deployment Time	The time it takes to move a new or updated model from development to production.
	User Adoption Rate	The percentage of target users/employees actively using the AI tool.

Module Review Summary

Short Summary of what we covered in this module



Module Review Summary

- ❑ **The four pillars of AI projects** are Vision, Value Realization, Risk Management, and Adoption Plan.
- ❑ A **business use case** is a document that presents the costs, risks, and benefits of a particular initiative, justifies the investment, etc.
- ❑ **Uplift Metrics** are what we use to “influence” the deal and are the foundation of the business case.
- ❑ **Capital Expenditures (Capex)** are investments made by an organization for long-term benefits in the future.
- ❑ **Operational Expenditures (Opex)** are the ongoing costs related to day-to-day operations
- ❑ The **primary cost drivers** for AI workloads in the cloud center around the specialized Compute resources needed for both training and inference (deployment), followed by costs for data storage and data transfer.
- ❑ Selecting the **right Artificial Intelligence (AI) vendor** is a crucial strategic decision that moves beyond simple features.
- ❑ Common **evaluation pillars** are Technical Capabilities, Data & Governance, Business Alignment and Partnership & Support.
- ❑ **Executive Messaging** is focused on how we address the C Suite and the board of directors.
- ❑ A **Key Performance Indicator (KPI)** for Cloud and Artificial Intelligence (AI) is essential for measuring effectiveness, financial efficiency, and alignment with business strategy.

Module Review Questions

Let's Test our knowledge



Module Review Questions

1. A global financial services firm is deploying a new AI-powered loan application system. The AI development team is currently focused on establishing an internal audit framework to regularly check the model's decisions for algorithmic bias across different demographic groups and implementing strict protocols to ensure all customer data processing adheres to GDPR and CCPA regulations.

Which of the four pillars of AI strategy are they primarily focused on in this scenario? (Select One)

- A. Value Realization
- B. Risk Management
- C. Vision
- D. Adoption Plans

Module Review Questions

1. A global financial services firm is deploying a new AI-powered loan application system. The AI development team is currently focused on establishing an internal audit framework to regularly check the model's decisions for algorithmic bias across different demographic groups and implementing strict protocols to ensure all customer data processing adheres to GDPR and CCPA regulations.

Which of the four pillars of AI strategy are they primarily focused on in this scenario? (Select One)

- A. Value Realization
- B. Risk Management**
- C. Vision
- D. Adoption Plans

Explanation: Risk Management, this pillar emphasizes the importance of proactively addressing the potential risks of AI implementation

Module Review Questions

2. An organization is planning its budget for a new on-premises AI deployment. Which of the following expenses would typically be classified as a Capital Expenditure (Capex)? (Select One)

- A. Monthly subscription fees
- B. Annual software licenses for GRC software
- C. The cost of electricity and cooling for the server room.
- D. The purchase of dedicated, high-performance GPU servers for model training.

Module Review Questions

2. An organization is planning its budget for a new on-premises AI deployment. Which of the following expenses would typically be classified as a Capital Expenditure (Capex)? (Select One)

- A. Monthly subscription fees
- B. Annual software licenses for GRC software
- C. The cost of electricity and cooling for the server room.
- D. The purchase of dedicated, high-performance GPU servers for model training.**

Explanation: Capex involves investments made for long-term benefits in the future, such as purchasing physical, long-lasting assets like servers.

Module Review Questions

3. For AI workloads hosted in the cloud, what is typically the primary cost driver, often overshadowing data storage and data transfer costs? (Select One)

- A. Specialized Compute resources for model training and inference
- B. Data storage on AWS
- C. Data transfer fees out of the cloud provider
- D. Salaries for MLOps Team

Module Review Questions

3. For AI workloads hosted in the cloud, what is typically the primary cost driver, often overshadowing data storage and data transfer costs? (Select One)

- A. Specialized Compute resources for model training and inference**
- B. Data storage on AWS
- C. Data transfer fees out of the cloud provider
- D. Salaries for MLOps Team

Explanation: The specialized Compute resources (like GPUs or TPUs) needed for intensive model training and deployment (inference) are the primary cost drivers for AI workloads in the cloud.

Domain 5 – Governance, Ethics and Responsible AI

Module 5– CSGAIP Exam

GRC and Ethical Concerns



Module 5

Sections

- 01 Section 5.1 – Identify Ethical, Legal and IP Considerations in AI Use
- 02 Section 5.2 – Implement Best Practices for Responsible AI

Identify Ethical, Legal and IP Considerations in AI Use

Section 5.1



Section 5.1 Lesson Overview

Defining AI
Ethics

Key AI
Principles

AI and Job
Displacement

Use Case - AI
Ethics in Facial
Recognition

Legal and
Professional
Accountability

Sources of AI
Bias

NIST AI RMF

Whiteboard -
Implement
NIST AI RMF



Defining AI Ethics

Fundamentals

Lesson Key Objectives

What is AI Ethics?

1

Why is AI Ethics important?

2

Defining AI Ethics

AI Ethics

AI ethics is a critical branch of applied ethics that focuses on the moral principles and guidelines for the responsible development and deployment of artificial intelligence.

Why AI Ethics are Important?

- ☐ Ensures that AI technologies are created and used in a manner that is beneficial to humanity, respects fundamental human rights, and minimizes the risk of harm.
- ☐ Ethical considerations surrounding AI use have become more pressing than ever.





Key AI Ethical Principles

Overview and Defining

Lesson Key Objectives

What are the key AI Principles?

1

What are the organizations that defined these principles?

2

Key AI Ethical Principles

Key AI Ethical Principles

The **key ethical principles** for Artificial Intelligence (AI) are a set of guidelines designed to ensure that AI systems are developed, deployed, and used in a way that respects human rights, values, and the law, while also benefiting society.

Various organizations have defined AI Ethical Principles all with the same focus but some with varying terms and influence.

These principles form the **foundation** for developing and deploying AI systems responsibly.



Key AI Ethical Principles

Key AI Ethical Principles

- ☐ Fairness and Non-discrimination
- ☐ Transparency and Explainability
- ☐ Accountability and Responsibility
- ☐ Privacy and Data Protection
- ☐ Human Oversight and Control
- ☐ Safety and Security
- ☐ Beneficence and Non-maleficence
- ☐ Reliability and Robustness



Key AI Ethical Principles

Principle	Core Concept	Why It Matters
Fairness & Non-Discrimination	AI systems must treat all individuals and groups equitably, avoiding systemic bias that could lead to discriminatory outcomes.	Bias in training data can perpetuate and amplify societal prejudices, leading to unfair decisions in areas like hiring, lending, or criminal justice.
Transparency & Explainability	Users and stakeholders should be able to understand <i>how</i> an AI system arrives at a decision or outcome (the "why" and "how").	Allows for scrutiny , auditing, and trust-building; essential for challenging decisions and identifying flaws.
Accountability & Governance	Clear assignment of responsibility for the design, deployment, and outcomes of AI systems. Humans must maintain oversight.	Ensures that when an AI system causes harm or makes an error, there are clear mechanisms and responsible parties for redress .
Privacy & Data Protection	Respecting and protecting user data, including personal information, through robust security measures and adherence to data governance laws.	AI relies heavily on data. This principle ensures individual rights and control over their own information are upheld.
Reliability & Safety	AI systems must function as intended, be robust against unintended manipulation (security), and operate safely and consistently in real-world conditions.	Ensures the technology is trustworthy and does not pose physical or psychological harm to users or the environment.
Human Agency & Control	AI should augment and empower human capabilities, not diminish them. Humans should retain the ultimate right to override automated decisions.	Maintains human dignity and decision-making authority, especially in high-stakes situations.

Key Organizations Defining Core AI Ethics Principles

Organization Type	Organization	Key Framework/Document	Core Principles Emphasized	Key Distinction
Intergovernmental	UNESCO (United Nations)	Recommendation on the Ethics of Artificial Intelligence (2021)	Human Rights (Cornerstone), Fairness, Transparency, Accountability, Privacy, Sustainability.	The first global standard, adopted by all 194 Member States, making it the most universally agreed-upon ethical framework.
Intergovernmental	OECD (Organization for Economic Co-operation and Development)	OECD AI Principles (2019)	Inclusive Growth, Respect for Human Rights & Values, Transparency, Robustness, Accountability.	The first intergovernmental standard, foundational for many national AI policies and used by the G20.
Government/Regulatory	European Union (EU)	EU AI Act (Proposed/Adopted)	Risk-based regulation (Unacceptable, High, Limited, Minimal risk), Safety, Fundamental Rights, Legal Compliance.	Shifts from voluntary principles to binding legal requirements based on the potential harm (risk level) of the AI system.
Standards Body	NIST (National Institute of Standards and Technology - U.S.)	AI Risk Management Framework (AI RMF) (2023)	Validity & Reliability, Safety, Security & Resilience, Accountability, Transparency, Explainability, Fairness.	Provides a voluntary, actionable toolkit for organizations to systematically govern and manage AI risks.
Industry/Tech	Leading Tech Companies (e.g., Microsoft, Google/DeepMind, IBM, OpenAI)	Internal AI Principles/Policies	Fairness, Reliability & Safety, Privacy & Security, Transparency, Accountability.	Translates global principles into operational standards and tools for AI development and deployment.



Seven Cs of Artificial Intelligence (AI)

Identify the C's

Lesson Key Objectives

What are the Seven C's of AI

1



Seven C's of AI

What are the Seven C's of AI?

- ❑ Coined by Dr. Chang - founder of the Medical Intelligence and Innovation Institute (MI3) at Children's Hospital of Orange County (CHOC) and the founder of the AIMed (Artificial Intelligence in Medicine) community.
- ❑ Focuses on ensuring the data that powers AI is of high quality and that the systems are trustworthy and ethical.
- ❑ The **most common framework**, particularly in the context of data management and responsible AI development.
- ❑ Frameworks can differ slightly by organization.

Seven C's of AI

Compliance

Confidence

Consolidation

Consistency

Clarity

Context

Causation

Seven C's of AI

What are the Seven C's of AI?

- ❑ **Compliance:** Adhering to legal, ethical, and regulatory standards (e.g., data protection, privacy).
- ❑ **Confidence:** Building trust in AI systems through reliability, accuracy, predictability, and explainability.
- ❑ **Consolidation:** Integrating diverse data sources effectively to create comprehensive and nuanced AI models.
- ❑ **Consistency:** Ensuring reliable and predictable performance across different scenarios and over time.





Seven C's of AI

What are the Seven C's of AI?(Continued)

- ❑ **Clarity:** Making complex AI decision-making processes understandable to humans through transparency and interpretability (Explainable AI - XAI).
- ❑ **Context:** Enabling AI to understand and interpret subtle environmental, cultural, and situational nuances.
- ❑ **Causation(Ethics):** Recognizing the limitations of AI in determining cause-and-effect relationships and the need for human oversight.



AI and Job Displacement

Concerns with AI and Jobs

Lesson Key Objectives

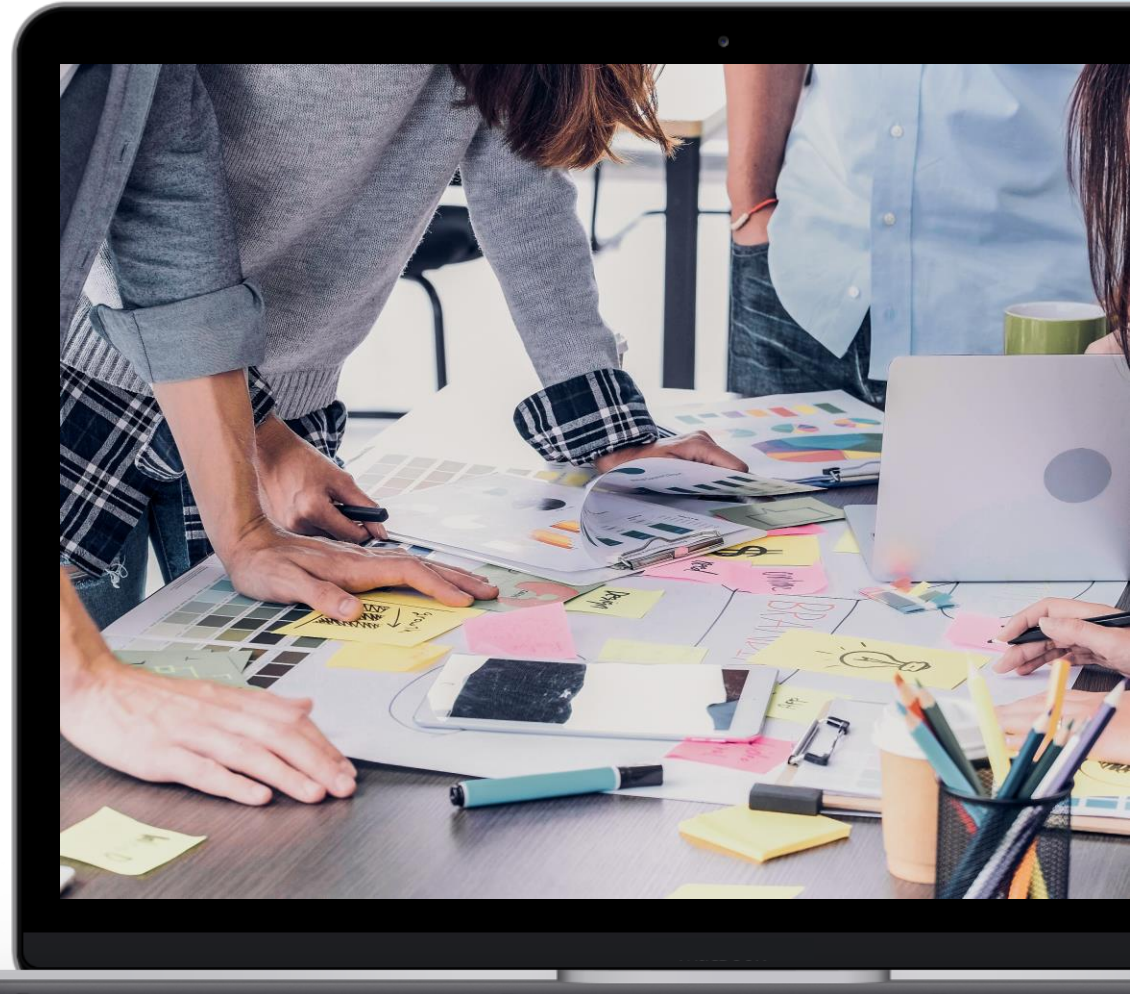
What are the concerns about AI and Jobs?

1

Automation, AI and Job Displacement

Concerns about Automation and AI displacing Jobs

- ❑ Concern: AI may displace 800M jobs by 2030 (McKinsey).
- ❑ By 2022, 133M new jobs and 75M predicted job displacements (World Economic Forum).
- ❑ Automation leads to job displacement and a shift towards higher-level skills.
- ❑ Need regulations for minimizing impact and maximizing benefits.
- ❑ New job opportunities in manufacturing, healthcare, and finance.
- ❑ Impact based on tech growth, worker adaptability, sectors.





Use Case

AI Ethics in Facial Recognition

Lesson Key Objectives

What is Enforcement and what are the stages?

1

Why are the effective mechanisms to minimize ethics concerns?

2

Use Case – Facial Recognition

A significant use case highlighting AI ethics in facial recognition is Mass Surveillance and Algorithmic Bias in Law Enforcement.

The ethical challenge arises when law enforcement agencies use AI-driven facial recognition systems for real-time or retroactive identification of individuals in public spaces.

Use Case: Mass Surveillance and Algorithmic Bias in Law Enforcement

The Scenario: Police departments in a major city deploy real-time facial recognition technology (FRT) across public spaces—using networks of street cameras—to automatically scan and compare faces against watchlists of known suspects.



Use Case – Facial Recognition

Scenario and possible issues and solutions

AI Ethics Principle	Use Case Scenario: Facial Recognition in Law Enforcement	Ethical Issues & Impact	Mitigation / Solution
Fairness & Equity	Law enforcement deploys real-time Facial Recognition Technology (FRT) in public spaces to match faces against a suspect watchlist.	Algorithmic Bias: Systems show significantly higher error rates (misidentification) for people with darker skin tones and women. This leads to wrongful stops, interrogations, and arrests for innocent individuals from already marginalized communities.	Mandatory Bias Audits: Prohibit the use of FRT systems that do not demonstrate fair and comparable accuracy rates across all demographic groups. Diverse Training Data: Require datasets that are representative of the target population.
Privacy & Autonomy	The FRT system scans and logs the biometric data (face geometry) of everyone in public, not just suspects, and stores it in a searchable database.	Mass Surveillance & Lack of Consent: Individuals are tracked without their knowledge or permission, chilling civil liberties (e.g., freedom of assembly/protest). This biometric data is unchangeable, posing a lifelong risk if breached.	Data Minimization: Restrict the collection and retention of data only to individuals who are actual targets of a specific, legitimate investigation. Public Notice & Consent: Mandate clear public signage and require legislative approval for deployment.
Transparency & Accountability	The underlying algorithms are proprietary "black boxes," and the policies for FRT usage are unclear or non-existent.	Lack of Redress: When a false arrest occurs, the lack of transparency makes it impossible for the individual to challenge the faulty AI evidence. The absence of clear policy prevents public oversight and accountability for system errors.	Transparency Requirements: Mandate disclosure of which FRT system is used, how it was tested, and the specific policies governing its deployment. Clear Accountability: Establish legal mechanisms for individuals to challenge and seek redress for harms caused by algorithmic errors.



Sources of AI Bias

Sources and Types

Lesson Key Objectives

Define Bias.

1

Identify potential sources of bias.

2



Sources of AI Bias

AI bias, also known as algorithmic bias or machine learning bias, is the systematic and unfair discrimination that can occur when an AI system produces results that unfairly favor or disfavor certain people or groups.

The sources of AI bias generally fall into three main categories, often occurring at different stages of the AI development pipeline:

- ☐ Data Bias
- ☐ Algorithmic/Model Bias
- ☐ Human/Cognitive Bias



AI Ethics Driven Strategy

Primary Source of AI Bias	Description	Example/How Bias is Introduced
1. Data Bias	Bias present in the data used to train the AI model. If the data is flawed, the model will learn the flaws.	Selection/Sampling Bias: Training a facial recognition system primarily on images of one demographic group, leading to poor performance on others.
	Historical Bias: The data reflects past societal inequalities (e.g., historical loan application data that discriminated against a minority group).	
	Labeling Bias: Human annotators subjectively or inconsistently label data, injecting their own biases into the training labels.	
2. Algorithmic/Model Bias	Bias introduced by the design, structure, or optimization of the algorithm itself.	Algorithmic Design: The choice of features or weighting that unintentionally causes the model to prioritize certain groups or outcomes.
	Confirmation Bias: The algorithm is designed to excessively favor existing patterns in the data, thus reinforcing and amplifying pre-existing stereotypes.	
3. Human/Cognitive Bias	Bias that stems from the conscious or unconscious decisions of the people involved in the AI's development and use.	Developer Bias: The personal beliefs and assumptions of the programmers influencing the design, data collection, or evaluation metrics.
	Deployment Bias (Feedback Loops): A biased model influences human decisions, and those decisions generate new biased data, which is then fed back into the model for re-training, reinforcing the original bias.	



NIST AI Framework

Important Frameworks to Know

Lesson Key Objectives

What is the NIST AI RMF?

1

What are the core integrated functions?

2

What were the consequences of the GDPR violation?

3

NIST AI RMF

The National Institute of Standards and Technology (NIST) **Artificial Intelligence Risk Management Framework (AI RMF)** is a voluntary framework designed to help organizations manage the risks associated with Artificial Intelligence systems.

Released in January 2023, its primary goal is to promote the development and use of **trustworthy and responsible AI** by providing a systematic approach to identifying, assessing, and mitigating AI risks throughout the entire AI lifecycle.



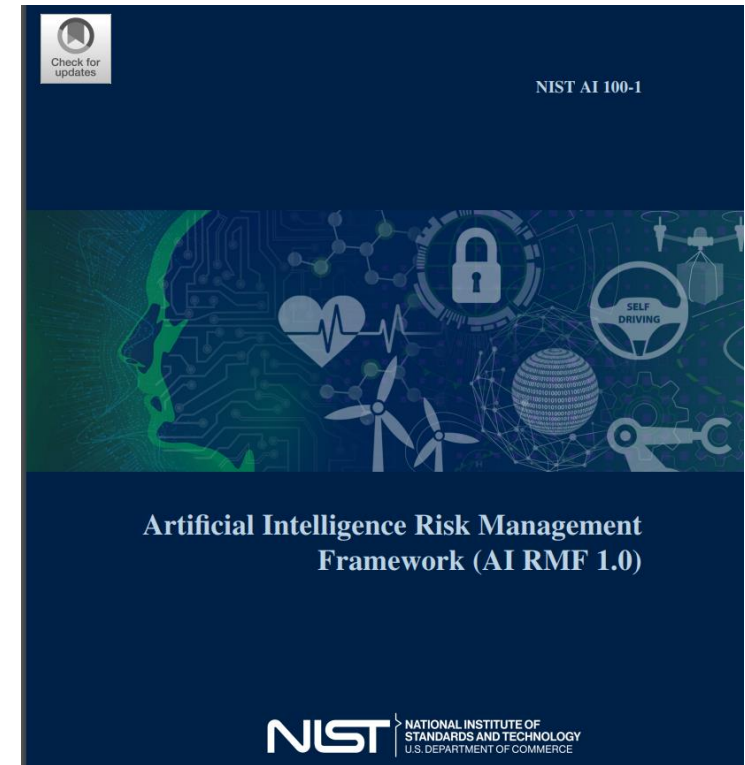
NIST AI RMF

The AI RMF emphasizes key **characteristics that contribute to the trustworthiness of an AI system**, which organizations should measure and manage risks against:

- ☐ **Valid and Reliable**
- ☐ **Safe, Secure, and Resilient**
- ☐ **Accountable and Transparent**
- ☐ **Explainable and Interpretable**
- ☐ **Privacy-Enhanced**
- ☐ **Fair, with Harmful Biases Managed**



NIST AI RMF Framework



NIST AI RMF Playbook

Risk Management Frameworks

NIST AI RMF Core Functions

Function	Purpose/Definition	Key Actions/Focus
Govern	Establish the foundational structure, policies, and culture for managing AI risks.	Governance structure, organizational policies, ethical culture, continuous oversight across the AI lifecycle.
Map	Establish the context for identifying and framing AI risks.	Understanding the AI system's purpose, stakeholders, expected benefits, costs, and potential negative impacts.
Measure	Analyze, assess, benchmark, and monitor identified AI risks and related impacts.	Employing quantitative, qualitative, or mixed-method tools for risk assessment and tracking.
Manage	Prioritize and act upon identified and measured risks.	Allocating resources, implementing mitigation strategies, ensuring continuous monitoring and improvement of risk controls.

Risk Management Frameworks

NIST AI RMF Implémentation Tiers

Tier	Name	Key Characteristics
1	Partial	Limited awareness of AI risks. Risk management is often reactive and lacks consistency. Governance structures for AI risk management may need to be established.
2	Risk-Informed	Recognizing the importance of AI risk management. Some formal processes are implemented. Risk management is becoming more proactive , with a developing understanding of AI risks and potential impacts.
3	Repeatable	Established and documented AI risk management practices. Practices are consistently applied throughout the organization. Governance structures are in place, and risk management is integrated into the overall strategy.
4	Adaptive	The highest level of maturity . Comprehensive and dynamic approach to managing AI risks. Practices are continuously monitored and adapted to emerging threats. Governance is robust, with a commitment to ethical AI and regulatory compliance.

Whiteboard Discussion



NIST AI Risk Management Framework (RMF)

Whiteboard

Lesson Key Objectives

What are the core steps in the NIST AI RMF?

1

What are the four implementation tiers ?

2

NIST RMF Framework



A **Risk Management Framework (RMF)** is essentially a structured, repeatable plan that organizations use to handle uncertainty.

Implementing the **NIST Risk Management Framework (RMF)** is a structured, seven-step process designed to integrate security, privacy, and risk management activities into the System Development Life Cycle (SDLC).

<https://www.nist.gov/news-events/news/2023/01/nist-risk-management-framework-aims-improve-trustworthiness-artificial>

Whiteboard Discussion

Implement Best Practices for Responsible AI

Section 5.2



Section 5.2 Lesson Overview

What is
Responsible AI
Ethics

Whiteboard -
Implement
Responsible AI
Principals

Seven C's of
Responsible AI

AI Center of
Excellence

Module Review
Questions

Module Review
Summary

Certification
Preparation

Resources and
Course Closeout



Responsible AI Ethics and Governance

Defining and placing into practice

Lesson Key Objectives

What is Responsible AI Ethics

1

How does AI Governance apply

2

Responsible AI Ethics and Governance

What is Responsible AI?

Responsible AI is an umbrella term for an approach to developing, assessing, and deploying AI systems that are trustworthy and align with societal principles.

- ❑ Set of core ethical principles that aim to **guide the creation and use of AI** in a way that respects human values and fundamental rights.
- ❑ Has **six core key principles**.



Responsible AI Ethics and Governance

Principle	Description
Fairness and Inclusiveness	Aims to ensure that AI systems treat all individuals and groups equitably, actively working to identify and mitigate unfair biases in data and algorithms.
Transparency and Explainability	Requires that the operations and decisions of AI systems are understandable to users and those affected by them, making the reasoning behind AI outcomes clear.
Accountability	Establishes clear lines of responsibility for the impacts of AI systems, ensuring that the organizations and individuals who design and deploy them are answerable.
Reliability and Safety	Ensures that AI systems perform consistently and as intended, are resilient to unexpected situations or malicious attacks, and include safeguards to prevent harm.
Privacy and Security	Mandates that AI systems must protect personal data and respect user privacy, with robust security measures to prevent unauthorized access and misuse.
Human-Centeredness	Puts human well-being at the forefront of AI design and development, ensuring that the ultimate purpose of AI is to augment and benefit humanity.

Responsible AI Ethics and Governance

AI Governance provides the structure and processes to operationalize the principles of Responsible AI within an organization.

- ❑ Set of **policies, procedures, and standards** that guide the development, deployment, and ongoing management of AI systems.
- ❑ Use a **framework** to manage. (e.g: NIST, EU)





Responsible AI Ethics and Governance

Responsible AI and Governance

- **Responsible AI Ethics** provides the "what" – the moral and ethical principles that should guide the development of AI.
- **AI Governance** provides the "how" – the concrete actions and structures that organizations put in place to ensure that those principles are consistently upheld.
- Together, they form a vital partnership for building a future where artificial intelligence is a force for good.

Module Review Summary

Short Summary of what we covered in this module



Module Review Summary

- ❑ **AI Ethics** is a critical branch of applied ethics that focuses on the moral principles and guidelines for the responsible development and deployment of artificial intelligence.
- ❑ **AI Ethical Principles** are to ensure we design and implement AI systems properly. These principles are Fairness and Non-discrimination, Transparency and Explainability, Accountability and Responsibility, Privacy and Data Protection, Human Oversight and Control, Safety and Security, Beneficence and Non-maleficence
- ❑ **Responsible AI** is an umbrella term for an approach to developing, assessing, and deploying AI systems that are trustworthy and align with societal principles and has six core key principles.
- ❑ **Responsible AI Ethics** provides the "what" – the moral and ethical principles that should guide the development of AI.
- ❑ **AI Governance** provides the concrete actions (framework, policies, etc) and structures that organizations put in place to ensure that those principles are consistently upheld.
- ❑ The **Seven C's of AI** are Compliance, Confidence, Consolidation, Consistency, Clarity, Context, Causation
- ❑ An **AI Center of Excellence** acts as a strategic hub of expertise, best practices, and resources to ensure that all AI initiatives align with the organization's strategic goals and deliver measurable business value.

Module Review Questions

Let's Test our knowledge



Module Review Questions

Question 1 - What is the main purpose of AI ethics?

- A. To ensure AI systems are cost-effective and provide a clear Return on Investment (ROI) for the enterprise.
- B. To create the fastest and most efficient machine learning models for immediate deployment.
- C. To define the specific technical requirements for the deployment of machine learning algorithms on cloud platforms.
- D. To ensure AI technologies are created and used in a manner that is beneficial to humanity, respects fundamental human rights, and minimizes the risk of harm.

Module Review Questions

Question 1 - What is the main purpose of AI ethics?

- A. To ensure AI systems are cost-effective and provide a clear Return on Investment (ROI) for the enterprise.
- B. To create the fastest and most efficient machine learning models for immediate deployment.
- C. To define the specific technical requirements for the deployment of machine learning algorithms on cloud platforms.
- D. To ensure AI technologies are created and used in a manner that is beneficial to humanity, respects fundamental human rights, and minimizes the risk of harm.

Explanation: AI ethics is a critical branch of applied ethics focused on the moral principles and guidelines for the responsible development and deployment of artificial intelligence. Its primary goal is to ensure that AI technologies are created and used in a manner that is beneficial to humanity, respects fundamental human rights, and minimizes the risk of harm.

Module Review Questions

Question 2 - Which of the following AI Ethical Principles requires clear lines of responsibility to be established to determine who is liable for the actions and outcomes of AI systems?

- A. Transparency and Explainability
- B. Fairness and Non-Discrimination
- C. Accountability and Responsibility
- D. Beneficence and Non-maleficence

Module Review Questions

Question 2 - Which of the following AI Ethical Principles requires clear lines of responsibility to be established to determine who is liable for the actions and outcomes of AI systems?

- A. Transparency and Explainability
- B. Fairness and Non-Discrimination
- C. Accountability and Responsibility**
- D. Beneficence and Non-maleficence

Explanation: The principle of **Accountability and Responsibility** mandates that clear lines of responsibility must be established to determine who is liable for the actions and outcomes of AI systems. This ensures that the organizations and individuals who design and deploy the systems are answerable for their impacts. Transparency and Explainability is about understanding the decision-making process , and Fairness is about avoiding bias

Module Review Questions

Question 3 - Within the "Seven C's of AI," which 'C' is specifically defined as:

"Making complex AI decision-making processes understandable to humans through transparency and interpretability (Explainable AI - XAI)"?

- A. Consistency
- B. Clarity
- C. Confidence
- D. Consolidation

Module Review Questions

Question 3 - Within the "Seven C's of AI," which 'C' is specifically defined as:

"Making complex AI decision-making processes understandable to humans through transparency and interpretability (Explainable AI - XAI)"?

- A. Consistency
- B. Clarity**
- C. Confidence
- D. Consolidation

Explanation: The "Clarity" principle within the Seven C's of AI is defined as making complex AI decision-making processes understandable to humans through transparency and interpretability. This concept is directly tied to Explainable AI (XAI). **Consistency** refers to reliable and predictable performance , and **Confidence** is about building trust in the system

Module Review Questions

Question 4 - The sources of AI bias generally fall into three main categories.

Which of the following is NOT listed as one of the three primary sources of bias?

- A. Data Bias
- B. Algorithmic/Model Bias
- C. Human/Cognitive Bias
- D. Organizational Bias

Module Review Questions

Question 4 - The sources of AI bias generally fall into three main categories.

Which of the following is NOT listed as one of the three primary sources of bias?

- A. Data Bias
- B. Algorithmic/Model Bias
- C. Human/Cognitive Bias
- D. **Organizational Bias**

Explanation: The sources of AI bias generally fall into **three main categories**: Data Bias: Bias present in the training data. Algorithmic/Model Bias: Bias introduced by the design, structure, or optimization of the algorithm itself. Human/Cognitive Bias: Bias that stems from the conscious or unconscious decisions of the people involved in the AI's development and use . "Organizational Bias" is not listed as one of the three main, primary categories.



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Resources and Course Closeout

Thanks for Joining!



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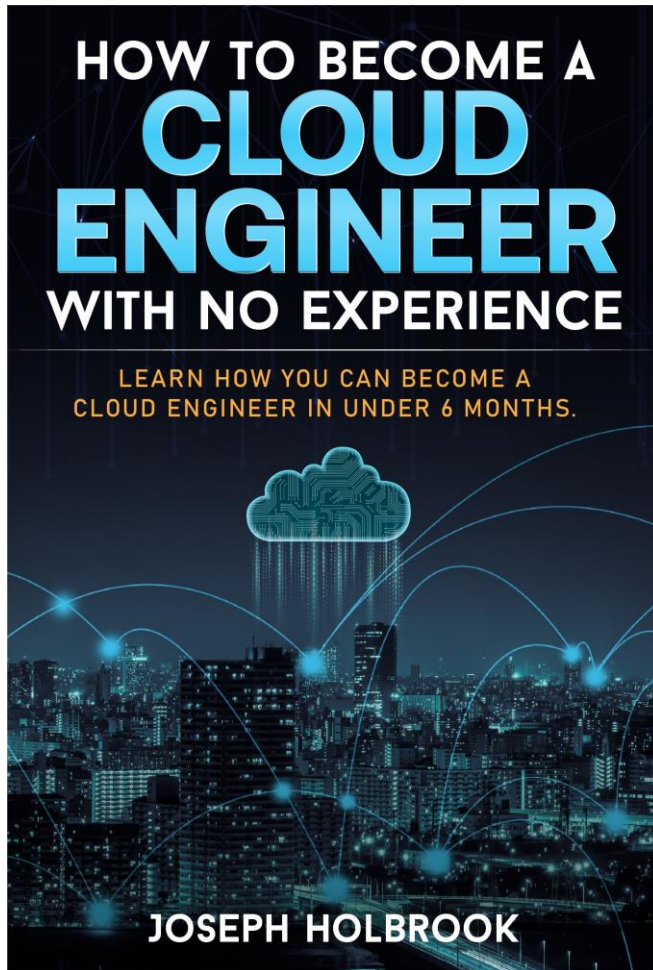
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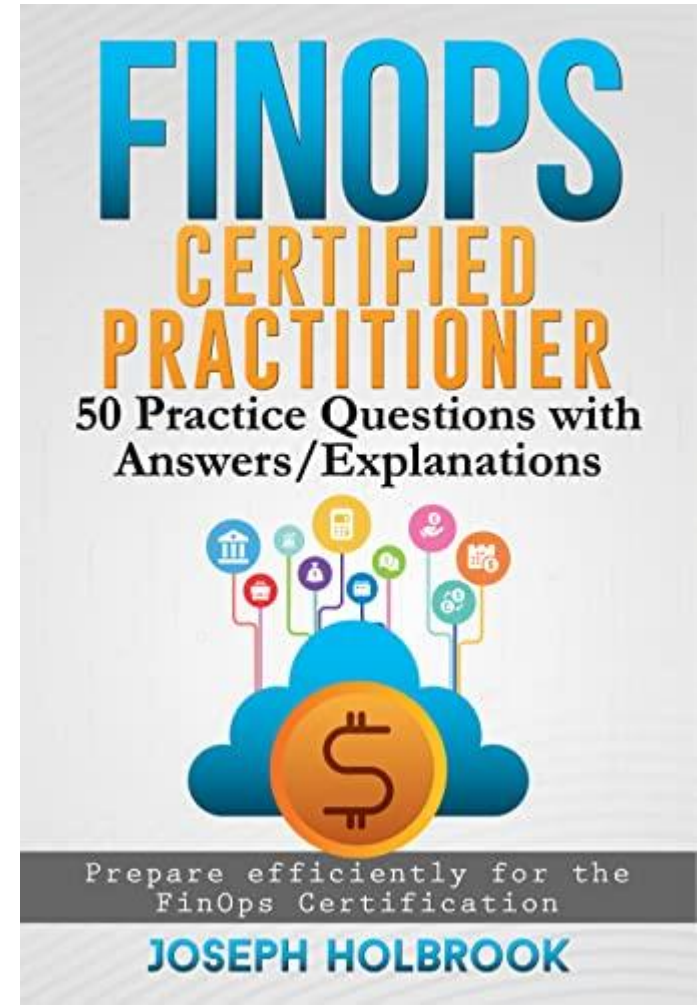


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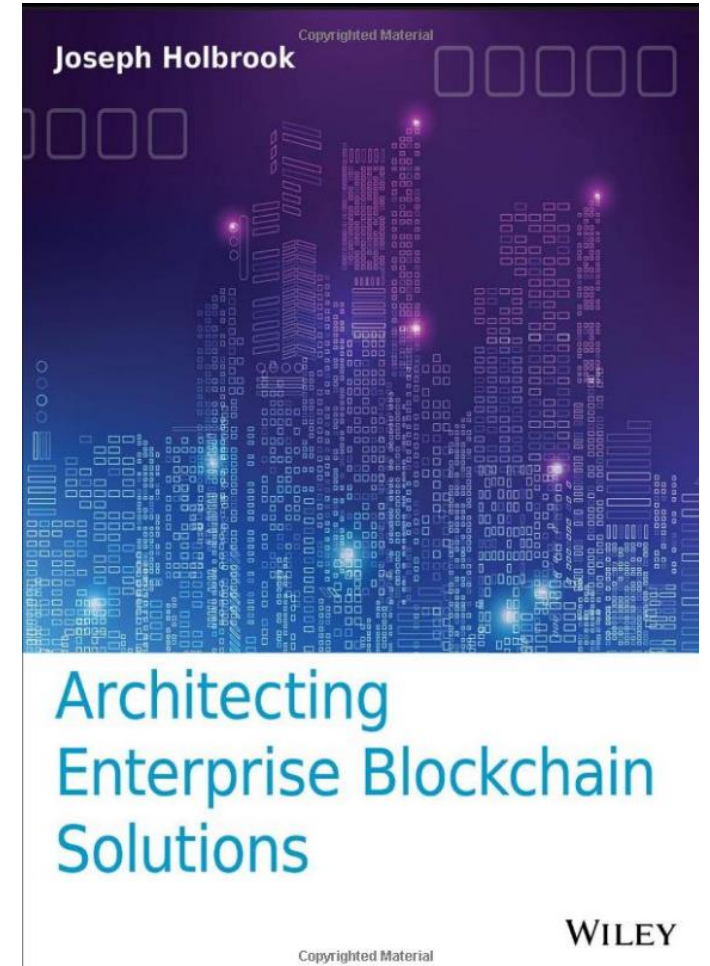
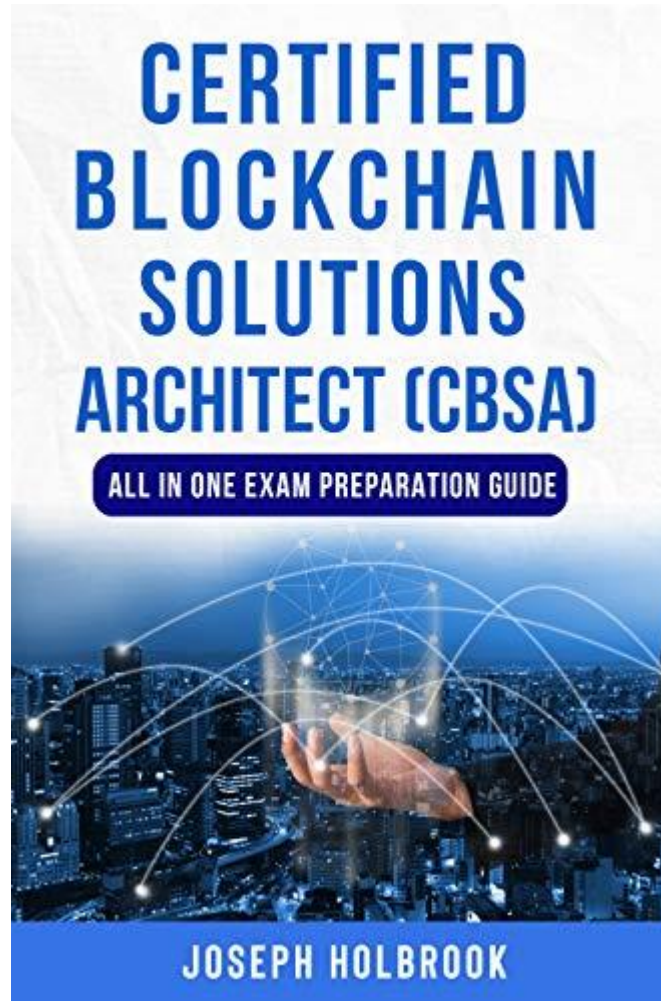
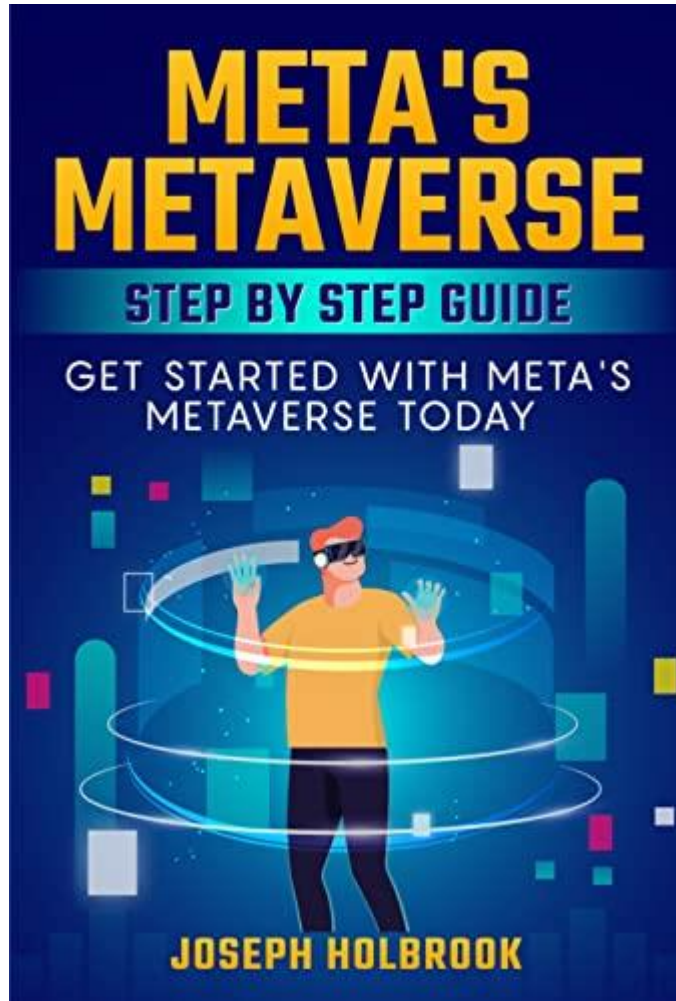


JOSEPH HOLBROOK



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Become a Cloud Architect

Cloud Architects are Highly Paid IT Professionals.
Here's how you can start in a new career!

The infographic is a vertical flowchart on a blue background. It starts with a box titled 'Determine Your Path' showing people at a table, with text 'AWS, Azure and GCP are the Top Clouds'. An arrow points down to 'Get Certified', which shows a person at a computer and text 'Choose Cloud Certifications at the Practitioner Level or the Associate Level'. Another arrow points down to 'Join a Bootcamp Cloud Courses', showing a laptop and text 'Consider a bootcamp and some pro help to increase your chances'. This leads to 'Update your LinkedIn Profile', showing a LinkedIn logo and text 'Update your LinkedIn Profile with new certifications/courses you completed to get noticed'. The next step is 'Apply for Jobs', showing a smartphone and text 'Insert some fast facts or links'. The final step is 'Coaching Options', showing a person at a desk and text 'Get some help with Mock Interviews and Career Advice from a coach'. The Tech Commanders logo is at the bottom.

Determine Your Path
AWS, Azure and GCP are the Top Clouds

Get Certified
Choose Cloud Certifications at the Practitioner Level or the Associate Level

Join a Bootcamp Cloud Courses
Consider a bootcamp and some pro help to increase your chances

Update your LinkedIn Profile
Update your LinkedIn Profile with new certifications/courses you completed to get noticed

Apply for Jobs
Insert some fast facts or links

Coaching Options
Get some help with Mock Interviews and Career Advice from a coach

TECH COMMANDERS
ADVANCING YOUR IT SKILLS

TOP REASONS TO BECOME A CLOUD ENGINEER

Cloud Engineers are well paid
Technical experts.

The infographic has a dark blue header and a light blue body. It lists four reasons: 1. Career Growth: 'Reason 1 - Career Growth is in the Top Ten Roles' with text 'LinkedIn, Robert Half, etc have great insight' and a lightbulb icon. 2. Reasonable Barrier to Entry: 'REASONABLE BARRIER TO ENTRY' with text 'Degree not needed and only 6 months of preparation may be needed with proper effort and coaching' and a key icon. 3. Technically Challenging: 'TECHNICALLY CHALLENGING' with text 'Did you know that a cloud engineer needs to have skills in coding, DevOps, Security, Networking, etc' and a handshake icon. 4. Generous Career Salaries: 'GENEROUS CAREER SALARIES' with text 'Did you know that a Cloud Engineer averages \$128,837 a year the US?' and a bar chart icon. The Tech Commanders logo and website are at the bottom.

Reason 1 - Career Growth is in the Top Ten Roles
LinkedIn, Robert Half, etc have great insight

REASONABLE BARRIER TO ENTRY
Degree not needed and only 6 months of preparation may be needed with proper effort and coaching.

TECHNICALLY CHALLENGING
Did you know that a cloud engineer needs to have skills in coding, DevOps, Security, Networking, etc

GENEROUS CAREER SALARIES
Did you know that a Cloud Engineer averages \$128,837 a year the US?

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TECH COMMANDERS
ADVANCING YOUR IT SKILLS

CLOUD INTERVIEWACE

How the program works over 7 weeks

The infographic has a dark blue background with a white title. It shows a 7-week program: Step 1 - Week 1: 'Welcome Introduction Call' with a person at a computer icon. Step 2 - Week 2-4: 'Take and Complete On Demand Training' with a person at a laptop icon. Step 3 - Week 4: 'Coaching Calls Setup' with a person at a laptop icon. Step 4 - Week 5: 'Attend Mock Interview with feedback' with a person at a laptop icon. Step 5 - Week 6-7: 'Resumes, Coaching, Courses' with a person at a laptop icon. A final box says 'Certificate DOWNLOAD YOUR CERTIFICATE OF COMPLETION' with a certificate icon. The Tech Commanders logo is at the bottom.

Step 1 - Week 1
Welcome Introduction Call

Step 2 - Week 2-4
Take and Complete On Demand Training

Step 3 - Week 4
Coaching Calls Setup

Step 4 - Week 5
Attend Mock Interview with feedback

Step 5 - Week 6-7
Resumes, Coaching, Courses

Certificate
DOWNLOAD YOUR CERTIFICATE OF COMPLETION

TECH COMMANDERS
ADVANCING YOUR IT SKILLS

TOP FIVE REASONS TO BECOME A FINOPS CERTIFIED PROFESSIONAL

What is FinOps? FinOps is the practice of solid cloud financial management. Cross-functional teams work to identify the optimal trade-off between cost, speed, and agility, which is a balance guided by the strategic and tactical needs of the business.

The infographic has a dark blue header and a light blue body. It lists five reasons: 1. Applicant Distinguishment: 'The FinOps Practitioner/Professional are non vendor specific certifications that have become the preeminent FinOps Certifications. You will likely stand out from other applicants' with a person holding a certificate icon. 2. Companies Need You: 'Cloud Cost Management is highly important skillset to companies' with a person at a laptop icon. 3. Generous Salary Range: 'The average salary in the US is \$115,000 /year according to ZipRecruiter 09/22. Certain metros having high demand in CA, WA, TX, MA, NY' with a stack of money icon. 4. Provides Validation: 'Passing the challenging exams will provide you a sense of accomplishment and will distinguish you as an expert in the field' with a person at a laptop icon. 5. Career Options: 'Roles can be both FinOps dedicated or hybrid e.g. Architect/Operations/management' with a person at a laptop icon. A final box says '6) BASE KNOWLEDGE OF CLOUD AND FINOPS' with text 'After passing the exams you will have proven skills in Cloud and Cost Management' and a cloud icon. The Tech Commanders logo and website are at the bottom.

1) APPLICANT DISTINGUISHMENT
The FinOps Practitioner/Professional are non vendor specific certifications that have become the preeminent FinOps Certifications. You will likely stand out from other applicants

2) COMPANIES NEED YOU.
Cloud Cost Management is highly important skillset to companies

3) GENEROUS SALARY RANGE
The average salary in the US is \$115,000 /year according to ZipRecruiter 09/22. Certain metros having high demand in CA, WA, TX, MA, NY

4) PROVIDES VALIDATION
Passing the challenging exams will provide you a sense of accomplishment and will distinguish you as an expert in the field.

5) CAREER OPTIONS
Roles can be both FinOps dedicated or hybrid e.g. Architect/Operations/management

6) BASE KNOWLEDGE OF CLOUD AND FINOPS
After passing the exams you will have proven skills in Cloud and Cost Management

TECH COMMANDERS
ADVANCING YOUR IT SKILLS

www.techcommanders.com/finopscourses

TOP REASONS TO BECOME A BLOCKCHAIN ARCHITECT

ENJOY A BOOMING CAREER

The infographic has a dark blue header and a light blue body. It lists four reasons: 1. Career Growth: 'CAREER GROWTH IN THE TOP TEN ROLES' with text 'LinkedIn, RHIL, Talent, have great insight into the demand' and a person at a laptop icon. 2. Reasonable Barrier to Entry: 'A degree is generally not needed and with only 6 months of preparation/coaching/training.' with a person at a laptop icon. 3. Technically Challenging: 'Did you know that a Blockchain Architect needs to be enabled in: Enterprise blockchains, cryptos, networking, security, sales, etc' with a person at a laptop icon. 4. Socially Rewarding: 'Blockchain Architects in a lot of roles are working on sales teams and meeting with different enterprise clients routinely' with a group of people icon. A final box says '5) CAREER OPTIONS' with text 'Roles can be both FinOps dedicated or hybrid e.g. Architect/Operations/management' and a person at a laptop icon. The Tech Commanders logo and website are at the bottom.

CAREER GROWTH IN THE TOP TEN ROLES
LinkedIn, RHIL, Talent, have great insight into the demand

REASONABLE BARRIER TO ENTRY
A degree is generally not needed and with only 6 months of preparation/coaching/training.

TECHNICALLY CHALLENGING
Did you know that a Blockchain Architect needs to be enabled in: Enterprise blockchains, cryptos, networking, security, sales, etc

SOCIALLY REWARDING
Blockchain Architects in a lot of roles are working on sales teams and meeting with different enterprise clients routinely

GENEROUS SALARIES
Did your know that a Cloud Architect averages \$175,000 a year in the USA - Talent.com

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Interview Coaching and Enablement

Get ready to interview and get the offer.

Cloud InterviewACE Program Overview



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Course Closeout

Thank you for joining the course.

I wish you much success in your career.

Please reach out if I can be of assistance.

